

ISC Class XII Mathematics

Chapter 1 Relations and Functions

138 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 1.1 – Types of Relations: Reflexive, Symmetric, Transitive & Equivalence Relations
- II. 1.2 – Functions: Domain and Range
- III. 1.3 – One-One, Onto and Bijective Functions
- IV. 1.4 – Composition of Functions
- V. 1.5 – Invertible Functions and Inverse of a Function

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Q. No.	Topic / Statement Summary	Type No.	Kind	Section
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Exercise 1.1 – Types of Relations: Reflexive, Symmetric, Transitive & Equivalence Relations

1	If $A = \{-1, 1, 3\}$, then what is the number of relations on A ?	1	Numerical	1.1
2	(i) If $R = \{(x, y) : x + 2y = 8\}$ is a relation on $\mathbb{N}$, write the range of ...	2	Numerical	1.1
3	Let A be any non-empty set. State true or false:	3	Numerical	1.1
(i)	Identity relation on A is reflexive.	3	Numerical	1.1
(ii)	Every reflexive relation on A is identity relation on A .	3	Numerical	1.1
(iii)	Identity relation on A is symmetric.	3	Numerical	1.1
(iv)	Identity relation on A is an equivalence relation.	3	Numerical	1.1
(v)	Universal relation on A is an equivalence relation.	3	Numerical	1.1
4	State the reason for the relation R in the set $\{1, 2, 3\}$ given by ...	4	Numerical	1.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
5	The relation R on $A = \{1, 2, 3\}$ is defined by $R = \{(1, 2), (2, 1)\}$. What kind of ...	4	Numerical	1.1
6	Write the smallest equivalence relation on A where $A = \{1, 2, 3\}$.	3	Numerical	1.1
7	If $A = \{0, 1, 2, \dots, 9\}$ and the relation R on A is defined by ...	4	Numerical	1.1
8	Is the relation R on the set $\mathbb{R}$ of real numbers defined by ...	5	Numerical	1.1
9	Is the relation R on the set $\mathbb{Q}$ of rational numbers defined by ...	5	Numerical	1.1
10	If a relation R on the set $\{a, b, c\}$ is defined by $R = \{(b, b)\}$, then classify ...	4	Numerical	1.1
11	If the relation R on the set $A = \{1, 2, 3\}$ is defined by ...	4	Numerical	1.1
12	How many reflexive relations are possible on a set A of order 3?	6	Numerical	1.1
13	If R be the relation in the set $\{1, 2, 3, 4\}$ given by ...	4	Numerical	1.1
(i)	R is reflexive and symmetric but not transitive.	4	Numerical	1.1
(ii)	R is reflexive and transitive but not symmetric.	4	Numerical	1.1
(iii)	R is symmetric and transitive but not reflexive.	4	Numerical	1.1
14	Determine whether each of the following relations are reflexive, symmetric and ...	5	Numerical	1.1
(i)	Relation R in the set $A = \{1, 2, 3, \dots, 10\}$ defined by ...	5	Numerical	1.1
(ii)	Relation R in the set $\mathbb{Z}$ of all integers defined by ...	5	Numerical	1.1
(iii)	Relation R on the set $A = \{1, 2, 3, 4, 5, 6\}$ defined by ...	5	Numerical	1.1
15	If the relation R in the set A , where $A = \{1, 2, 3, 4, 5, 6\}$, is defined by ...	4	Numerical	1.1
16	If R is the relation defined on the set of natural numbers $\mathbb{N}$ as follows: ...	2	Numerical	1.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
17	Determine whether each of the following relations in the set A of human beings in a ...	5	Numerical	1.1
(i)	$R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$	5	Numerical	1.1
(ii)	$R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$	5	Numerical	1.1
(iii)	$R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$	5	Numerical	1.1
(iv)	$R = \{(x, y) : x \text{ and } y \text{ live within 2 kilometres}\}$	5	Numerical	1.1
18	Determine whether each of the following relations in the set A of students at a ...	5	Numerical	1.1
(i)	$R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are honest}\}$	5	Numerical	1.1
(ii)	$R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are obedient}\}$	5	Numerical	1.1
(iii)	$R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are hardworking}\}$	5	Numerical	1.1
19	Show that the relation R in the set $A = \{1, 2, 3\}$ given by ...	4	Proof	1.1
20	Explain why $x + 4y = 12$, $x, y \in N$ is NOT a symmetric relation.	4	Numerical	1.1
21	Let R be a relation defined over N , where N is the set of natural numbers, defined ...	4	Numerical	1.1
22	Let R be the relation on W (set of whole numbers) defined by ...	2	Numerical	1.1
23	Let R be the relation defined on $\mathbb{Q}_0$ (set of non-zero rational numbers) ...	5	Numerical	1.1
24	A relation R is defined on a set of real numbers $\mathbb{R}$ as ...	5	Numerical	1.1
25	Check whether the relation R on the set $\mathbb{N}$ of natural numbers given by ...	5	Numerical	1.1
26	Let R be the relation on $\mathbb{R}$ (set of real numbers) defined by ...	5	Proof	1.1
27	Determine whether each of the following relations are reflexive, symmetric and ...	5	Numerical	1.1
(i)	$R = \{(a, b) : a, b \in \mathbb{N}, a < b\}$	5	Numerical	1.1
(ii)	$R = \{(a, b) : a, b \in \mathbb{R}, a \leq b\}$	5	Numerical	1.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
28	Let T be the set of all triangles drawn in a plane with R as a relation in T given ...	7	Proof	1.1
29	Show that the relation R in the set A of all books in a library of a school, given by ...	7	Proof	1.1
30	Show that the relation R on the set $\mathbb{Z}$ of all integers defined by ...	7	Proof	1.1
31	Show that the relation R on the set $\mathbb{Z}$ of all integers defined by ...	8	Numerical	1.1
32	Prove that the relation R in the set $A = \{5, 6, 7, 8, 9\}$, given by ...	8	Numerical	1.1
33	Give examples of relations which are	9	Numerical	1.1
(i)	symmetric but neither reflexive nor transitive.	9	Numerical	1.1
(ii)	transitive but neither reflexive nor symmetric.	9	Numerical	1.1
(iii)	symmetric and transitive but not reflexive.	9	Numerical	1.1
(iv)	symmetric and reflexive but not transitive.	9	Numerical	1.1
(v)	reflexive and transitive but not symmetric.	9	Numerical	1.1
34	Give an example of a relation R on $A = \{a, b, c\}$ which is	9	Numerical	1.1
(i)	neither reflexive nor symmetric but transitive.	9	Numerical	1.1
(ii)	neither symmetric nor transitive but reflexive.	9	Numerical	1.1
(iii)	neither transitive nor reflexive but symmetric.	9	Numerical	1.1
35	Show that the relation S in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ given by ...	8	Numerical	1.1
36	Show that the relation R in the set $A = \{x \in \mathbb{W}, 0 \leq x \leq 17\}$...	8	Numerical	1.1
(i)	$R = \{(a, b) : a - b \text{ is a multiple of } 5\}$	8	Numerical	1.1
(ii)	$R = \{(a, b) : a = b\}$	8	Numerical	1.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
37	Show that the relation R defined by $(a, b) R (c, d) \iff a + d = b + c$ on ...	10	Numerical	1.1
38	Show that the relation R in the set $\mathbb{N} \times \mathbb{N}$ defined by ...	10	Proof	1.1
39	The Earth has 24 time zones, defined by dividing the Earth into 24 equal longitudinal ...	5	Numerical	1.1
(i)	Check whether the relation R is reflexive, symmetric and transitive.	5	Numerical	1.1
(ii)	Is relation R an equivalence relation?	5	Numerical	1.1

Exercise 1.2 – Functions: Domain and Range

1	A function is defined by $f(x) = \frac{3x^2 + 2x - 1}{x + 1}$...	11	Numerical	1.2
2	(i) Is the relation f defined by $f(x) =$ (piecewise definition) a function? ...	12	Numerical	1.2
3	A real function f is defined by $f(x) = 3x - 2, x \in \mathbb{R}$, find the value of ...	11	Numerical	1.2
4	If a real function f is defined by $f(x) = 2x^2 - 3$, find x such that $f(x) = 15$.	13	Numerical	1.2
5	Given $f(x) = x^3 - 1$, find x if $f(x)$ is 215.	13	Numerical	1.2
6	A function is given by the formula $f(x) = 144 - 16x^2$. Calculate $f(2)$. Also find the ...	13	Numerical	1.2
7	A function ' f ' is defined by $f(x) = 2x^2 + 3$, for all $x \in \mathbb{R}$. Find (i) ...	13	Numerical	1.2
8	If $f(x) = 2x^2 + 1$ and domain of $f = \{-2, -1, 0, 1, 3\}$, find (i) range of f (ii) ...	14	Numerical	1.2
9	A function ' f ' is defined by $f(x) = x^2 + 3, x \in \mathbb{N}$ and $x \leq 5$. (i) Find ...	14	Numerical	1.2
10	Find the domain of the following real functions: (i) $f(x) = x + 2 $ (ii) ...	15	Numerical	1.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
11	Find the range of the following functions: (i) $f(x) = x^2 + 2, x \in \mathbb{R}$ (ii) ...	16	Numerical	1.2
12	Find the domain and the range of the function $f(x) = 2x^2 + 1$. Also find $f(-2)$ and the ...	16	Numerical	1.2
13	Find the domain and the range of the following functions: (i) $f(x) = \sqrt{x+2}$ (ii) ...	17	Numerical	1.2
14	Find the domain and the range of the following functions: (i) ...	18	Numerical	1.2
15	Find the domain and the range of the following functions: (i) $f(x) = \sqrt{16 - x^2}$...	17	Numerical	1.2
16	Find the domain and the range of the following functions: (i) $f(x) = x - 3 $ (ii) ...	15	Numerical	1.2
17	If a real function f is defined by $f(x) = \frac{ x - x}{2x}$, then find its domain ...	15	Numerical	1.2
18	Find the domain and the range of the function f defined by ...	15	Numerical	1.2
19	If $f(x) = \log\left(\frac{1+x}{1-x}\right)$, then show that ...	19	Proof	1.2
20	Find the domain of the following functions: (i) $\log(4x - 3)$ (ii) ...	20	Numerical	1.2
21	Find the domain of the following functions: (i) $f(x) = \frac{1}{3 - 2\sin x}$ (ii) ...	21	Numerical	1.2
22	Find the range of the following functions: (i) $f(x) = 2 - 3\cos x$ (ii) ...	21	Numerical	1.2

Exercise 1.3 – One-One, Onto and Bijective Functions

1	Show that the function $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(x) = 3x - 2$ is ...	22	Proof	1.3
2	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 2x - 1$ is not ...	22	Proof	1.3
3	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n^2$ is ...	22	Proof	1.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
4	Is the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(m) = m^2 + m + 2$ one-one? ...	22	Numerical	1.3
5	What is the range of the function $f(x) = \frac{ x-1 }{x-1}$?	15	Numerical	1.3
6	Find the domain of the function $f(x) = \frac{1}{5-3\sin x}$.	21	Numerical	1.3
7	Find the range of the function $f(x) = \frac{1}{2-\cos x}$.	21	Numerical	1.3
8	Find the range of the following functions:	21	Numerical	1.3
(i)	$f(x) = \sin x + \cos x$	21	Numerical	1.3
(ii)	$f(x) = 3\sin x + 4\cos x$	21	Numerical	1.3
9	The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = 2x-3 $.	22	Numerical	1.3
(i)	Is f one-one?	22	Numerical	1.3
(ii)	Is f onto?	22	Numerical	1.3
10	If $A = \{1, 2, 3\}$ and $B = \{3, 5\}$, then write the number of functions from A to ...	23	Numerical	1.3
11	If $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$, then write the number of functions from A ...	23	Numerical	1.3
12	If $A = \{a, b\}$ and $B = \{1, 3, 5\}$, then write the number of one-one functions from ...	23	Numerical	1.3
13	If $A = \{a, b, c\}$ and $B = \{p, q\}$, then write the number of onto functions from A ...	23	Numerical	1.3
14	If $A = \{1, 2, 3, 4\}$ and $B = \{-1, 3\}$, then write the number of onto functions from ...	23	Numerical	1.3
15	If $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, then write the number of one-one onto functions from A to B	23	Numerical	1.3
(i)	one-one onto functions from A to B	23	Numerical	1.3
(ii)	bijections from A to B .	23	Numerical	1.3
16	The given function $f : \mathbb{R} \rightarrow \mathbb{R}$ is not 'onto' function. Give reason.	22	Proof	1.3
17	Let $A = \{1, 2\}$, $B = \{3, 6\}$, function $f : A \rightarrow B$ be defined by $f(x) = 3x$ and ...	24	Proof	1.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
18	Check the injectivity and surjectivity of the following functions:	22	Numerical	1.3
(i)	$f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2$	22	Numerical	1.3
(ii)	$f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^2$	22	Numerical	1.3
(iii)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$	22	Numerical	1.3
(iv)	$f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^3$	22	Numerical	1.3
(v)	$f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^3$	22	Numerical	1.3
(vi)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$	22	Numerical	1.3
19	Let $\mathbb{I}$ be the set of all integers. Show that the function ...	22	Proof	1.3
20	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x^2 - 2$ is ...	22	Proof	1.3
21	Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and $f = \{(1, 4), (2, 5), (3, 6)\}$ be a ...	22	Proof	1.3
22	In each of the following cases, state whether the function is bijective or not ...	22	Numerical	1.3
(i)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 1$	22	Numerical	1.3
(ii)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3 - 4x$	22	Numerical	1.3
(iii)	$f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1 + x^2$	22	Numerical	1.3
23	Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^4$. Show that f is neither ...	22	Proof	1.3
24	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$ is ...	22	Proof	1.3
25	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by ...	22	Proof	1.3
26	Show that the function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = x^2 + x$ is ...	22	Proof	1.3
27	If the function $f : \mathbb{I} \rightarrow \mathbb{I}$ is defined by ...	25	Numerical	1.3
(i)	Find $f(3)$.	25	Numerical	1.3
(ii)	Find $f(4)$.	25	Numerical	1.3
(iii)	Find x , if $f(x) = -9$.	25	Numerical	1.3
28	See parts below:	25	Proof	1.3
(i)	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by ...	25	Proof	1.3
(ii)	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by ...	25	Proof	1.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
29	Find the number of all one-one functions from the set $\{1, 2, 3, 4\}$ to itself.	23	Numerical	1.3
30	Show that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{5x-3}{4} \dots$	22	Proof	1.3
31	Let $f : \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ be a function $\dots$	22	Numerical	1.3
32	If $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$, then show that the function $\dots$	22	Proof	1.3
33	(a) If a real-valued function is given by $f(x) = \sqrt{25 - x^2}$ is an onto function $\dots$	22	Numerical	1.3

Exercise 1.4 – Composition of Functions

1	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = (3 - x^3)^{1/3}$, then find $\dots$	26	Numerical	1.4
2	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = (5 - x^5)^{1/5}$, then find $\dots$	26	Numerical	1.4
3	If $f(x) = x $ and $g(x) = [x]$ for all $x \in \mathbb{R}$, find	27	Numerical	1.4
(i)	$(g \circ f)\left(-\frac{5}{3}\right)$	27	Numerical	1.4
(ii)	$(f \circ g)\left(-\frac{5}{3}\right)$	27	Numerical	1.4
4	If $f(x) = x + 7$ and $g(x) = x - 7$, find $(f \circ g)(7)$.	27	Numerical	1.4
5	If $f : \{2, 3, 4, 5\} \rightarrow \{3, 4, 5, 9\}$ and $g : \{3, 4, 5, 9\} \rightarrow \{7, 11, 15\}$ are $\dots$	28	Numerical	1.4
6	Let $A = \{3, 4, 5, 6\}$, $B = \{13, 14, 15, 16\}$ and $C = \{23, 24, 25\}$. If $\dots$	28	Numerical	1.4
7	Let $A = \{1, 2, 3, 4\}$. If the functions $f : A \rightarrow A$ and $g : A \rightarrow A$ are defined $\dots$	28	Numerical	1.4
(i)	$g \circ f$	28	Numerical	1.4
(ii)	$f \circ g$	28	Numerical	1.4
(iii)	$f \circ f$	28	Numerical	1.4
8	If the functions f and g are given by $f = \{(1, 2), (3, 5), (4, 1)\}$ and $\dots$	28	Numerical	1.4
(i)	find the domain of f and g	28	Numerical	1.4
(ii)	find the range of f and g	28	Numerical	1.4
(iii)	write $f \circ g$ and $g \circ f$ as sets of ordered pairs	28	Numerical	1.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
9	Let $\mathbb{R}$ be the set of all real numbers. Find $g \circ f$ and $f \circ g \dots$	26	Numerical	1.4
(i)	$f(x) = 2x - 1$ and $g(x) = x^2 + 3$ for all $x \in \mathbb{R}$	26	Numerical	1.4
(ii)	$f(x) = x + 1$ and $g(x) = x $ for all $x \in \mathbb{R}$	26	Numerical	1.4
(iii)	$f(x) = x^3$ and $g(x) = 2x^2 + 1$ for all $x \in \mathbb{R}$	26	Numerical	1.4
(iv)	$f(x) = \sin x$ and $g(x) = 5x^2$ for all $x \in \mathbb{R}$	26	Numerical	1.4
10	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x^2 - 3x + 2$ for all $x \in \mathbb{R}$	26	Numerical	1.4
11	Let $\mathbb{R}$ be the set of all real numbers. If the functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^2 + 1$ and $g(x) = x^2 - 1$	26	Numerical	1.4
(i)	$f \circ g$	26	Numerical	1.4
(ii)	$g \circ f$	26	Numerical	1.4
(iii)	$f \circ f$	26	Numerical	1.4
(iv)	$g \circ g$	26	Numerical	1.4
12	If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^2 + 1$ and $g(x) = x^2 - 1$	26	Proof	1.4
13	If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^2 + 1$ and $g(x) = x^2 - 1$	26	Proof	1.4
14	If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^2 + 1$ and $g(x) = x^2 - 1$	26	Numerical	1.4
(i)	$g \circ f$	26	Numerical	1.4
(ii)	$f \circ g$	26	Numerical	1.4
(iii)	$f \circ f$	26	Numerical	1.4
15	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2 + 1$ is not invertible	22	Numerical	1.4

Exercise 1.5 – Invertible Functions and Inverse of a Function

1	If $f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$, find f^{-1} .	29	Numerical	1.5
2	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = 3x + 5$, then find f^{-1} .	30	Numerical	1.5
3	If $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{3x-2}{5}$ is invertible $\dots$	30	Numerical	1.5
4	If $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x-7}{4}$ is invertible $\dots$	30	Numerical	1.5
5	If f is an invertible function defined as $f(x) = \frac{3x-4}{5}$, write $f^{-1}(x)$.	30	Numerical	1.5
6	If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = ax - b \dots$	30	Numerical	1.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
7	If $f : \mathbb{R} - \left\{-\frac{3}{5}\right\} \rightarrow \mathbb{R}$ is a function defined by ...	30	Numerical	1.5
8	If $f : \mathbb{R} - \left\{-\frac{4}{3}\right\} \rightarrow \mathbb{R}$ is a function defined by ...	30	Numerical	1.5
9	If $f(x) = [4 - (x - 7)^3]^{1/5}$ is a real invertible function, then find $f^{-1}(x)$.	30	Numerical	1.5
10	The function $f : X \rightarrow Y$ is not invertible. State the reason.	31	Numerical	1.5
11	Which one of the following graphs is a function of x ?	32	Numerical	1.5
12	Let $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$ be a function defined by $f(1) = a$, $f(2) = b$ and ...	33a	Proof	1.5
13	Let $S = \{1, 2, 3\}$. Determine whether the functions $f : S \rightarrow S$ defined as below ...	33a	Numerical	1.5
(i)	$f = \{(1, 1), (2, 2), (3, 3)\}$	33a	Numerical	1.5
(ii)	$f = \{(1, 2), (2, 1), (3, 1)\}$	33a	Numerical	1.5
(iii)	$f = \{(1, 3), (3, 2), (2, 1)\}$	33a	Numerical	1.5
(iv)	$f = \{(1, 3), (2, 2), (3, 1)\}$	33a	Numerical	1.5
14	Explain why for the following functions $f : X \rightarrow Y$, we cannot define f^{-1} :	31	Proof	1.5
(i)	$X = \mathbb{R} - \{0\}, Y = \mathbb{R}$ and $f(x) = \frac{1}{x}$ for all ...	31	Proof	1.5
(ii)	$X = \{-1, 0, 1, 2\}, Y = \{2, 3\}$ and $f(x) = 2$ for $x = -1, 0, 1$ and ...	31	Proof	1.5
(iii)	$X = \mathbb{R} = Y$ and $f(x) = x^2 + 1$ for all $x \in \mathbb{R}$	31	Proof	1.5
15	Let $S = \{a, b, c\}$ and $T = \{1, 2, 3\}$. Find f^{-1} of the following functions ...	29	Numerical	1.5
(i)	$f = \{(a, 3), (b, 2), (c, 1)\}$	29	Numerical	1.5
(ii)	$f = \{(a, 2), (b, 1), (c, 1)\}$	29	Numerical	1.5
16	Let $\mathbb{R}$ be the set of all real numbers. Show that the function ...	33b	Proof	1.5
17	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by ...	33b	Proof	1.5
18	Consider a bijective function $f : \mathbb{R}_0^+ \rightarrow (7, \infty)$ given by ...	33b	Numerical	1.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
19	If $A = \mathbb{R} - \{-3\}$ and $B = \mathbb{R} - \{2\}$ and a function $f : A \rightarrow B$ be ...	33b	Proof	1.5
20	If $f : A \rightarrow A$ and $A = \mathbb{R} - \{\frac{8}{5}\}$, show that the function ...	33b	Proof	1.5
21	Consider the function $f : \mathbb{R}_+ \rightarrow [9, \infty)$ defined by $f(x) = x^2 + 9$...	33b	Proof	1.5
22	If $\mathbb{R}_0$ is the set of all non-zero real numbers, show that the function ...	33b	Proof	1.5
23	Prove that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2 + x + 1$ is ...	33b	Proof	1.5
24	Let $f : \mathbb{N} \rightarrow S$ be a function defined by $f(x) = 4x^2 + 12x + 15$, where S is ...	33b	Proof	1.5
25	Let $f : \mathbb{N} \rightarrow S$ be a function defined by $f(x) = 9x^2 + 6x - 5$, where S is ...	33b	Numerical	1.5
26	If function $f : \mathbf{R} \rightarrow \mathbf{R}$ is defined by $f(x) = 5x - 3$, then find ...	34	Numerical	1.5
27	A teacher wrote $f(x) = x^9$ on the board and asked her student to examine whether the ...	35	Numerical	1.5
28	A part of the graph of the function $f(x) = 2x^3 - 3x^2 - 12x + 8, x \in \mathbf{R}$...	36	Numerical	1.5
(i)	Explain why 'f' does not have an inverse.	36	Numerical	1.5
(ii)	The domain of 'f' is now restricted to $a \leq x \leq b$ where $a < 0$ and ...	36	Numerical	1.5
29	"A function f is called a self-inverse function if $f^{-1}(x) = f(x)$ for all values of ...	37	Proof	1.5

Formula & Standard Results Sheet

I. Relations – Types and Equivalence Relations (Ex 1.1, 1.2)

Number of Relations

$$n(A \times B) = n(A) \cdot n(B), \quad \text{Total relations from } A \text{ to } B = 2^{n(A) \cdot n(B)}$$

On a set A alone, total relations on $A = 2^{n(A)^2}$

Domain and Range of a Relation

$$\text{Domain}(R) = \{x : (x, y) \in R\}, \quad \text{Range}(R) = \{y : (x, y) \in R\}$$

Domain is the set of first components; Range is the set of second components

Reflexive Relation

$$(a, a) \in R \quad \forall a \in A$$

Every element of A must be related to itself

Symmetric Relation

$$(a, b) \in R \implies (b, a) \in R$$

Must hold for every pair in R , not just some

Transitive Relation

$$(a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R$$

Check every chain of two related pairs

Equivalence Relation

R is reflexive + symmetric + transitive

A relation is an equivalence relation only if all three properties hold simultaneously

Identity and Universal Relations

$$I_A = \{(a, a) : a \in A\}, \quad \text{Universal relation} = A \times A$$

I_A is always reflexive, symmetric, transitive; the universal relation is always an equivalence relation

Equivalence Class

$$[a] = \{x \in A : (x, a) \in R\}$$

The equivalence class of a collects every element related to a ; equivalence classes partition A

II. Functions – Domain and Range (Ex 1.2)**Value of a Function**

$$f(a) = f(x)|_{x=a}$$

Substitute the given input directly into the rule for f

Range of a Function

$$\text{Range}(f) = \{f(x) : x \in \text{Domain}(f)\}$$

Found by expressing x in terms of $y = f(x)$ and finding the valid values of y

Modulus Function

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Domain is all of $\mathbb{R}$; always defined and non-negative

Square-Root Type Functions

$$\sqrt{g(x)} \text{ needs } g(x) \geq 0, \quad \frac{1}{\sqrt{g(x)}} \text{ needs } g(x) > 0$$

A root in the denominator forces the strict inequality since division by zero is undefined

Rational Function

$$\frac{P(x)}{Q(x)}, \quad Q(x) \neq 0$$

Domain excludes every root of the denominator $Q(x)$

Logarithmic Function

$$\log(g(x)) \text{ is defined only when } g(x) > 0$$

Also exclude any point where a denominator inside $g(x)$ vanishes

Range of Trigonometric Functions

$$-1 \leq \sin x \leq 1, \quad -1 \leq \cos x \leq 1$$

Used to bound the range of any function built from $\sin x$ or $\cos x$

Range of $a \sin x + b \cos x$

$$-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$$

Standard bound for a linear combination of sine and cosine

III. One-One, Onto and Bijective Functions (Ex 1.3)**One-One (Injective) Function**

$$f(x_1) = f(x_2) \implies x_1 = x_2, \quad \forall x_1, x_2 \in \text{Domain}(f)$$

Equivalently: distinct inputs always give distinct outputs

Onto (Surjective) Function

$$\forall y \in \text{Codomain}, \exists x \in \text{Domain such that } f(x) = y$$

Equivalently: $\text{Range}(f) = \text{Codomain}(f)$

Bijective Function

$$f \text{ is bijective} \iff f \text{ is both one-one and onto}$$

A bijective function is precisely the kind of function that is invertible

Number of Functions from A to B

$$n(B)^{n(A)}$$

Every element of A can independently map to any of the $n(B)$ elements of B

Number of One-One Functions

$$n(B)P_{n(A)} = \frac{n(B)!}{(n(B) - n(A))!}, \quad \text{valid only when } n(A) \leq n(B)$$

Equals 0 when $n(A) > n(B)$, since no injective map can exist

Number of Bijections

$$n! \quad \text{when } n(A) = n(B) = n$$

Every bijection between two equal-sized finite sets is a permutation of one onto the other

IV. Composition of Functions (Ex 1.4)**Composition of Functions**

$$(g \circ f)(x) = g(f(x))$$

Apply f first, then apply g to the result; $g \circ f$ is defined only where $\text{Range}(f) \subseteq \text{Domain}(g)$

Composition on Finite Sets

$(g \circ f)(x) = g(f(x))$ computed pair-by-pair from the ordered-pair definitions of f, g
 Trace each element of the domain through f first, then through g

V. Invertible Functions and Inverse of a Function (Ex 1.5)**Invertibility Criterion**

f is invertible $\iff f$ is bijective (one-one and onto)
 This is the single test to apply before attempting to find f^{-1}

Finding the Inverse from Ordered Pairs

$f = \{(x, y)\} \implies f^{-1} = \{(y, x) : (x, y) \in f\}$
 Reverse every ordered pair; valid only once bijectivity (or a roster-level one-one/onto check) is confirmed

Inverse via Equation Solving

$y = f(x) \iff x = f^{-1}(y)$
 Solve the equation $y = f(x)$ for x in terms of y , then relabel $y \rightarrow x$ to state $f^{-1}(x)$

Composition Test for Inverse Functions

$g \circ f = I_{\text{domain}}$ and $f \circ g = I_{\text{codomain}} \iff g = f^{-1}$
 The standard way to verify a claimed inverse is correct

Inverse of an Inverse

$(f^{-1})^{-1} = f$
 Reversing the reverse of a bijection returns the original function

Self-Inverse Function

$f^{-1} = f \iff (f \circ f)(x) = x \quad \forall x \in \text{Domain}(f)$
 Check by composing f with itself and confirming it returns the identity

Monotonicity Test for Injectivity

$f'(x) \geq 0 \quad \forall x$ (or $f'(x) \leq 0 \quad \forall x$) $\implies f$ is monotonic $\implies f$ is one-one
 Useful for proving injectivity of continuous algebraic functions without testing $f(x_1) = f(x_2)$ directly

1.1 – Types of Relations: Reflexive, Symmetric, Transitive & Equivalence Relations

Question 1

If $A = \{-1, 1, 3\}$, then what is the number of relations on A ?

TYPE 1 Total Number of Relations

Given: $A = \{-1, 1, 3\}$

Concept / Formula Used: $n(A) = m$ $A \times A$ m^2 A 2^{m^2}

Solution:

The set A contains 3 elements:

$$n(A) = 3$$

The number of elements in the Cartesian product $A \times A$ is given by:

$$\begin{aligned} n(A \times A) &= n(A) \times n(A) \\ &= 3 \times 3 \\ &= 9 \end{aligned}$$

Since any relation on A is a subset of $A \times A$, the total number of relations on A is the total number of subsets of $A \times A$:

$$\begin{aligned} \text{Total number of relations} &= 2^{n(A \times A)} \\ &= 2^9 \\ &= 512 \end{aligned}$$

Final Answer

The total number of relations on A is 512.

Question 2

- (i) If $R = \{(x, y) : x + 2y = 8\}$ is a relation on $\mathbb{N}$, write the range of R .
 (ii) Let $R = \{(a, a^3) : a \text{ is a prime number less than } 5\}$. Find the range of R .

TYPE 2 Domain and Range of a Relation

Given: $R_1 = \{(x, y) : x + 2y = 8, x, y \in \mathbb{N}\}$, $R_2 = \{(a, a^3) : a \text{ is prime, } a < 5\}$

Concept / Formula Used: $R \quad R \quad \text{Range}(R) = \{y : (x, y) \in R\}$

Solution:

Part (i): Given equation:

$$\begin{aligned}x + 2y &= 8 \\2y &= 8 - x \\y &= \frac{8 - x}{2}\end{aligned}$$

Since $x, y \in \mathbb{N} = \{1, 2, 3, \dots\}$, we substitute positive integer values for x :

$$\begin{aligned}\text{For } x = 1 : \quad y &= \frac{8 - 1}{2} = \frac{7}{2} \notin \mathbb{N} \\ \text{For } x = 2 : \quad y &= \frac{8 - 2}{2} = 3 \in \mathbb{N} \implies (2, 3) \in R \\ \text{For } x = 3 : \quad y &= \frac{8 - 3}{2} = \frac{5}{2} \notin \mathbb{N} \\ \text{For } x = 4 : \quad y &= \frac{8 - 4}{2} = 2 \in \mathbb{N} \implies (4, 2) \in R \\ \text{For } x = 5 : \quad y &= \frac{8 - 5}{2} = \frac{3}{2} \notin \mathbb{N} \\ \text{For } x = 6 : \quad y &= \frac{8 - 6}{2} = 1 \in \mathbb{N} \implies (6, 1) \in R \\ \text{For } x \geq 7 : \quad y &\leq 0 \notin \mathbb{N}\end{aligned}$$

Therefore, in roster form:

$$R = \{(2, 3), (4, 2), (6, 1)\}$$

The range of R consists of the second elements:

$$\text{Range}(R) = \{1, 2, 3\}$$

Part (ii): Prime numbers less than 5 are 2 and 3. Calculating a^3 for each prime a :

$$\begin{aligned}\text{For } a = 2 : \quad a^3 &= 2^3 = 8 \implies (2, 8) \in R \\ \text{For } a = 3 : \quad a^3 &= 3^3 = 27 \implies (3, 27) \in R\end{aligned}$$

In roster form:

$$R = \{(2, 8), (3, 27)\}$$

The range of R consists of the second elements:

$$\text{Range}(R) = \{8, 27\}$$

Final Answer

- (i) Range of $R = \{1, 2, 3\}$
(ii) Range of $R = \{8, 27\}$

Question 3

Let A be any non-empty set. State true or false:

- (i) Identity relation on A is reflexive.
- (ii) Every reflexive relation on A is identity relation on A .
- (iii) Identity relation on A is symmetric.
- (iv) Identity relation on A is an equivalence relation.
- (v) Universal relation on A is an equivalence relation.

TYPE 3 Properties of Special Relations

Given: $A \neq \emptyset$

Concept / Formula Used: $I_A = \{(a, a) : a \in A\}$ $U = A \times A$

Solution:

- (i) **True.** The identity relation $I_A = \{(a, a) : a \in A\}$ contains (a, a) for every element $a \in A$. Hence, it satisfies the definition of reflexivity.
- (ii) **False.** A reflexive relation must contain all pairs (a, a) , but it can also contain other additional pairs (a, b) with $a \neq b$. An identity relation can contain *only* pairs of the form (a, a) .
- (iii) **True.** For every $(a, a) \in I_A$, reversing the order gives (a, a) , which is also in I_A . Hence, it is symmetric.
- (iv) **True.** I_A is reflexive, symmetric, and transitive, so it is an equivalence relation.
- (v) **True.** The universal relation $U = A \times A$ contains all possible ordered pairs of elements of A , so it is reflexive, symmetric, and transitive, making it an equivalence relation.

Final Answer

(i) True (ii) False (iii) True (iv) True (v) True

Question 4

State the reason for the relation R in the set $\{1, 2, 3\}$ given by $R = \{(1, 2), (2, 1)\}$ not to be transitive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 1)\}$

Concept / Formula Used: $R \quad (a, b) \in R \quad (b, c) \in R \implies (a, c) \in R$

Solution:

We examine the ordered pairs present in R :

$$(1, 2) \in R \quad \text{and} \quad (2, 1) \in R$$

For the relation R to be transitive, the pair formed by the first element of the first pair and the second element of the second pair, which is $(1, 1)$, must belong to R .

Checking the set R :

$$(1, 1) \notin R$$

Since $(1, 2) \in R$ and $(2, 1) \in R$ but $(1, 1) \notin R$, R is not transitive.

Final Answer

R is not transitive because $(1, 2) \in R$ and $(2, 1) \in R$, but $(1, 1) \notin R$.

Question 5

The relation R on $A = \{1, 2, 3\}$ is defined by $R = \{(1, 2), (2, 1)\}$. What kind of the relation R is?

TYPE 4 Verifying Relation Properties on Finite Sets

★ Likely Exam Question

Given: $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 1)\}$

Concept / Formula Used: $R \text{ is transitive} \iff (a, a) \in R \quad a \in A \quad (a, b) \in R \implies (b, a) \in R \quad (a, b) \in R \quad (b, c) \in R \implies (a, c) \in R$

Solution:

1. **Reflexivity:**

$$1 \in A \quad \text{but} \quad (1, 1) \notin R$$

Since $(a, a) \notin R$ for $a = 1$, R is not reflexive.

2. **Symmetry:**

$$\begin{aligned} (1, 2) \in R &\implies (2, 1) \in R \\ (2, 1) \in R &\implies (1, 2) \in R \end{aligned}$$

Since for every $(a, b) \in R$, $(b, a) \in R$, the relation R is symmetric.

3. **Transitivity:**

$$(1, 2) \in R \quad \text{and} \quad (2, 1) \in R \quad \text{but} \quad (1, 1) \notin R$$

Hence, R is not transitive.

Final Answer

R is a symmetric relation, but neither reflexive nor transitive.

Question 6

Write the smallest equivalence relation on A where $A = \{1, 2, 3\}$.

TYPE 3 Properties of Special Relations

★ **Likely Exam Question**

Given: $A = \{1, 2, 3\}$

Concept / Formula Used: $A \quad I_A = \{(a, a) : a \in A\}$

Solution:

For a relation R on $A = \{1, 2, 3\}$ to be an equivalence relation, it must be reflexive, symmetric, and transitive.

To satisfy reflexivity, R must contain:

$$(1, 1), (2, 2), \text{ and } (3, 3)$$

Consider the set:

$$R_0 = \{(1, 1), (2, 2), (3, 3)\}$$

Checking properties of R_0 : 1. Reflexive: $(a, a) \in R_0$ for all $a \in \{1, 2, 3\}$. 2. Symmetric: For each $(a, a) \in R_0$, reversing it gives $(a, a) \in R_0$. 3. Transitive: $(a, a) \in R_0$ and $(a, a) \in R_0 \implies (a, a) \in R_0$.

Thus, R_0 is an equivalence relation with the minimum possible number of elements (3 elements).

Final Answer

$$R = \{(1, 1), (2, 2), (3, 3)\}$$

Question 7

If $A = \{0, 1, 2, \dots, 9\}$ and the relation R on A is defined by $R = \{(x, y) : x, y \in A, y = 2x + 1\}$, then determine whether the relation R is (i) reflexive (ii) symmetric (iii) transitive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $y = 2x + 1$

Solution:

First, we list the elements of relation R by choosing $x \in A$ such that $y = 2x + 1 \in A$:

$$x = 0 \implies y = 2(0) + 1 = 1 \in A \implies (0, 1) \in R$$

$$x = 1 \implies y = 2(1) + 1 = 3 \in A \implies (1, 3) \in R$$

$$x = 2 \implies y = 2(2) + 1 = 5 \in A \implies (2, 5) \in R$$

$$x = 3 \implies y = 2(3) + 1 = 7 \in A \implies (3, 7) \in R$$

$$x = 4 \implies y = 2(4) + 1 = 9 \in A \implies (4, 9) \in R$$

$$x = 5 \implies y = 2(5) + 1 = 11 \notin A$$

Thus, in roster form:

$$R = \{(0, 1), (1, 3), (2, 5), (3, 7), (4, 9)\}$$

(i) **Reflexivity:**

$$0 \in A \quad \text{but} \quad (0, 0) \notin R$$

Hence, R is not reflexive.

(ii) **Symmetry:**

$$(0, 1) \in R \quad \text{but} \quad (1, 0) \notin R$$

Hence, R is not symmetric.

(iii) **Transitivity:**

$$(0, 1) \in R \quad \text{and} \quad (1, 3) \in R \quad \text{but} \quad (0, 3) \notin R$$

Hence, R is not transitive.

Final Answer

R is neither reflexive, nor symmetric, nor transitive.

Question 8

Is the relation R on the set $\mathbb{R}$ of real numbers defined by $R = \{(a, b) : a, b \in \mathbb{R}, 1 + ab \geq 0\}$ transitive? Justify your answer.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $R = \{(a, b) \in \mathbb{R} \times \mathbb{R} : 1 + ab \geq 0\}$

Concept / Formula Used: $a, b, c \quad (a, b) \in R \quad (b, c) \in R \quad (a, c) \notin R$

Solution:

Let us test the relation with specific real numbers. Choose $a = 1$, $b = -\frac{1}{2}$, and $c = -3$.

1. Check if $(a, b) = \left(1, -\frac{1}{2}\right) \in R$:

$$\begin{aligned} 1 + ab &= 1 + (1) \left(-\frac{1}{2}\right) \\ &= 1 - \frac{1}{2} \\ &= \frac{1}{2} \geq 0 \end{aligned}$$

Thus, $\left(1, -\frac{1}{2}\right) \in R$.

2. Check if $(b, c) = \left(-\frac{1}{2}, -3\right) \in R$:

$$\begin{aligned} 1 + bc &= 1 + \left(-\frac{1}{2}\right)(-3) \\ &= 1 + \frac{3}{2} \\ &= \frac{5}{2} \geq 0 \end{aligned}$$

Thus, $\left(-\frac{1}{2}, -3\right) \in R$.

3. Check if $(a, c) = (1, -3) \in R$:

$$\begin{aligned} 1 + ac &= 1 + (1)(-3) \\ &= 1 - 3 \\ &= -2 < 0 \end{aligned}$$

Thus, $(1, -3) \notin R$.

Since $\left(1, -\frac{1}{2}\right) \in R$ and $\left(-\frac{1}{2}, -3\right) \in R$, but $(1, -3) \notin R$, the relation R fails transitivity.

Final Answer

No, R is not transitive because $\left(1, -\frac{1}{2}\right) \in R$ and $\left(-\frac{1}{2}, -3\right) \in R$, but $(1, -3) \notin R$.

Question 9

Is the relation R on the set $\mathbb{Q}$ of rational numbers defined by $R = \{(x, y) : x, y \in \mathbb{Q}, x < y^2\}$, symmetric? Justify your answer.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $R = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} : x < y^2\}$

Concept / Formula Used: $R \quad x, y \in \mathbb{Q} \quad (x, y) \in R \quad (y, x) \notin R$

Solution:

Let us choose a counterexample with rational numbers. Take $x = 1$ and $y = 5$.

1. Check if $(x, y) = (1, 5) \in R$:

$$y^2 = 5^2 = 25$$

Since $1 < 25$, the condition $x < y^2$ holds true. Thus, $(1, 5) \in R$.

2. Check if $(y, x) = (5, 1) \in R$:

$$x^2 = 1^2 = 1$$

Since $5 \not< 1$, the condition $y < x^2$ is false. Thus, $(5, 1) \notin R$.

Since $(1, 5) \in R$ but $(5, 1) \notin R$, the relation R is not symmetric.

Final Answer

No, R is not symmetric because $(1, 5) \in R$ as $1 < 5^2$, but $(5, 1) \notin R$ as $5 \not< 1^2$.

Question 10

If a relation R on the set $\{a, b, c\}$ is defined by $R = \{(b, b)\}$, then classify the relation.

TYPE 4 Verifying Relation Properties on Finite Sets

★ Likely Exam Question

Given: $A = \{a, b, c\}$, $R = \{(b, b)\}$

Concept / Formula Used: R

Solution:

1. **Reflexivity:**

$$a \in A \quad \text{and} \quad c \in A \quad \text{but} \quad (a, a) \notin R, (c, c) \notin R$$

Hence, R is not reflexive.

2. **Symmetry:** The only pair in R is (b, b) . Reversing it gives (b, b) , which is in R .

$$(b, b) \in R \implies (b, b) \in R$$

Hence, R is symmetric.

3. **Transitivity:** The only chain of pairs is $(b, b) \in R$ and $(b, b) \in R$, which gives $(b, b) \in R$. Hence, R is transitive.

Final Answer

R is symmetric and transitive, but not reflexive.

Question 11

If the relation R on the set $A = \{1, 2, 3\}$ is defined by $R = \{(1, 1), (2, 2), (3, 3), (2, 1), (3, 2)\}$, then determine whether the relation R is (i) reflexive (ii) symmetric (iii) transitive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (2, 1), (3, 2)\}$

Concept / Formula Used: $A = \{1, 2, 3\}$

Solution:

(i) **Reflexivity:** For every $x \in A$:

$$(1, 1) \in R, \quad (2, 2) \in R, \quad (3, 3) \in R$$

Since $(x, x) \in R$ for all $x \in A$, R is reflexive.

(ii) **Symmetry:**

$$(2, 1) \in R \quad \text{but} \quad (1, 2) \notin R$$

Hence, R is not symmetric.

(iii) **Transitivity:** We observe:

$$(3, 2) \in R \quad \text{and} \quad (2, 1) \in R$$

For transitivity, we require $(3, 1) \in R$. Checking R :

$$(3, 1) \notin R$$

Hence, R is not transitive.

Final Answer

R is reflexive, but neither symmetric nor transitive.

Question 12

How many reflexive relations are possible on a set A of order 3?

TYPE 6 Counting Specific Types of Relations

Given: $n(A) = 3$

Concept / Formula Used: $n \quad 2^{n(n-1)} = 2^{n^2-n}$

Solution:

Given the number of elements in set A :

$$n = 3$$

The Cartesian product $A \times A$ contains $n^2 = 3^2 = 9$ ordered pairs.

For a relation R on A to be reflexive, it must contain all 3 diagonal pairs $(a_1, a_1), (a_2, a_2), (a_3, a_3)$.
The number of non-diagonal pairs is:

$$\begin{aligned} n^2 - n &= 9 - 3 \\ &= 6 \end{aligned}$$

Each non-diagonal pair can either be included or excluded from the relation independently.

$$\begin{aligned} \text{Total reflexive relations} &= 2^{n^2-n} \\ &= 2^6 \\ &= 64 \end{aligned}$$

Final Answer

64

Question 13

If R be the relation in the set $\{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2)\}$, then determine whether

- (i) R is reflexive and symmetric but not transitive.
- (ii) R is reflexive and transitive but not symmetric.
- (iii) R is symmetric and transitive but not reflexive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{1, 2, 3, 4\}$, $R = \{(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2)\}$

Concept / Formula Used: R

Solution:

1. **Reflexivity:**

$$(1, 1) \in R, \quad (2, 2) \in R, \quad (3, 3) \in R, \quad (4, 4) \in R$$

Since $(a, a) \in R$ for every $a \in A$, R is reflexive.

2. **Symmetry:**

$$(1, 2) \in R \quad \text{but} \quad (2, 1) \notin R$$

Hence, R is not symmetric.

3. **Transitivity:** Checking non-trivial chains:

$$\begin{aligned} (1, 3) \in R \text{ and } (3, 2) \in R &\implies (1, 2) \in R \\ (1, 1) \in R \text{ and } (1, 2) \in R &\implies (1, 2) \in R \\ (1, 1) \in R \text{ and } (1, 3) \in R &\implies (1, 3) \in R \\ (3, 3) \in R \text{ and } (3, 2) \in R &\implies (3, 2) \in R \end{aligned}$$

All pair combinations satisfy transitivity. Hence, R is transitive.

Thus, R is reflexive and transitive but not symmetric.

Final Answer

Option (ii) is correct: R is reflexive and transitive but not symmetric.

Question 14

Determine whether each of the following relations are reflexive, symmetric and transitive:

- (i) Relation R in the set $A = \{1, 2, 3, \dots, 10\}$ defined by $R = \{(x, y) : 2x - y = 0\}$
- (ii) Relation R in the set $\mathbb{Z}$ of all integers defined by $R = \{(x, y) : x - y \text{ is an integer}\}$
- (iii) Relation R on the set $A = \{1, 2, 3, 4, 5, 6\}$ defined by $R = \{(a, b) : b = a + 1\}$

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: (i) $A = \{1, 2, \dots, 10\}, 2x - y = 0$, (ii) $x, y \in \mathbb{Z}, x - y \in \mathbb{Z}$, (iii) $A = \{1, \dots, 6\}, b = a + 1$

Solution:

Part (i): $A = \{1, 2, \dots, 10\}, 2x - y = 0 \implies y = 2x$.

$$R = \{(1, 2), (2, 4), (3, 6), (4, 8), (5, 10)\}$$

- Reflexive: $1 \in A$ but $(1, 1) \notin R$. Not reflexive. - Symmetric: $(1, 2) \in R$ but $(2, 1) \notin R$. Not symmetric. - Transitive: $(1, 2) \in R$ and $(2, 4) \in R$, but $(1, 4) \notin R$. Not transitive.

Part (ii): $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x - y \in \mathbb{Z}\}$. - Reflexive: For any $x \in \mathbb{Z}$:

$$x - x = 0 \in \mathbb{Z} \implies (x, x) \in R$$

Hence, R is reflexive. - Symmetric: Let $(x, y) \in R \implies x - y = k \in \mathbb{Z}$.

$$y - x = -(x - y) = -k \in \mathbb{Z} \implies (y, x) \in R$$

Hence, R is symmetric. - Transitive: Let $(x, y) \in R$ and $(y, z) \in R$. Then $x - y = k_1 \in \mathbb{Z}$ and $y - z = k_2 \in \mathbb{Z}$. Adding the equations:

$$\begin{aligned}(x - y) + (y - z) &= k_1 + k_2 \\ x - z &= k_1 + k_2 \in \mathbb{Z} \implies (x, z) \in R\end{aligned}$$

Hence, R is transitive.

Part (iii): $A = \{1, 2, 3, 4, 5, 6\}, b = a + 1$.

$$R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}$$

- Reflexive: $(1, 1) \notin R$. Not reflexive. - Symmetric: $(1, 2) \in R$ but $(2, 1) \notin R$. Not symmetric. - Transitive: $(1, 2) \in R$ and $(2, 3) \in R$, but $(1, 3) \notin R$. Not transitive.

Final Answer

- (i) Neither reflexive, nor symmetric, nor transitive.
- (ii) Equivalence relation (reflexive, symmetric, and transitive).
- (iii) Neither reflexive, nor symmetric, nor transitive.

Question 15

If the relation R in the set A , where $A = \{1, 2, 3, 4, 5, 6\}$, is defined by $R = \{(x, y) : y \text{ is divisible by } x\}$, then express R in the roster form. Also determine whether the relation R is (i) reflexive (ii) symmetric (iii) transitive.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $A = \{1, 2, 3, 4, 5, 6\}$, $R = \{(x, y) : y \text{ is divisible by } x\}$

Concept / Formula Used: $y = kx$ $k \in \mathbb{Z}^+$

Solution:

1. **Roster Form:** Finding all pairs (x, y) such that y is divisible by x :

$$x = 1 \implies y \in \{1, 2, 3, 4, 5, 6\} \implies (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)$$

$$x = 2 \implies y \in \{2, 4, 6\} \implies (2, 2), (2, 4), (2, 6)$$

$$x = 3 \implies y \in \{3, 6\} \implies (3, 3), (3, 6)$$

$$x = 4 \implies y \in \{4\} \implies (4, 4)$$

$$x = 5 \implies y \in \{5\} \implies (5, 5)$$

$$x = 6 \implies y \in \{6\} \implies (6, 6)$$

Combining all pairs:

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$

2. **Properties of R :** (i) **Reflexivity:** For every $x \in A$, x is divisible by x (since $x = 1 \cdot x$). Thus, $(x, x) \in R$ for all $x \in A$. R is reflexive.

(ii) **Symmetry:** $(1, 2) \in R$ since 2 is divisible by 1. But 1 is not divisible by 2, so $(2, 1) \notin R$. R is not symmetric.

(iii) **Transitivity:** Let $(x, y) \in R$ and $(y, z) \in R$. Then:

$$\begin{aligned} y &= k_1 x \quad \text{for some } k_1 \in \mathbb{Z}^+ \\ z &= k_2 y \quad \text{for some } k_2 \in \mathbb{Z}^+ \\ \implies z &= k_2(k_1 x) = (k_1 k_2)x \end{aligned}$$

Since $k_1 k_2 \in \mathbb{Z}^+$, z is divisible by x , so $(x, z) \in R$. R is transitive.

Final Answer

Roster form: $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$
 R is reflexive and transitive, but not symmetric.

Question 16

If R is the relation defined on the set of natural numbers $\mathbb{N}$ as follows: $R = \{(x, y) : x, y \in \mathbb{N}, 2x + y = 41\}$, find the domain and the range of the relation R . Determine whether the relation is reflexive, symmetric and transitive.

TYPE 2 Domain and Range of a Relation

Given: $R = \{(x, y) \in \mathbb{N} \times \mathbb{N} : 2x + y = 41\}$

Solution:

Given equation:

$$\begin{aligned} 2x + y &= 41 \\ y &= 41 - 2x \end{aligned}$$

Since $x, y \in \mathbb{N} = \{1, 2, 3, \dots\}$, $y \geq 1$:

$$\begin{aligned} 41 - 2x &\geq 1 \\ 2x &\leq 40 \\ x &\leq 20 \end{aligned}$$

Calculating values of y for $x \in \{1, 2, \dots, 20\}$:

$$\begin{aligned} x = 1 &\implies y = 41 - 2(1) = 39 \\ x = 2 &\implies y = 41 - 2(2) = 37 \\ &\vdots \\ x = 20 &\implies y = 41 - 2(20) = 1 \end{aligned}$$

$$\text{Domain of } R = \{1, 2, 3, \dots, 20\}$$

$$\text{Range of } R = \{1, 3, 5, \dots, 39\}$$

Checking relation properties: 1. **Reflexivity:** Take $x = 1$.

$$2(1) + 1 = 3 \neq 41 \implies (1, 1) \notin R$$

Hence, R is not reflexive.

2. **Symmetry:** Take $(1, 39) \in R$ (since $2(1) + 39 = 41$).

$$2(39) + 1 = 78 + 1 = 79 \neq 41 \implies (39, 1) \notin R$$

Hence, R is not symmetric.

3. **Transitivity:** Take $(11, 19) \in R$ (since $2(11) + 19 = 41$) and $(19, 3) \in R$ (since $2(19) + 3 = 41$).

$$2(11) + 3 = 22 + 3 = 25 \neq 41 \implies (11, 3) \notin R$$

Hence, R is not transitive.

Final Answer

Domain = $\{1, 2, 3, \dots, 20\}$, Range = $\{1, 3, 5, \dots, 39\}$.

R is neither reflexive, nor symmetric, nor transitive.

Question 17

Determine whether each of the following relations in the set A of human beings in a city at a particular time are reflexive, symmetric and transitive:

- (i) $R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$
- (ii) $R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$
- (iii) $R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$
- (iv) $R = \{(x, y) : x \text{ and } y \text{ live within 2 kilometres}\}$

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: A = set of human beings in a city

Solution:

Part (i): $R = \{(x, y) : x \text{ and } y \text{ work at the same place}\}$ - Reflexive: Every person x works at the same place as x . $(x, x) \in R$. - Symmetric: If x and y work at the same place, then y and x work at the same place. $(x, y) \in R \implies (y, x) \in R$. - Transitive: If x and y work at the same place, and y and z work at the same place, then x and z work at the same place. $(x, y), (y, z) \in R \implies (x, z) \in R$. Thus, R is reflexive, symmetric, and transitive (an equivalence relation).

Part (ii): $R = \{(x, y) : x \text{ and } y \text{ live in the same locality}\}$ - Following the exact same logic as (i), R is reflexive, symmetric, and transitive (an equivalence relation).

Part (iii): $R = \{(x, y) : x \text{ is exactly 7 cm taller than } y\}$ - Reflexive: A person x cannot be 7 cm taller than himself/herself. $(x, x) \notin R$. - Symmetric: If x is 7 cm taller than y , then y is 7 cm shorter than x . $(y, x) \notin R$. - Transitive: If x is 7 cm taller than y and y is 7 cm taller than z , then x is 14 cm taller than z . $(x, z) \notin R$. Thus, R is neither reflexive, nor symmetric, nor transitive.

Part (iv): $R = \{(x, y) : x \text{ and } y \text{ live within 2 kilometres}\}$ - Reflexive: The distance between x and x is $0 \text{ km} \leq 2 \text{ km}$. $(x, x) \in R$. - Symmetric: If distance between x and y is $\leq 2 \text{ km}$, distance between y and x is $\leq 2 \text{ km}$. $(x, y) \in R \implies (y, x) \in R$. - Transitive: Let x live at point 0 km, y at 1.5 km, and z at 3 km.

$$\text{dist}(x, y) = 1.5 \text{ km} \leq 2 \text{ km} \implies (x, y) \in R$$

$$\text{dist}(y, z) = 1.5 \text{ km} \leq 2 \text{ km} \implies (y, z) \in R$$

$$\text{dist}(x, z) = 3.0 \text{ km} > 2 \text{ km} \implies (x, z) \notin R$$

Thus, R is not transitive.

Final Answer

- (i) Equivalence relation (reflexive, symmetric, transitive).
- (ii) Equivalence relation (reflexive, symmetric, transitive).
- (iii) Neither reflexive, nor symmetric, nor transitive.
- (iv) Reflexive and symmetric, but not transitive.

Question 18

Determine whether each of the following relations in the set A of students at a particular time are symmetric and transitive:

- (i) $R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are honest}\}$
- (ii) $R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are obedient}\}$
- (iii) $R = \{(x, y) : x, y \in A, x \text{ and } y \text{ are hardworking}\}$

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: A = set of students at a particular time

Concept / Formula Used: Let P denote the attribute in question (honest / obedient / hardworking). If $H \subseteq A$ is the subset of students possessing P , then $R = \{(x, y) : x, y \in H\} = H \times H$

Solution:

Let $H \subseteq A$ be the subset of students who possess the given attribute (honest, obedient, or hardworking, according to the part). Then

$$(x, y) \in R \iff x \in H \text{ and } y \in H$$

Symmetry: Let $(x, y) \in R$.

$$\begin{aligned} (x, y) \in R &\implies x \in H \text{ and } y \in H \\ &\implies y \in H \text{ and } x \in H \\ &\implies (y, x) \in R \end{aligned}$$

Hence, R is symmetric in all three cases (i), (ii), and (iii).

Transitivity: Let $(x, y) \in R$ and $(y, z) \in R$.

$$\begin{aligned} (x, y) \in R \text{ and } (y, z) \in R &\implies (x \in H \text{ and } y \in H) \text{ and } (y \in H \text{ and } z \in H) \\ &\implies x \in H \text{ and } z \in H \\ &\implies (x, z) \in R \end{aligned}$$

Hence, R is transitive in all three cases (i), (ii), and (iii).

Final Answer

The relations in (i), (ii) and (iii) are all symmetric and transitive.

Common Mistake: Testing symmetry/transitivity with a concrete numbered example (e.g. students labelled 1, 2, 3) is not valid here – A is the set of *all* students, and R has the special structure $H \times H$, which forces both symmetry and transitivity to hold in general. A single counter-example only disproves a property; it cannot be used to prove one holds only sometimes when the underlying structure guarantees it always.

Question 19

Show that the relation R in the set $A = \{1, 2, 3\}$ given by $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$ is reflexive but neither symmetric nor transitive. (NCERT)

TYPE 4 Verifying Relation Properties on Finite Sets

To Prove: R is reflexive but neither symmetric nor transitive

Solution:

Reflexivity: For R to be reflexive, $(a, a) \in R$ for every $a \in A = \{1, 2, 3\}$.

$$(1, 1) \in R, \quad (2, 2) \in R, \quad (3, 3) \in R$$

All three pairs are present, so R is reflexive.

Symmetry: Consider the pair $(1, 2) \in R$.

$$(1, 2) \in R \quad \text{but} \quad (2, 1) \notin R$$

Since $(1, 2) \in R$ does not imply $(2, 1) \in R$, R is not symmetric.

Transitivity: Consider the pairs $(1, 2) \in R$ and $(2, 3) \in R$.

$$(1, 2) \in R \text{ and } (2, 3) \in R \quad \text{but} \quad (1, 3) \notin R$$

Since $(1, 2) \in R$ and $(2, 3) \in R$ does not imply $(1, 3) \in R$, R is not transitive.

Hence Proved

R is reflexive but neither symmetric nor transitive.

Question 20

Explain why $x + 4y = 12$, $x, y \in N$ is NOT a symmetric relation.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $R = \{(x, y) : x, y \in N, x + 4y = 12\}$

Solution:

List the pairs (x, y) satisfying $x + 4y = 12$ with $x, y \in N$:

$$y = 1 \implies x = 8 \implies (8, 1) \in R$$

$$y = 2 \implies x = 4 \implies (4, 2) \in R$$

(For $y \geq 3$, $x = 12 - 4y \leq 0$, which is not a natural number, so no further pairs exist.)

Consider the pair $(4, 2) \in R$. For R to be symmetric, $(2, 4)$ must also belong to R . Check:

$$2 + 4(4) = 2 + 16 = 18 \neq 12 \implies (2, 4) \notin R$$

Final Answer

$(4, 2) \in R$ but $(2, 4) \notin R$, so R is not symmetric.

Question 21

Let R be a relation defined over N , where N is the set of natural numbers, defined as “ $m R n$ if and only if m is a multiple of n , where $m, n \in N$.” Find whether R is reflexive, symmetric and transitive or not.

TYPE 4 Verifying Relation Properties on Finite Sets

Given: $R = \{(m, n) : m, n \in N, m \text{ is a multiple of } n\}$

Solution:

Reflexivity: Every natural number m is a multiple of itself, since $m = 1 \cdot m$.

$$\implies (m, m) \in R \text{ for all } m \in N$$

Hence, R is reflexive.

Symmetry: Take $m = 4$, $n = 2$. Since $4 = 2 \times 2$, 4 is a multiple of 2, so $(4, 2) \in R$. Check whether $(2, 4) \in R$, i.e. whether 2 is a multiple of 4:

$$2 \neq 4k \text{ for any } k \in N \implies (2, 4) \notin R$$

Hence, R is not symmetric.

Transitivity: Let $(m, n) \in R$ and $(n, p) \in R$, i.e. m is a multiple of n and n is a multiple of p .

$$\begin{aligned} m &= kn \text{ for some } k \in N \quad \text{and} \quad n = lp \text{ for some } l \in N \\ \implies m &= k(lp) = (kl)p, \quad kl \in N \end{aligned}$$

So m is a multiple of p , i.e. $(m, p) \in R$. Hence, R is transitive.

Final Answer

R is reflexive and transitive but not symmetric.

Question 22

Let R be the relation on W (set of whole numbers) defined by

$$R = \{(x, y) : x, y \in W, 3x + 2y = 12\}$$

Check the relation R for reflexivity, symmetricity and transitivity.

TYPE 2 Domain and Range of a Relation

Given: $R = \{(x, y) : x, y \in W, 3x + 2y = 12\}$

Solution:

First, list all pairs (x, y) with $x, y \in W = \{0, 1, 2, 3, \dots\}$ satisfying $3x + 2y = 12$:

$$x = 0 \implies 2y = 12 \implies y = 6 \implies (0, 6) \in R$$

$$x = 2 \implies 6 + 2y = 12 \implies y = 3 \implies (2, 3) \in R$$

$$x = 4 \implies 12 + 2y = 12 \implies y = 0 \implies (4, 0) \in R$$

(For $x = 1$ or $x = 3$, y works out to a non-integer, so these are excluded.)

$$R = \{(0, 6), (2, 3), (4, 0)\}$$

Reflexivity: Check if $(0, 0) \in R$: $3(0) + 2(0) = 0 \neq 12$. So $(0, 0) \notin R$. Hence, R is not reflexive.

Symmetry: Consider $(0, 6) \in R$. Check whether $(6, 0) \in R$:

$$3(6) + 2(0) = 18 \neq 12 \implies (6, 0) \notin R$$

Hence, R is not symmetric.

Transitivity: Since no two pairs in R share a common middle element in the required chain form (e.g. $(4, 0)$ and $(0, 6)$ are both in R), check whether $(4, 6) \in R$:

$$3(4) + 2(6) = 12 + 12 = 24 \neq 12 \implies (4, 6) \notin R$$

Hence, R is not transitive.

Final Answer

R is neither reflexive, nor symmetric, nor transitive.

Common Mistake: The printed answer key for this question states “symmetric,” which does not hold for the actual relation $R = \{(0, 6), (2, 3), (4, 0)\}$ – verify by direct enumeration, as shown above, rather than accepting a stated answer without checking it against the listed ordered pairs.

Question 23

Let R be the relation defined on $\mathbb{Q}_0$ (set of non-zero rational numbers) defined by $R = \left\{ (a, b) : a, b \in \mathbb{Q}_0, b = \frac{1}{a} \right\}$. Check the relation R for reflexivity, symmetry and transitivity.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $\mathbb{Q}_0 = \mathbb{Q} \setminus \{0\}$, $R = \{(a, b) \in \mathbb{Q}_0 \times \mathbb{Q}_0 : b = \frac{1}{a}\}$

Concept / Formula Used: $b = \frac{1}{a} \iff ab = 1$

Solution:

1. **Reflexivity:** For $(a, a) \in R$, we require:

$$\begin{aligned} a &= \frac{1}{a} \\ a^2 &= 1 \\ a &= \pm 1 \end{aligned}$$

For any $a \in \mathbb{Q}_0$ such that $a \neq \pm 1$ (for instance, $a = 2$):

$$2 \neq \frac{1}{2} \implies (2, 2) \notin R$$

Hence, R is not reflexive.

2. **Symmetry:** Let $(a, b) \in R$.

$$(a, b) \in R \implies b = \frac{1}{a}$$

Taking the reciprocal on both sides (since $a, b \neq 0$):

$$a = \frac{1}{b} \implies (b, a) \in R$$

Hence, R is symmetric.

3. **Transitivity:** Let $(a, b) \in R$ and $(b, c) \in R$.

$$b = \frac{1}{a} \quad \text{and} \quad c = \frac{1}{b}$$

Substituting $b = \frac{1}{a}$ into $c = \frac{1}{b}$:

$$c = \frac{1}{\frac{1}{a}} = a$$

For $(a, c) = (a, a) \in R$, we would require $a = \frac{1}{a} \implies a = \pm 1$. Counterexample: Take $a = 2, b = \frac{1}{2}, c = 2$:

$$\left(2, \frac{1}{2}\right) \in R \quad \text{and} \quad \left(\frac{1}{2}, 2\right) \in R \quad \text{but} \quad (2, 2) \notin R$$

Hence, R is not transitive.

Final Answer

R is symmetric, but neither reflexive nor transitive.

Question 24

A relation R is defined on a set of real numbers $\mathbb{R}$ as

$$R = \{(x, y) : xy \text{ is an irrational number}\}.$$

Check whether R is reflexive, symmetric and transitive.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : xy \in \mathbb{R} \setminus \mathbb{Q}\}$

Concept / Formula Used: $R \quad A \quad (x, x) \in R \quad x \in A \quad (x, y) \in R \implies (y, x) \in R \quad (x, y) \in R \quad (y, z) \in R \implies (x, z) \in R$

Solution:

Reflexivity: Let $x = 0 \in \mathbb{R}$. We evaluate $x \cdot x$:

$$\begin{aligned} x \cdot x &= 0 \cdot 0 \\ &= 0 \end{aligned}$$

Since $0 \in \mathbb{Q}$ (a rational number), $x \cdot x$ is not an irrational number. Thus, $(0, 0) \notin R$. Therefore, R is not reflexive.

Symmetry: Let $(x, y) \in R$. By definition of relation R :

$$xy \in \mathbb{R} \setminus \mathbb{Q}$$

Using the commutative property of real number multiplication:

$$yx = xy$$

Therefore, yx is also an irrational number. This implies $(y, x) \in R$. Thus, R is symmetric.

Transitivity: Consider $x = \sqrt{2} \in \mathbb{R}$, $y = \sqrt{3} \in \mathbb{R}$, and $z = \sqrt{2} \in \mathbb{R}$. Evaluating xy and yz :

$$\begin{aligned} xy &= \sqrt{2} \cdot \sqrt{3} = \sqrt{6} \quad (\text{irrational}) \implies (x, y) \in R \\ yz &= \sqrt{3} \cdot \sqrt{2} = \sqrt{6} \quad (\text{irrational}) \implies (y, z) \in R \end{aligned}$$

Now, examine xz :

$$\begin{aligned} xz &= \sqrt{2} \cdot \sqrt{2} \\ &= 2 \quad (\text{rational}) \end{aligned}$$

Since $2 \in \mathbb{Q}$, $(x, z) \notin R$. Thus, R is not transitive.

Final Answer

R is symmetric, but neither reflexive nor transitive.

Question 25

Check whether the relation R on the set $\mathbb{N}$ of natural numbers given by $R = \{(a, b) : a, b \in \mathbb{N}, a \text{ is a divisor of } b\}$ is reflexive, symmetric or transitive. Also determine whether R is an equivalence relation.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $R = \{(a, b) : a, b \in \mathbb{N}, a \mid b\}$

Solution:

Reflexivity: Let $a \in \mathbb{N}$. Every natural number divides itself, since:

$$a = 1 \cdot a$$

Thus, $a \mid a$ for all $a \in \mathbb{N}$, which means $(a, a) \in R$. Therefore, R is reflexive.

Symmetry: Consider $a = 2 \in \mathbb{N}$ and $b = 4 \in \mathbb{N}$. Since 2 divides 4, we have $(2, 4) \in R$. However, 4 does not divide 2, so:

$$(4, 2) \notin R$$

Therefore, R is not symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$ for $a, b, c \in \mathbb{N}$. Then, there exist positive integers $k, m \in \mathbb{N}$ such that:

$$b = ka$$

$$c = mb$$

Substituting $b = ka$ into $c = mb$:

$$c = m(ka)$$

$$c = (mk)a$$

Since $mk \in \mathbb{N}$, a divides c , which means $(a, c) \in R$. Therefore, R is transitive.

Equivalence Relation Determination: Since R is not symmetric, it is not an equivalence relation.

Final Answer

R is reflexive and transitive, but not symmetric. Hence, R is not an equivalence relation.

Question 26

Let R be the relation on $\mathbb{R}$ (set of real numbers) defined by $R = \{(a, b) : a, b \in \mathbb{R}, a^2 + b^2 = 1\}$. Show that R is symmetric but neither reflexive nor transitive.

TYPE 5 Verifying Relation Properties on Infinite Sets

To Prove: R is symmetric, but R is neither reflexive nor transitive.

Concept / Formula Used: $R \quad (a, b) \in R \implies (b, a) \in R \quad R \quad (a, a) \in R$
 $R \quad a \in \mathbb{R} \quad R \quad (a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R$

Solution:

Symmetry: Let $(a, b) \in R$. Then,

$$a^2 + b^2 = 1$$

By commutativity of addition in $\mathbb{R}$:

$$b^2 + a^2 = 1$$

This implies $(b, a) \in R$. Hence, R is symmetric.

Reflexivity: Let $a = 2 \in \mathbb{R}$. We check if $(2, 2) \in R$:

$$\begin{aligned} 2^2 + 2^2 &= 4 + 4 \\ &= 8 \neq 1 \end{aligned}$$

Since $a^2 + a^2 \neq 1$ for $a = 2$, $(2, 2) \notin R$. Hence, R is not reflexive.

Transitivity: Consider $a = 1$, $b = 0$, and $c = 1$. Checking $(a, b) \in R$:

$$1^2 + 0^2 = 1 + 0 = 1 \implies (1, 0) \in R$$

Checking $(b, c) \in R$:

$$0^2 + 1^2 = 0 + 1 = 1 \implies (0, 1) \in R$$

Now checking $(a, c) \in R$:

$$1^2 + 1^2 = 1 + 1 = 2 \neq 1 \implies (1, 1) \notin R$$

Hence, R is not transitive.

Hence Proved

R is symmetric, but neither reflexive nor transitive.

Question 27

Determine whether each of the following relations are reflexive, symmetric and transitive:

- (i) $R = \{(a, b) : a, b \in \mathbb{N}, a < b\}$
- (ii) $R = \{(a, b) : a, b \in \mathbb{R}, a \leq b\}$

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: Part (i): $R = \{(a, b) : a, b \in \mathbb{N}, a < b\}$; Part (ii): $R = \{(a, b) : a, b \in \mathbb{R}, a \leq b\}$

Concept / Formula Used: $(a, a) \in R \quad (a, b) \in R \implies (b, a) \in R \quad (a, b) \in R$
 $(b, c) \in R \implies (a, c) \in R$

Solution:

Part (i): $R = \{(a, b) : a, b \in \mathbb{N}, a < b\}$

Reflexivity: For any $a \in \mathbb{N}$, the statement $a < a$ is false. Thus, $(a, a) \notin R$. Hence, R is not reflexive.

Symmetry: Let $1, 2 \in \mathbb{N}$. Since $1 < 2$, we have $(1, 2) \in R$. However, $2 < 1$ is false, so $(2, 1) \notin R$. Hence, R is not symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then,

$$a < b \quad \text{and} \quad b < c$$

By the transitive property of order in natural numbers:

$$a < c$$

Thus, $(a, c) \in R$. Hence, R is transitive.

Part (ii): $R = \{(a, b) : a, b \in \mathbb{R}, a \leq b\}$

Reflexivity: For any $a \in \mathbb{R}$, $a \leq a$ is true (since $a = a$). Thus, $(a, a) \in R$ for all $a \in \mathbb{R}$. Hence, R is reflexive.

Symmetry: Let $1, 2 \in \mathbb{R}$. Since $1 \leq 2$, we have $(1, 2) \in R$. However, $2 \leq 1$ is false, so $(2, 1) \notin R$. Hence, R is not symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then,

$$a \leq b \quad \text{and} \quad b \leq c$$

By the transitive property of inequalities:

$$a \leq c$$

Thus, $(a, c) \in R$. Hence, R is transitive.

Final Answer

- (i) R is transitive, but neither reflexive nor symmetric.
- (ii) R is reflexive and transitive, but not symmetric.

Question 28

Let T be the set of all triangles drawn in a plane with R as a relation in T given by $R = \{(T_1, T_2) : T_1 \cong T_2\}$. Show that R is an equivalence relation.

TYPE 7 Proving Equivalence Relations

To Prove: $R = \{(T_1, T_2) : T_1 \cong T_2\}$ is an equivalence relation.

Solution:

Reflexivity: Let $T_1 \in T$. Every triangle is congruent to itself.

$$T_1 \cong T_1$$

Thus, $(T_1, T_1) \in R$ for all $T_1 \in T$. Hence, R is reflexive.

Symmetry: Let $(T_1, T_2) \in R$. By definition of relation R :

$$T_1 \cong T_2$$

If triangle T_1 is congruent to triangle T_2 , then triangle T_2 is congruent to triangle T_1 :

$$T_2 \cong T_1$$

Thus, $(T_2, T_1) \in R$. Hence, R is symmetric.

Transitivity: Let $(T_1, T_2) \in R$ and $(T_2, T_3) \in R$. Then,

$$T_1 \cong T_2$$

$$T_2 \cong T_3$$

If T_1 is congruent to T_2 , and T_2 is congruent to T_3 , then T_1 is congruent to T_3 :

$$T_1 \cong T_3$$

Thus, $(T_1, T_3) \in R$. Hence, R is transitive.

Conclusion: Since R is reflexive, symmetric, and transitive, R is an equivalence relation on T .

Hence Proved

R is an equivalence relation on the set of triangles T .

Question 29

Show that the relation R in the set A of all books in a library of a school, given by $R = \{(x, y) : x \text{ and } y \text{ have same number of pages}\}$, is an equivalence relation.

TYPE 7 Proving Equivalence Relations

To Prove: R is an equivalence relation on the set A of all books.

Concept / Formula Used: R

Solution:

Reflexivity: Let $x \in A$. Book x has the same number of pages as itself.

$$(x, x) \in R \quad \forall x \in A$$

Hence, R is reflexive.

Symmetry: Let $(x, y) \in R$. Then, book x and book y have the same number of pages. This implies book y and book x have the same number of pages.

$$(y, x) \in R$$

Hence, R is symmetric.

Transitivity: Let $(x, y) \in R$ and $(y, z) \in R$. Then,

$$\text{Number of pages in } x = \text{Number of pages in } y$$

$$\text{Number of pages in } y = \text{Number of pages in } z$$

Equating the expressions:

$$\text{Number of pages in } x = \text{Number of pages in } z$$

Thus, $(x, z) \in R$. Hence, R is transitive.

Conclusion: Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

Hence Proved

R is an equivalence relation on set A .

Question 30

Show that the relation R on the set $\mathbb{Z}$ of all integers defined by $R = \{(a, b) : a - b \text{ is divisible by } 3, a, b \in \mathbb{Z}\}$ is an equivalence relation.

TYPE 7 Proving Equivalence Relations

To Prove: $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : 3 \mid (a - b)\}$ is an equivalence relation.

Concept / Formula Used: $3 \mid (a - b) \iff a - b = 3k \quad k \in \mathbb{Z}$

Solution:

Reflexivity: Let $a \in \mathbb{Z}$.

$$a - a = 0$$

Since $0 = 3 \times 0$, $a - a$ is divisible by 3. Thus, $(a, a) \in R$ for all $a \in \mathbb{Z}$. Hence, R is reflexive.

Symmetry: Let $(a, b) \in R$. Then,

$$a - b = 3k \quad \text{for some } k \in \mathbb{Z}$$

Multiplying both sides by -1 :

$$\begin{aligned} -(a - b) &= -3k \\ b - a &= 3(-k) \end{aligned}$$

Since $-k \in \mathbb{Z}$, $b - a$ is divisible by 3. Thus, $(b, a) \in R$. Hence, R is symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then,

$$\begin{aligned} a - b &= 3k \quad \text{for some } k \in \mathbb{Z} \\ b - c &= 3m \quad \text{for some } m \in \mathbb{Z} \end{aligned}$$

Adding the two equations:

$$\begin{aligned}(a - b) + (b - c) &= 3k + 3m \\ a - c &= 3(k + m)\end{aligned}$$

Since $k + m \in \mathbb{Z}$, $a - c$ is divisible by 3. Thus, $(a, c) \in R$. Hence, R is transitive.

Conclusion: Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

Hence Proved

R is an equivalence relation on $\mathbb{Z}$.

Question 31

Show that the relation R on the set $\mathbb{Z}$ of all integers defined by $R = \{(a, b) : a - b \text{ is divisible by } 2, a, b \in \mathbb{Z}\}$ is an equivalence relation. Also, write the equivalence class of 0.

TYPE 8 Equivalence Classes and Related Elements

★ Likely Exam Question

Given: $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : 2 \mid (a - b)\}$

Concept / Formula Used: $x \quad [x] = \{y \in \mathbb{Z} : (y, x) \in R\}$

Solution:

Reflexivity: Let $a \in \mathbb{Z}$.

$$a - a = 0 = 2 \times 0$$

Since 0 is divisible by 2, $(a, a) \in R$ for all $a \in \mathbb{Z}$. Hence, R is reflexive.

Symmetry: Let $(a, b) \in R$. Then,

$$a - b = 2k \quad \text{for some } k \in \mathbb{Z}$$

Rearranging:

$$\begin{aligned}b - a &= -(a - b) \\ &= -2k \\ &= 2(-k)\end{aligned}$$

Since $-k \in \mathbb{Z}$, $b - a$ is divisible by 2. Thus, $(b, a) \in R$. Hence, R is symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then,

$$\begin{aligned}a - b &= 2k \quad \text{for some } k \in \mathbb{Z} \\ b - c &= 2m \quad \text{for some } m \in \mathbb{Z}\end{aligned}$$

Adding the two equations:

$$\begin{aligned}(a - b) + (b - c) &= 2k + 2m \\ a - c &= 2(k + m)\end{aligned}$$

Since $k + m \in \mathbb{Z}$, $a - c$ is divisible by 2. Thus, $(a, c) \in R$. Hence, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

Finding the Equivalence Class of 0: By definition:

$$\begin{aligned}[0] &= \{a \in \mathbb{Z} : (a, 0) \in R\} \\ &= \{a \in \mathbb{Z} : a - 0 \text{ is divisible by } 2\} \\ &= \{a \in \mathbb{Z} : a = 2k, k \in \mathbb{Z}\} \\ &= \{\dots, -4, -2, 0, 2, 4, \dots\}\end{aligned}$$

Final Answer

R is an equivalence relation on $\mathbb{Z}$, and the equivalence class of 0 is $[0] = \{\dots, -4, -2, 0, 2, 4, \dots\}$ (the set of all even integers).

Question 32

Prove that the relation R in the set $A = \{5, 6, 7, 8, 9\}$, given by $R = \{(a, b) : |a - b| \text{ is divisible by } 2\}$, is an equivalence relation. Find all elements related to element 6.

TYPE 8 Equivalence Classes and Related Elements

★ Likely Exam Question

Given: $A = \{5, 6, 7, 8, 9\}$ and $R = \{(a, b) \in A \times A : |a - b| \text{ is even}\}$

Concept / Formula Used: $x \in A \quad \{y \in A : (y, x) \in R\}$

Solution:

Reflexivity: Let $a \in A$.

$$\begin{aligned}|a - a| &= |0| \\ &= 0\end{aligned}$$

Since 0 is divisible by 2, $(a, a) \in R$ for all $a \in A$. Hence, R is reflexive.

Symmetry: Let $(a, b) \in R$. Then,

$$|a - b| = 2k \quad \text{for some } k \in \mathbb{W}$$

Since $|b - a| = |-(a - b)| = |a - b|$:

$$|b - a| = 2k$$

Thus, $(b, a) \in R$. Hence, R is symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then $a - b$ is even and $b - c$ is even. Let $a - b = 2k$ and $b - c = 2m$ where $k, m \in \mathbb{Z}$. Adding these equations:

$$\begin{aligned}(a - b) + (b - c) &= 2k + 2m \\ a - c &= 2(k + m)\end{aligned}$$

Thus, $|a - c| = |2(k + m)| = 2|k + m|$, which is divisible by 2. Hence, $(a, c) \in R$. Therefore, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation.

Elements related to element 6: Let $x \in A$ be related to 6. Then $|x - 6|$ must be divisible by 2. Testing each element of $A = \{5, 6, 7, 8, 9\}$:

$$\begin{aligned}x = 5 &\implies |5 - 6| = 1 \quad (\text{not divisible by } 2) \\ x = 6 &\implies |6 - 6| = 0 \quad (\text{divisible by } 2) \\ x = 7 &\implies |7 - 6| = 1 \quad (\text{not divisible by } 2) \\ x = 8 &\implies |8 - 6| = 2 \quad (\text{divisible by } 2) \\ x = 9 &\implies |9 - 6| = 3 \quad (\text{not divisible by } 2)\end{aligned}$$

Thus, the elements related to 6 are 6 and 8.

Final Answer

R is an equivalence relation, and the set of elements related to 6 is $\{6, 8\}$.

Question 33

Give examples of relations which are

- (i) symmetric but neither reflexive nor transitive.
- (ii) transitive but neither reflexive nor symmetric.
- (iii) symmetric and transitive but not reflexive.
- (iv) symmetric and reflexive but not transitive.
- (v) reflexive and transitive but not symmetric.

TYPE 9 Constructing Examples of Relations

Solution:

Part (i): Symmetric but neither reflexive nor transitive. Let L be the set of all lines in a plane, and relation R on L defined by:

$$R = \{(l_1, l_2) : l_1 \text{ is perpendicular to } l_2\}$$

- Reflexive: A line l_1 cannot be perpendicular to itself, so $(l_1, l_1) \notin R$. - Symmetric: If $l_1 \perp l_2$, then $l_2 \perp l_1$, so $(l_1, l_2) \in R \implies (l_2, l_1) \in R$. - Transitive: If $l_1 \perp l_2$ and $l_2 \perp l_3$, then $l_1 \parallel l_3$, so $(l_1, l_3) \notin R$.

Part (ii): Transitive but neither reflexive nor symmetric. Let R be defined on $\mathbb{N}$ by:

$$R = \{(a, b) : a < b\}$$

- Reflexive: $a < a$ is false, so $(a, a) \notin R$. - Symmetric: $1 < 2 \implies (1, 2) \in R$, but $2 < 1$ is false, so $(2, 1) \notin R$. - Transitive: $a < b$ and $b < c \implies a < c$, so $(a, c) \in R$.

Part (iii): Symmetric and transitive but not reflexive. Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$. - Reflexive: $(3, 3) \notin R$, so R is not reflexive on A . - Symmetric: For every $(a, b) \in R$, $(b, a) \in R$. - Transitive: If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

Part (iv): Symmetric and reflexive but not transitive. Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$. - Reflexive: $(1, 1), (2, 2), (3, 3) \in R$. - Symmetric: All pairs are symmetric. - Transitive: $(1, 2) \in R$ and $(2, 3) \in R$, but $(1, 3) \notin R$.

Part (v): Reflexive and transitive but not symmetric. Let R be defined on $\mathbb{R}$ by:

$$R = \{(a, b) : a \leq b\}$$

- Reflexive: $a \leq a$ is true for all $a \in \mathbb{R}$. - Symmetric: $1 \leq 2 \implies (1, 2) \in R$, but $2 \leq 1$ is false $\implies (2, 1) \notin R$. - Transitive: $a \leq b$ and $b \leq c \implies a \leq c$.

Final Answer

- (i) $R = \{(l_1, l_2) : l_1 \perp l_2\}$ on lines in a plane.
- (ii) $R = \{(a, b) : a < b\}$ on $\mathbb{N}$.
- (iii) $R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ on $A = \{1, 2, 3\}$.
- (iv) $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$ on $A = \{1, 2, 3\}$.
- (v) $R = \{(a, b) : a \leq b\}$ on $\mathbb{R}$.

Question 34

Give an example of a relation R on $A = \{a, b, c\}$ which is

- (i) neither reflexive nor symmetric but transitive.
- (ii) neither symmetric nor transitive but reflexive.
- (iii) neither transitive nor reflexive but symmetric.

TYPE 9 Constructing Examples of Relations

Given: $A = \{a, b, c\}$

Concept / Formula Used: $A \times A$

Solution:

Part (i): Neither reflexive nor symmetric but transitive. Let $R = \{(a, b)\}$. - Reflexivity: $(a, a) \notin R$, so not reflexive. - Symmetry: $(a, b) \in R$ but $(b, a) \notin R$, so not symmetric. - Transitivity: There are no pairs (x, y) and (y, z) to test non-vacuously; hence it is vacuously transitive.

Part (ii): Neither symmetric nor transitive but reflexive. Let $R = \{(a, a), (b, b), (c, c), (a, b), (b, c)\}$. - Reflexivity: Contains $(a, a), (b, b), (c, c)$, so reflexive. - Symmetry: $(a, b) \in R$ but $(b, a) \notin R$, so not symmetric. - Transitivity: $(a, b) \in R$ and $(b, c) \in R$, but $(a, c) \notin R$, so not transitive.

Part (iii): Neither transitive nor reflexive but symmetric. Let $R = \{(a, b), (b, a)\}$. - Reflexivity: $(a, a) \notin R$, so not reflexive. - Symmetry: $(a, b) \in R \implies (b, a) \in R$, so symmetric. - Transitivity: $(a, b) \in R$ and $(b, a) \in R$, but $(a, a) \notin R$, so not transitive.

Final Answer

- (i) $R = \{(a, b)\}$
- (ii) $R = \{(a, a), (b, b), (c, c), (a, b), (b, c)\}$
- (iii) $R = \{(a, b), (b, a)\}$

Question 35

Show that the relation S in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ given by

$$S = \{(a, b) : a, b \in A, |a - b| \text{ is divisible by } 4\}$$

is an equivalence relation. Find the set of all elements related to 1.

TYPE 8 Equivalence Classes and Related Elements

★ Likely Exam Question

Given: $A = \{0, 1, 2, \dots, 12\}$ and $S = \{(a, b) \in A \times A : 4 \mid |a - b|\}$

Concept / Formula Used: $1 \quad [1] = \{x \in A : (x, 1) \in S\}$

Solution:

Reflexivity: Let $a \in A$.

$$|a - a| = 0$$

Since $0 = 4 \times 0$, $|a - a|$ is divisible by 4. Thus, $(a, a) \in S$ for all $a \in A$. Hence, S is reflexive.

Symmetry: Let $(a, b) \in S$. Then,

$$|a - b| = 4k \quad \text{for some } k \in \mathbb{W}$$

Since $|b - a| = |-(a - b)| = |a - b|$:

$$|b - a| = 4k$$

Thus, $(b, a) \in S$. Hence, S is symmetric.

Transitivity: Let $(a, b) \in S$ and $(b, c) \in S$. Then,

$$a - b = 4k \quad \text{for some } k \in \mathbb{Z}$$

$$b - c = 4m \quad \text{for some } m \in \mathbb{Z}$$

Adding the two equations:

$$(a - b) + (b - c) = 4k + 4m$$

$$a - c = 4(k + m)$$

Thus, $|a - c| = 4|k + m|$, which is divisible by 4. Hence, $(a, c) \in S$. Therefore, S is transitive.

Since S is reflexive, symmetric, and transitive, S is an equivalence relation.

Finding elements related to 1:

$$[1] = \{x \in A : |x - 1| \text{ is a multiple of } 4\}$$

For $x \in \{0, 1, 2, \dots, 12\}$:

$$|x - 1| = 0 \implies x = 1$$

$$|x - 1| = 4 \implies x - 1 = 4 \implies x = 5$$

$$|x - 1| = 8 \implies x - 1 = 8 \implies x = 9$$

Thus, the elements related to 1 are $\{1, 5, 9\}$.

Final Answer

S is an equivalence relation, and the set of elements related to 1 is $\{1, 5, 9\}$.

Question 36

Show that the relation R in the set $A = \{x \in \mathbb{W}, 0 \leq x \leq 17\}$ given by

(i) $R = \{(a, b) : |a - b| \text{ is a multiple of } 5\}$

(ii) $R = \{(a, b) : a = b\}$

are equivalence relations. Find the set of all elements related to 2 in each case.

TYPE 8 Equivalence Classes and Related Elements

Given: $A = \{0, 1, 2, \dots, 17\}$

Concept / Formula Used: $[2] = \{x \in A : (x, 2) \in R\}$

Solution:

Part (i): $R = \{(a, b) : |a - b| \text{ is a multiple of } 5\}$

Reflexivity: For any $a \in A$, $|a - a| = 0 = 5 \times 0$. Thus, $(a, a) \in R$. R is reflexive.

Symmetry: Let $(a, b) \in R$. Then $|a - b| = 5k$ for $k \in \mathbb{W}$. Since $|b - a| = |a - b| = 5k$, $(b, a) \in R$. R is symmetric.

Transitivity: Let $(a, b) \in R$ and $(b, c) \in R$. Then $a - b = 5k$ and $b - c = 5m$ for $k, m \in \mathbb{Z}$. Adding them:

$$a - c = (a - b) + (b - c) = 5(k + m)$$

Thus $|a - c|$ is a multiple of 5, so $(a, c) \in R$. R is transitive.

Hence, R is an equivalence relation.

Elements related to 2 in part (i):

$$[2] = \{x \in A : |x - 2| \text{ is a multiple of } 5\}$$

Testing values in $A = \{0, 1, \dots, 17\}$:

$$\begin{aligned}x = 2 &\implies |2 - 2| = 0 \\x = 7 &\implies |7 - 2| = 5 \\x = 12 &\implies |12 - 2| = 10 \\x = 17 &\implies |17 - 2| = 15\end{aligned}$$

Thus, the set of elements related to 2 is $\{2, 7, 12, 17\}$.

Part (ii): $R = \{(a, b) : a = b\}$

Reflexivity: For any $a \in A$, $a = a$ is true, so $(a, a) \in R$. R is reflexive.

Symmetry: If $(a, b) \in R$, then $a = b \implies b = a \implies (b, a) \in R$. R is symmetric.

Transitivity: If $(a, b) \in R$ and $(b, c) \in R$, then $a = b$ and $b = c \implies a = c \implies (a, c) \in R$. R is transitive.

Hence, R is an equivalence relation.

Elements related to 2 in part (ii):

$$[2] = \{x \in A : x = 2\} = \{2\}$$

Final Answer

- (i) R is an equivalence relation; set of elements related to 2 is $\{2, 7, 12, 17\}$.
- (ii) R is an equivalence relation; set of elements related to 2 is $\{2\}$.

Question 37

Show that the relation R defined by $(a, b) R (c, d) \iff a + d = b + c$ on $A \times A$ where $A = \{1, 2, 3, \dots, 10\}$ is an equivalence relation. Hence, write the equivalence class $[(3, 4)]$.

TYPE 10 Equivalence Relations on Product Sets

★ Likely Exam Question

Given: $A = \{1, 2, 3, \dots, 10\}$ and relation R on $A \times A$ defined by $(a, b) R (c, d) \iff a + d = b + c$

Concept / Formula Used: $[(a, b)] = \{(x, y) \in A \times A : (x, y) R (a, b)\}$

Solution:

Reflexivity: Let $(a, b) \in A \times A$.

$$a + b = b + a$$

This satisfies the definition $(a, b) R (a, b)$. Thus, R is reflexive.

Symmetry: Let $(a, b)R(c, d)$. Then,

$$a + d = b + c$$

Rearranging the equality:

$$c + b = d + a$$

By definition of R , this means:

$$(c, d)R(a, b)$$

Thus, R is symmetric.

Transitivity: Let $(a, b)R(c, d)$ and $(c, d)R(e, f)$. Then,

$$a + d = b + c \quad \text{--- (1)}$$

$$c + f = d + e \quad \text{--- (2)}$$

Adding equation (1) and equation (2):

$$\begin{aligned}(a + d) + (c + f) &= (b + c) + (d + e) \\ a + d + c + f &= b + c + d + e\end{aligned}$$

Subtracting $c + d$ from both sides:

$$a + f = b + e$$

This implies $(a, b)R(e, f)$. Thus, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation on $A \times A$.

Equivalence Class $[(3, 4)]$:

$$\begin{aligned}[(3, 4)] &= \{(x, y) \in A \times A : (x, y)R(3, 4)\} \\ &= \{(x, y) \in A \times A : x + 4 = y + 3\} \\ &= \{(x, y) \in A \times A : y = x + 1\}\end{aligned}$$

Since $x, y \in \{1, 2, 3, \dots, 10\}$, we evaluate $y = x + 1$ for $x \in \{1, 2, \dots, 9\}$:

$$x = 1 \implies y = 2 \implies (1, 2)$$

$$x = 2 \implies y = 3 \implies (2, 3)$$

$$x = 3 \implies y = 4 \implies (3, 4)$$

$$x = 4 \implies y = 5 \implies (4, 5)$$

$$x = 5 \implies y = 6 \implies (5, 6)$$

$$x = 6 \implies y = 7 \implies (6, 7)$$

$$x = 7 \implies y = 8 \implies (7, 8)$$

$$x = 8 \implies y = 9 \implies (8, 9)$$

$$x = 9 \implies y = 10 \implies (9, 10)$$

Final Answer

R is an equivalence relation, and $[(3, 4)] = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8), (8, 9), (9, 10)\}$.

Question 38

Show that the relation R in the set $\mathbb{N} \times \mathbb{N}$ defined by $(a, b) R (c, d)$ iff $a^2 + d^2 = b^2 + c^2$ for all $a, b, c, d \in \mathbb{N}$ is an equivalence relation.

TYPE 10 Equivalence Relations on Product Sets

To Prove: R is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Concept / Formula Used: $(a, b) R (c, d) \iff a^2 + d^2 = b^2 + c^2$

Solution:

Reflexivity: Let $(a, b) \in \mathbb{N} \times \mathbb{N}$. We check if $(a, b) R (a, b)$:

$$a^2 + b^2 = b^2 + a^2$$

By commutativity of addition in $\mathbb{N}$, this statement holds. Thus, $(a, b) R (a, b)$ for all $(a, b) \in \mathbb{N} \times \mathbb{N}$. Hence, R is reflexive.

Symmetry: Let $(a, b) R (c, d)$. Then,

$$a^2 + d^2 = b^2 + c^2$$

Swapping left and right sides:

$$b^2 + c^2 = a^2 + d^2$$

By commutativity of addition:

$$c^2 + b^2 = d^2 + a^2$$

This implies $(c, d) R (a, b)$. Hence, R is symmetric.

Transitivity: Let $(a, b) R (c, d)$ and $(c, d) R (e, f)$. Then,

$$a^2 + d^2 = b^2 + c^2 \quad \text{--- (1)}$$

$$c^2 + f^2 = d^2 + e^2 \quad \text{--- (2)}$$

Adding equation (1) and equation (2):

$$\begin{aligned} (a^2 + d^2) + (c^2 + f^2) &= (b^2 + c^2) + (d^2 + e^2) \\ a^2 + d^2 + c^2 + f^2 &= b^2 + c^2 + d^2 + e^2 \end{aligned}$$

Subtracting $c^2 + d^2$ from both sides:

$$a^2 + f^2 = b^2 + e^2$$

This implies $(a, b) R (e, f)$. Hence, R is transitive.

Conclusion: Since R is reflexive, symmetric, and transitive, R is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Hence Proved

R is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Question 39

The Earth has 24 time zones, defined by dividing the Earth into 24 equal longitudinal segments. These are the regions on Earth that have the same standard time. For example, USA and India fall in different time zones, but Sri Lanka and India are in the same time zone.

A relation R is defined on the set $U = \{\text{All people on the Earth}\}$ such that $R = \{(x, y) \mid \text{the time difference between the time zones } x \text{ and } y \text{ reside in is 6 hours}\}$.

(i) Check whether the relation R is reflexive, symmetric and transitive.

(ii) Is relation R an equivalence relation?

Show your work.

TYPE 5 Verifying Relation Properties on Infinite Sets

Given: $U = \{\text{All people on the Earth}\}$, $R = \{(x, y) \in U \times U : \text{time difference between zones of } x \text{ and } y \text{ is 6 hours}\}$

Concept / Formula Used: $(x, x) \in R$ x $(x, y) \in R \implies (y, x) \in R$
 $(x, y) \in R \text{ and } (y, z) \in R \implies (x, z) \in R$

Solution:

Part (i): Checking Reflexivity, Symmetry, and Transitivity

Reflexivity: Let $x \in U$ be any person on Earth. Person x resides in a specific time zone. The time difference between x 's time zone and x 's own time zone is:

$$|t_x - t_x| = 0 \text{ hours}$$

Since $0 \neq 6$ hours, $(x, x) \notin R$ for any $x \in U$. Thus, R is not reflexive.

Symmetry: Let $(x, y) \in R$. This means the time difference between the time zones of x and y is 6 hours:

$$|t_x - t_y| = 6 \text{ hours}$$

Since $|t_y - t_x| = |t_x - t_y|$:

$$|t_y - t_x| = 6 \text{ hours}$$

This implies $(y, x) \in R$. Thus, R is symmetric.

Transitivity: Suppose $(x, y) \in R$ and $(y, z) \in R$. This means:

$$|t_x - t_y| = 6 \text{ hours}$$

$$|t_y - t_z| = 6 \text{ hours}$$

Evaluating the possible values for $t_x - t_z$:

$$t_x - t_z = (t_x - t_y) + (t_y - t_z) = 6 + 6 = 12 \text{ hours}$$

$$\text{or } t_x - t_z = (t_x - t_y) + (t_y - t_z) = 6 + (-6) = 0 \text{ hours}$$

In either case (12 hours or 0 hours), the time difference $|t_x - t_z| \neq 6$ hours. Thus, $(x, z) \notin R$. Hence, R is not transitive.

Part (ii): Is R an equivalence relation?

Since R is neither reflexive nor transitive, R fails to satisfy the definition of an equivalence relation.

Final Answer

- (i) R is symmetric, but neither reflexive nor transitive.
- (ii) No, R is not an equivalence relation.

1.2 – Functions: Domain and Range

Question 1

A function is defined by $f(x) = \frac{3x^2 + 2x - 1}{x + 1}$, $x \in \mathbb{R}, x \neq -1$. Find the value of $f(-3) + 1$.

TYPE 11 Evaluating Function Values

Given: $f(x) = \frac{3x^2 + 2x - 1}{x + 1}$, $x \in \mathbb{R}, x \neq -1$

To Find: $f(-3) + 1$

Concept / Formula Used: $f(a) = f(x)|_{x=a}$

Solution:

Substitute $x = -3$ into the given function $f(x)$:

$$\begin{aligned} f(-3) &= \frac{3(-3)^2 + 2(-3) - 1}{(-3) + 1} \\ &= \frac{3(9) - 6 - 1}{-2} \\ &= \frac{27 - 7}{-2} \\ &= \frac{20}{-2} \\ &= -10 \end{aligned}$$

Now, add 1 to $f(-3)$:

$$\begin{aligned} f(-3) + 1 &= -10 + 1 \\ &= -9 \end{aligned}$$

Final Answer

$$f(-3) + 1 = -9$$

Question 2

- (i) Is the relation f defined by $f(x) = \begin{cases} x^2, & 0 \leq x \leq 3 \\ 2x, & 3 \leq x \leq 5 \end{cases}$ a function? Justify your answer.
- (ii) A relation R in the set of real numbers $\mathbb{R}$ defined as $R = \{(a, b) : \sqrt{a} = b\}$ is a function or not. Justify.

TYPE 12 Testing Definition of a Function

Given: (i) $f(x) = \begin{cases} x^2, & 0 \leq x \leq 3 \\ 2x, & 3 \leq x \leq 5 \end{cases}$, (ii) $R = \{(a, b) \in \mathbb{R} \times \mathbb{R} : b = \sqrt{a}\}$

To Find: Determine whether f and R are functions with justification

Concept / Formula Used: $f : A \rightarrow B \quad x \in A \quad B$

Solution:

- (i) Check the value of $f(x)$ at the point of overlap $x = 3$:

$$\text{For } 0 \leq x \leq 3 : \quad f(3) = 3^2 = 9$$

$$\text{For } 3 \leq x \leq 5 : \quad f(3) = 2(3) = 6$$

Since $f(3)$ yields two distinct values (9 and 6), the domain element 3 does not have a unique image. Therefore, f is not a function.

- (ii) For $R = \{(a, b) \in \mathbb{R} \times \mathbb{R} : b = \sqrt{a}\}$: Consider any negative real number $a \in \mathbb{R}$ such that $a < 0$, e.g., $a = -4$. The expression $\sqrt{-4}$ does not produce a real number. Hence, negative real numbers in the domain do not have any image in $\mathbb{R}$. Therefore, R is not a function on $\mathbb{R}$.

Final Answer

- (i) No, f is not a function because $f(3)$ has two images (9 and 6).
(ii) No, R is not a function because $\sqrt{a}$ is undefined for negative real numbers $a < 0$.

Common Mistake: Assuming piecewise expressions are always functions without checking the boundary values for consistency is a common oversight.

Question 3

A real function f is defined by $f(x) = 3x - 2, x \in \mathbb{R}$, find the value of $f(-3) + f(0) \times f(7)$.

TYPE 11 Evaluating Function Values**Given:** $f(x) = 3x - 2, x \in \mathbb{R}$ **To Find:** $f(-3) + f(0) \times f(7)$ **Concept / Formula Used:** $f(a) = 3a - 2$ **Solution:**

Evaluate each term separately:

$$f(-3) = 3(-3) - 2 = -9 - 2 = -11$$

$$f(0) = 3(0) - 2 = 0 - 2 = -2$$

$$f(7) = 3(7) - 2 = 21 - 2 = 19$$

Substitute these calculated values into the main expression:

$$\begin{aligned} f(-3) + f(0) \times f(7) &= -11 + (-2) \times 19 \\ &= -11 - 38 \\ &= -49 \end{aligned}$$

Final Answer

$$f(-3) + f(0) \times f(7) = -49$$

Question 4

If a real function f is defined by $f(x) = 2x^2 - 3$, find x such that $f(x) = 15$.

TYPE 13 Finding Pre-images of Function Values**Given:** $f(x) = 2x^2 - 3$ and $f(x) = 15$ **To Find:** Values of x **Concept / Formula Used:** $f(x) = y \implies 2x^2 - 3 = y$ **Solution:**

Set $f(x) = 15$:

$$\begin{aligned}2x^2 - 3 &= 15 \\2x^2 &= 15 + 3 \\2x^2 &= 18 \\x^2 &= \frac{18}{2} \\x^2 &= 9 \\x &= \pm\sqrt{9} \\x &= \pm 3\end{aligned}$$

Final Answer

$$x = 3 \text{ or } x = -3$$

Question 5

Given $f(x) = x^3 - 1$, find x if $f(x)$ is 215.

TYPE 13 Finding Pre-images of Function Values

Given: $f(x) = x^3 - 1$ and $f(x) = 215$

To Find: Value of x

Concept / Formula Used: $x^3 - 1 = y \implies x = \sqrt[3]{y+1}$

Solution:

Set $f(x) = 215$:

$$\begin{aligned}x^3 - 1 &= 215 \\x^3 &= 215 + 1 \\x^3 &= 216 \\x &= \sqrt[3]{216} \\x &= 6\end{aligned}$$

Final Answer

$$x = 6$$

Question 6

A function is given by the formula $f(x) = 144 - 16x^2$. Calculate $f(2)$. Also find the values of x when $f(x) = 0$.

TYPE 13 Finding Pre-images of Function Values

Given: $f(x) = 144 - 16x^2$

To Find: $f(2)$ and x such that $f(x) = 0$

Concept / Formula Used: $f(a) = 144 - 16a^2$

Solution:

Calculate $f(2)$:

$$\begin{aligned}f(2) &= 144 - 16(2)^2 \\&= 144 - 16(4) \\&= 144 - 64 \\&= 80\end{aligned}$$

Now set $f(x) = 0$ to find the roots:

$$\begin{aligned}144 - 16x^2 &= 0 \\16x^2 &= 144 \\x^2 &= \frac{144}{16} \\x^2 &= 9 \\x &= \pm 3\end{aligned}$$

Final Answer

$f(2) = 80$, and $x = \pm 3$ when $f(x) = 0$

Question 7

A function ' f ' is defined by $f(x) = 2x^2 + 3$, for all $x \in \mathbb{R}$. Find (i) image of -1 under f (ii) element (elements) of the domain which has image 35

TYPE 13 Finding Pre-images of Function Values

Given: $f(x) = 2x^2 + 3, x \in \mathbb{R}$

To Find: (i) $f(-1)$, (ii) x such that $f(x) = 35$

Concept / Formula Used: $f(a) = 2a^2 + 3$

Solution:

(i) Image of -1 under f :

$$\begin{aligned}f(-1) &= 2(-1)^2 + 3 \\&= 2(1) + 3 \\&= 2 + 3 \\&= 5\end{aligned}$$

(ii) Elements of the domain with image 35:

$$\begin{aligned}f(x) &= 35 \\2x^2 + 3 &= 35 \\2x^2 &= 35 - 3 \\2x^2 &= 32 \\x^2 &= 16 \\x &= \pm 4\end{aligned}$$

Final Answer

(i) Image of -1 is 5 , (ii) Domain elements are 4 and -4

Question 8

If $f(x) = 2x^2 + 1$ and domain of $f = \{-2, -1, 0, 1, 3\}$, find (i) range of f (ii) $f(3) \times f(-2)$ (iii) x if $f(x) = 9$

TYPE 14 Domain and Range of Functions on Finite Sets

Given: $f(x) = 2x^2 + 1$, Domain = $\{-2, -1, 0, 1, 3\}$

To Find: (i) Range of f , (ii) $f(3) \times f(-2)$, (iii) x if $f(x) = 9$

Concept / Formula Used: $\text{Range}(f) = \{f(x) : x \in \text{Domain}\}$

Solution:

(i) Evaluate $f(x)$ for each element in the domain:

$$\begin{aligned}f(-2) &= 2(-2)^2 + 1 = 2(4) + 1 = 9 \\f(-1) &= 2(-1)^2 + 1 = 2(1) + 1 = 3 \\f(0) &= 2(0)^2 + 1 = 0 + 1 = 1 \\f(1) &= 2(1)^2 + 1 = 2(1) + 1 = 3 \\f(3) &= 2(3)^2 + 1 = 2(9) + 1 = 19\end{aligned}$$

Gathering unique outputs gives the range:

$$\text{Range of } f = \{1, 3, 9, 19\}$$

(ii) Calculate $f(3) \times f(-2)$:

$$\begin{aligned}f(3) \times f(-2) &= 19 \times 9 \\&= 171\end{aligned}$$

(iii) Find $x \in \text{Domain}$ such that $f(x) = 9$:

$$\begin{aligned} 2x^2 + 1 &= 9 \\ 2x^2 &= 8 \\ x^2 &= 4 \\ x &= \pm 2 \end{aligned}$$

Cross-referencing with $\text{Domain} = \{-2, -1, 0, 1, 3\}$, the values belonging to the domain are $x = -2$. (Note: $x = 2$ is not in the given domain).

Final Answer

(i) $\text{Range} = \{1, 3, 9, 19\}$, (ii) 171, (iii) $x = -2$

Question 9

A function ' f ' is defined by $f(x) = x^2 + 3, x \in \mathbb{N}$ and $x \leq 5$. (i) Find the range of f (ii) Find $f(2) \times f(5)$ (iii) Does $f(-3)$ exist? (iv) Find x when $f(x) = 7$

TYPE 14 Domain and Range of Functions on Finite Sets

Given: $f(x) = x^2 + 3$, $\text{Domain} = \{x \in \mathbb{N} : x \leq 5\} = \{1, 2, 3, 4, 5\}$

To Find: (i) Range of f , (ii) $f(2) \times f(5)$, (iii) Existence of $f(-3)$, (iv) x when $f(x) = 7$

Concept / Formula Used: $\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$

Solution:

(i) Calculate $f(x)$ for all elements in the domain $\{1, 2, 3, 4, 5\}$:

$$\begin{aligned} f(1) &= 1^2 + 3 = 4 \\ f(2) &= 2^2 + 3 = 7 \\ f(3) &= 3^2 + 3 = 12 \\ f(4) &= 4^2 + 3 = 19 \\ f(5) &= 5^2 + 3 = 28 \end{aligned}$$

Thus, $\text{Range}(f) = \{4, 7, 12, 19, 28\}$.

(ii) Calculate $f(2) \times f(5)$:

$$\begin{aligned} f(2) \times f(5) &= 7 \times 28 \\ &= 196 \end{aligned}$$

(iii) Check if $f(-3)$ exists: Since $-3 \notin \mathbb{N}$, -3 is not in the specified domain of f . Thus, $f(-3)$ does not exist.

(iv) Solve for x when $f(x) = 7$:

$$x^2 + 3 = 7$$

$$x^2 = 4$$

$$x = 2 \quad (\text{since } x \in \mathbb{N}, x > 0)$$

Final Answer

(i) Range = $\{4, 7, 12, 19, 28\}$, (ii) 196, (iii) No, $f(-3)$ does not exist, (iv) $x = 2$

Question 10

Find the domain of the following real functions: (i) $f(x) = |x + 2|$ (ii) $f(x) = -|x - 2|$
(iii) $f(x) = 2 - |x - 1|$

TYPE 15 Finding Domain and Range of Modulus Functions

Given: (i) $f(x) = |x + 2|$, (ii) $f(x) = -|x - 2|$, (iii) $f(x) = 2 - |x - 1|$

To Find: Domain of each function

Concept / Formula Used: $|g(x)| \in \mathbb{R} \quad x \in \mathbb{R}$

Solution:

(i) For $f(x) = |x + 2|$: The modulus expression $|x + 2|$ is defined and returns a real value for every real number x .

$$\text{Domain}(f) = \mathbb{R} = (-\infty, \infty)$$

(ii) For $f(x) = -|x - 2|$: The expression $-|x - 2|$ is defined for all real values of x .

$$\text{Domain}(f) = \mathbb{R} = (-\infty, \infty)$$

(iii) For $f(x) = 2 - |x - 1|$: The expression $2 - |x - 1|$ is defined for all real values of x .

$$\text{Domain}(f) = \mathbb{R} = (-\infty, \infty)$$

Final Answer

(i) Domain = $\mathbb{R}$, (ii) Domain = $\mathbb{R}$, (iii) Domain = $\mathbb{R}$

Question 11

Find the range of the following functions: (i) $f(x) = x^2 + 2, x \in \mathbb{R}$ (ii) $f(x) = 3 - 2x, x \in \mathbb{R}, x \geq 1$

TYPE 16 Finding Domain and Range of Polynomial Functions

Given: (i) $f(x) = x^2 + 2, x \in \mathbb{R}$, (ii) $f(x) = 3 - 2x, x \in \mathbb{R}, x \geq 1$

To Find: Range of each function

Concept / Formula Used: $x^2 \geq 0 \quad x \in \mathbb{R}$

Solution:

(i) For $f(x) = x^2 + 2, x \in \mathbb{R}$: For any real number x :

$$x^2 \geq 0$$

Add 2 to both sides:

$$\begin{aligned} x^2 + 2 &\geq 2 \\ f(x) &\geq 2 \end{aligned}$$

Hence, $\text{Range}(f) = [2, \infty)$.

(ii) For $f(x) = 3 - 2x, x \geq 1$: Starting from the domain condition $x \geq 1$: Multiply by -2 (reversing the inequality direction):

$$-2x \leq -2$$

Add 3 to both sides:

$$\begin{aligned} 3 - 2x &\leq 3 - 2 \\ f(x) &\leq 1 \end{aligned}$$

Hence, $\text{Range}(f) = (-\infty, 1]$.

Final Answer

(i) $\text{Range} = [2, \infty)$, (ii) $\text{Range} = (-\infty, 1]$

Question 12

Find the domain and the range of the function $f(x) = 2x^2 + 1$. Also find $f(-2)$ and the numbers which are associated with the number 51 in its range.

TYPE 16 Finding Domain and Range of Polynomial Functions

Given: $f(x) = 2x^2 + 1$

To Find: Domain, Range, $f(-2)$, and numbers associated with 51

Concept / Formula Used: $x^2 \geq 0 \implies 2x^2 + 1 \geq 1$

Solution:

1. Domain: Since $2x^2 + 1$ is defined for every real number x :

$$\text{Domain}(f) = \mathbb{R}$$

2. Range: For all $x \in \mathbb{R}$:

$$\begin{aligned}x^2 &\geq 0 \\2x^2 &\geq 0 \\2x^2 + 1 &\geq 1 \\f(x) &\geq 1\end{aligned}$$

Thus, $\text{Range}(f) = [1, \infty)$.

3. Calculate $f(-2)$:

$$\begin{aligned}f(-2) &= 2(-2)^2 + 1 \\&= 2(4) + 1 \\&= 8 + 1 \\&= 9\end{aligned}$$

4. Find numbers associated with 51: Set $f(x) = 51$:

$$\begin{aligned}2x^2 + 1 &= 51 \\2x^2 &= 50 \\x^2 &= 25 \\x &= \pm 5\end{aligned}$$

Final Answer

Domain = $\mathbb{R}$, Range = $[1, \infty)$, $f(-2) = 9$, associated numbers = 5 and -5

Question 13

Find the domain and the range of the following functions: (i) $f(x) = \sqrt{x+2}$ (ii) $f(x) = \sqrt{3-2x}$ (iii) $f(x) = \frac{1}{\sqrt{x-5}}$

TYPE 17 Finding Domain and Range of Irrational Functions

★ Likely Exam Question

Given: (i) $f(x) = \sqrt{x+2}$, (ii) $f(x) = \sqrt{3-2x}$, (iii) $f(x) = \frac{1}{\sqrt{x-5}}$

To Find: Domain and Range of each function

Concept / Formula Used: $\sqrt{g(x)}$ $g(x) \geq 0$ $\frac{1}{\sqrt{g(x)}}$ $g(x) > 0$

Solution:

(i) For $f(x) = \sqrt{x+2}$: Domain condition:

$$\begin{aligned}x+2 &\geq 0 \\x &\geq -2\end{aligned}$$

Thus, $\text{Domain}(f) = [-2, \infty)$. Since square root outputs non-negative real numbers, as x increases from -2 to ∞ , $f(x)$ increases from 0 to ∞ :

$$\text{Range}(f) = [0, \infty)$$

(ii) For $f(x) = \sqrt{3-2x}$: Domain condition:

$$\begin{aligned}3-2x &\geq 0 \\-2x &\geq -3 \\x &\leq \frac{3}{2}\end{aligned}$$

Thus, $\text{Domain}(f) = (-\infty, \frac{3}{2}]$. Square root values are non-negative, so:

$$\text{Range}(f) = [0, \infty)$$

(iii) For $f(x) = \frac{1}{\sqrt{x-5}}$: Domain condition (strictly positive in denominator):

$$\begin{aligned}x-5 &> 0 \\x &> 5\end{aligned}$$

Thus, $\text{Domain}(f) = (5, \infty)$. As $x \rightarrow 5^+$, $\sqrt{x-5} \rightarrow 0^+$, so $f(x) \rightarrow \infty$. As $x \rightarrow \infty$, $f(x) \rightarrow 0^+$. Since $\sqrt{x-5} > 0$, $f(x) > 0$:

$$\text{Range}(f) = (0, \infty)$$

Final Answer

- (i) $\text{Domain} = [-2, \infty)$, $\text{Range} = [0, \infty)$
(ii) $\text{Domain} = (-\infty, \frac{3}{2}]$, $\text{Range} = [0, \infty)$
(iii) $\text{Domain} = (5, \infty)$, $\text{Range} = (0, \infty)$

Question 14

Find the domain and the range of the following functions: (i) $f(x) = \frac{4-x}{x-4}$ (ii) $f(x) = \frac{x^2-9}{x-3}$ (iii) $f(x) = \frac{x+1}{2x+1}$ (iv) $f(x) = \frac{4+x}{4-x}$

TYPE 18 Finding Domain and Range of Rational Functions

Given: (i) $f(x) = \frac{4-x}{x-4}$, (ii) $f(x) = \frac{x^2-9}{x-3}$, (iii) $f(x) = \frac{x+1}{2x+1}$, (iv) $f(x) = \frac{4+x}{4-x}$

To Find: Domain and Range of each function

Concept / Formula Used: $\frac{P(x)}{Q(x)} \quad Q(x) \neq 0$

Solution:

(i) For $f(x) = \frac{4-x}{x-4}$: Denominator condition: $x-4 \neq 0 \implies x \neq 4$.

$$\text{Domain}(f) = \mathbb{R} \setminus \{4\}$$

Simplifying $f(x)$ for $x \neq 4$:

$$f(x) = \frac{-(x-4)}{x-4} = -1$$

$$\text{Range}(f) = \{-1\}$$

(ii) For $f(x) = \frac{x^2-9}{x-3}$: Denominator condition: $x-3 \neq 0 \implies x \neq 3$.

$$\text{Domain}(f) = \mathbb{R} \setminus \{3\}$$

Simplifying $f(x)$ for $x \neq 3$:

$$f(x) = \frac{(x-3)(x+3)}{x-3} = x+3$$

Since $x \neq 3$, $f(x) \neq 3+3=6$.

$$\text{Range}(f) = \mathbb{R} \setminus \{6\}$$

(iii) For $f(x) = \frac{x+1}{2x+1}$: Denominator condition: $2x+1 \neq 0 \implies x \neq -\frac{1}{2}$.

$$\text{Domain}(f) = \mathbb{R} \setminus \left\{-\frac{1}{2}\right\}$$

To find the range, set $y = \frac{x+1}{2x+1}$ and express x in terms of y :

$$y(2x+1) = x+1$$

$$2xy + y = x+1$$

$$2xy - x = 1 - y$$

$$x(2y-1) = 1-y$$

$$x = \frac{1-y}{2y-1}$$

For x to be real, $2y-1 \neq 0 \implies y \neq \frac{1}{2}$.

$$\text{Range}(f) = \mathbb{R} \setminus \left\{\frac{1}{2}\right\}$$

(iv) For $f(x) = \frac{4+x}{4-x}$: Denominator condition: $4-x \neq 0 \implies x \neq 4$.

$$\text{Domain}(f) = \mathbb{R} \setminus \{4\}$$

Set $y = \frac{4+x}{4-x}$ and express x in terms of y :

$$y(4-x) = 4+x$$

$$4y - xy = 4+x$$

$$4y - 4 = x + xy$$

$$x(1+y) = 4y-4$$

$$x = \frac{4y-4}{y+1}$$

For x to be real, $y+1 \neq 0 \implies y \neq -1$.

$$\text{Range}(f) = \mathbb{R} \setminus \{-1\}$$

Final Answer

(i) Domain = $\mathbb{R} \setminus \{4\}$, Range = $\{-1\}$

(ii) Domain = $\mathbb{R} \setminus \{3\}$, Range = $\mathbb{R} \setminus \{6\}$

(iii) Domain = $\mathbb{R} \setminus \left\{-\frac{1}{2}\right\}$, Range = $\mathbb{R} \setminus \left\{\frac{1}{2}\right\}$

(iv) Domain = $\mathbb{R} \setminus \{4\}$, Range = $\mathbb{R} \setminus \{-1\}$

Question 15

Find the domain and the range of the following functions: (i) $f(x) = \sqrt{16-x^2}$ (ii) $f(x) = \sqrt{x^2-9}$ (iii) $f(x) = \frac{1}{\sqrt{4-x^2}}$

TYPE 17 Finding Domain and Range of Irrational Functions

★ Likely Exam Question

Given: (i) $f(x) = \sqrt{16-x^2}$, (ii) $f(x) = \sqrt{x^2-9}$, (iii) $f(x) = \frac{1}{\sqrt{4-x^2}}$

To Find: Domain and Range of each function

Concept / Formula Used: $a^2 - x^2 \geq 0 \implies -a \leq x \leq a$

Solution:

(i) For $f(x) = \sqrt{16-x^2}$: Domain condition:

$$16 - x^2 \geq 0$$

$$x^2 \leq 16$$

$$-4 \leq x \leq 4$$

$$\text{Domain}(f) = [-4, 4]$$

Range: For $-4 \leq x \leq 4$, $0 \leq x^2 \leq 16$.

$$0 \leq 16 - x^2 \leq 16$$

$$0 \leq \sqrt{16 - x^2} \leq 4$$

$$\text{Range}(f) = [0, 4]$$

(ii) For $f(x) = \sqrt{x^2 - 9}$: Domain condition:

$$x^2 - 9 \geq 0$$

$$(x - 3)(x + 3) \geq 0$$

$$x \leq -3 \text{ or } x \geq 3$$

$$\text{Domain}(f) = (-\infty, -3] \cup [3, \infty)$$

Range: Since $x^2 \geq 9$, $x^2 - 9 \geq 0$, giving $\sqrt{x^2 - 9} \geq 0$. As $|x| \rightarrow \infty$, $f(x) \rightarrow \infty$.

$$\text{Range}(f) = [0, \infty)$$

(iii) For $f(x) = \frac{1}{\sqrt{4 - x^2}}$: Domain condition:

$$4 - x^2 > 0$$

$$x^2 < 4$$

$$-2 < x < 2$$

$$\text{Domain}(f) = (-2, 2)$$

Range: For $-2 < x < 2$, $0 < x^2 < 4$.

$$0 < 4 - x^2 \leq 4$$

$$0 < \sqrt{4 - x^2} \leq 2$$

$$\frac{1}{\sqrt{4 - x^2}} \geq \frac{1}{2}$$

As $x \rightarrow \pm 2$, $4 - x^2 \rightarrow 0^+$, so $f(x) \rightarrow \infty$.

$$\text{Range}(f) = \left[\frac{1}{2}, \infty \right)$$

Final Answer

(i) Domain = $[-4, 4]$, Range = $[0, 4]$

(ii) Domain = $(-\infty, -3] \cup [3, \infty)$, Range = $[0, \infty)$

(iii) Domain = $(-2, 2)$, Range = $\left[\frac{1}{2}, \infty \right)$

Question 16

Find the domain and the range of the following functions: (i) $f(x) = |x - 3|$ (ii) $f(x) = 3 - |x - 2|$

TYPE 15 Finding Domain and Range of Modulus Functions

Given: (i) $f(x) = |x - 3|$, (ii) $f(x) = 3 - |x - 2|$

To Find: Domain and Range of each function

Concept / Formula Used: $|g(x)| \geq 0 \quad x \in \mathbb{R}$

Solution:

(i) For $f(x) = |x - 3|$: The expression $|x - 3|$ is defined for all real numbers.

$$\text{Domain}(f) = \mathbb{R}$$

Since absolute values are non-negative:

$$|x - 3| \geq 0$$

$$\text{Range}(f) = [0, \infty)$$

(ii) For $f(x) = 3 - |x - 2|$: Defined for all real numbers:

$$\text{Domain}(f) = \mathbb{R}$$

Since $|x - 2| \geq 0$:

$$-|x - 2| \leq 0$$

$$3 - |x - 2| \leq 3$$

$$f(x) \leq 3$$

$$\text{Range}(f) = (-\infty, 3]$$

Final Answer

(i) Domain = $\mathbb{R}$, Range = $[0, \infty)$

(ii) Domain = $\mathbb{R}$, Range = $(-\infty, 3]$

Question 17

If a real function f is defined by $f(x) = \frac{|x| - x}{2x}$, then find its domain and range.

TYPE 15 Finding Domain and Range of Modulus Functions

Given: $f(x) = \frac{|x| - x}{2x}$

To Find: Domain and Range of f

Concept / Formula Used: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

Solution:

1. Domain: The denominator must not vanish:

$$2x \neq 0 \implies x \neq 0$$

Thus, $\text{Domain}(f) = \mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$.

2. Range: Consider the two cases for $x \in \text{Domain}(f)$: Case 1: When $x > 0$: Here, $|x| = x$.

$$\begin{aligned} f(x) &= \frac{x - x}{2x} \\ &= \frac{0}{2x} \\ &= 0 \end{aligned}$$

Case 2: When $x < 0$: Here, $|x| = -x$.

$$\begin{aligned} f(x) &= \frac{-x - x}{2x} \\ &= \frac{-2x}{2x} \\ &= -1 \end{aligned}$$

Thus, the function takes only two constant values: 0 and -1 .

$$\text{Range}(f) = \{-1, 0\}$$

Final Answer

Domain = $\mathbb{R} \setminus \{0\}$, Range = $\{-1, 0\}$

Question 18

Find the domain and the range of the function f defined by $f(x) = \frac{|x - 4|}{x - 4}$.

TYPE 15 Finding Domain and Range of Modulus Functions

★ Likely Exam Question

Given: $f(x) = \frac{|x - 4|}{x - 4}$

To Find: Domain and Range of f

Concept / Formula Used: $|x - 4| = \begin{cases} x - 4, & x > 4 \\ -(x - 4), & x < 4 \end{cases}$

Solution:

1. Domain: The denominator must not be zero:

$$x - 4 \neq 0 \implies x \neq 4$$

Thus, $\text{Domain}(f) = \mathbb{R} \setminus \{4\} = (-\infty, 4) \cup (4, \infty)$.

2. Range: Case 1: For $x > 4$: Since $x - 4 > 0$, we have $|x - 4| = x - 4$.

$$f(x) = \frac{x - 4}{x - 4} = 1$$

Case 2: For $x < 4$: Since $x - 4 < 0$, we have $|x - 4| = -(x - 4)$.

$$f(x) = \frac{-(x - 4)}{x - 4} = -1$$

The function yields only two values: 1 and -1 .

$$\text{Range}(f) = \{-1, 1\}$$

Final Answer

Domain = $\mathbb{R} \setminus \{4\}$, Range = $\{-1, 1\}$

Question 19

If $f(x) = \log \left(\frac{1+x}{1-x} \right)$, then show that $f(x) + f(y) = f \left(\frac{x+y}{1+xy} \right)$.

TYPE 19 Proving Functional Identities

★ Likely Exam Question

Given: $f(x) = \log \left(\frac{1+x}{1-x} \right)$

To Prove: $f(x) + f(y) = f \left(\frac{x+y}{1+xy} \right)$

Concept / Formula Used: $\log A + \log B = \log(A \cdot B)$

Solution:

Start with the Left-Hand Side (LHS):

$$\begin{aligned}\text{LHS} &= f(x) + f(y) \\ &= \log\left(\frac{1+x}{1-x}\right) + \log\left(\frac{1+y}{1-y}\right)\end{aligned}$$

Using the logarithmic property $\log A + \log B = \log(A \cdot B)$:

$$\begin{aligned}\text{LHS} &= \log\left[\left(\frac{1+x}{1-x}\right) \cdot \left(\frac{1+y}{1-y}\right)\right] \\ &= \log\left[\frac{(1+x)(1+y)}{(1-x)(1-y)}\right] \\ &= \log\left[\frac{1+x+y+xy}{1-x-y+xy}\right] \\ &= \log\left[\frac{(1+xy) + (x+y)}{(1+xy) - (x+y)}\right]\end{aligned}$$

Divide both numerator and denominator inside the log by $(1+xy)$:

$$\begin{aligned}\text{LHS} &= \log\left[\frac{\frac{(1+xy)+(x+y)}{1+xy}}{\frac{(1+xy)-(x+y)}{1+xy}}\right] \\ &= \log\left[\frac{1 + \frac{x+y}{1+xy}}{1 - \frac{x+y}{1+xy}}\right]\end{aligned}$$

By the given definition of the function $f(t) = \log\left(\frac{1+t}{1-t}\right)$ where $t = \frac{x+y}{1+xy}$:

$$\text{LHS} = f\left(\frac{x+y}{1+xy}\right) = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 20

Find the domain of the following functions: (i) $\log(4x - 3)$ (ii) $\frac{1}{\log(9 - x^2)}$

TYPE 20 Finding Domain and Range of Logarithmic Functions

Given: (i) $f(x) = \log(4x - 3)$, (ii) $f(x) = \frac{1}{\log(9 - x^2)}$

To Find: Domain of each function

Concept / Formula Used: $\log(g(x))$ $g(x) > 0$ $\frac{1}{h(x)}$ $h(x) \neq 0$

Solution:

(i) For $f(x) = \log(4x - 3)$: Logarithm is defined only when its argument is positive:

$$4x - 3 > 0$$

$$4x > 3$$

$$x > \frac{3}{4}$$

$$\text{Domain}(f) = \left(\frac{3}{4}, \infty\right)$$

(ii) For $f(x) = \frac{1}{\log(9 - x^2)}$: We apply two separate constraints: 1. Positivity of log argument:

$$9 - x^2 > 0$$

$$x^2 < 9$$

$$-3 < x < 3$$

2. Non-zero denominator:

$$\log(9 - x^2) \neq 0$$

$$9 - x^2 \neq 1 \quad (\text{since } \log 1 = 0)$$

$$x^2 \neq 8$$

$$x \neq \pm\sqrt{8} = \pm 2\sqrt{2}$$

Combining both conditions:

$$\text{Domain}(f) = (-3, 3) \setminus \{-2\sqrt{2}, 2\sqrt{2}\}$$

Final Answer

(i) Domain = $\left(\frac{3}{4}, \infty\right)$

(ii) Domain = $(-3, 3) \setminus \{-2\sqrt{2}, 2\sqrt{2}\}$

Question 21

Find the domain of the following functions: (i) $f(x) = \frac{1}{3 - 2\sin x}$ (ii) $f(x) = \frac{1}{\sqrt{1 - \cos x}}$

TYPE 21 Finding Domain and Range of Trigonometric Functions

Given: (i) $f(x) = \frac{1}{3 - 2\sin x}$, (ii) $f(x) = \frac{1}{\sqrt{1 - \cos x}}$

To Find: Domain of each function

Concept / Formula Used: $-1 \leq \sin x \leq 1$ $-1 \leq \cos x \leq 1$

Solution:

(i) For $f(x) = \frac{1}{3 - 2\sin x}$: Denominator must not be zero:

$$3 - 2\sin x \neq 0$$

$$2\sin x \neq 3$$

$$\sin x \neq \frac{3}{2}$$

Since $-1 \leq \sin x \leq 1$ for all real numbers x , $\sin x$ can never equal $\frac{3}{2}$. Therefore, $3 - 2\sin x > 0$ for all $x \in \mathbb{R}$.

$$\text{Domain}(f) = \mathbb{R}$$

(ii) For $f(x) = \frac{1}{\sqrt{1 - \cos x}}$: The term inside the square root in the denominator must be strictly positive:

$$1 - \cos x > 0$$

$$\cos x < 1$$

Since $\cos x \leq 1$ for all real numbers x , $\cos x < 1$ holds everywhere except where $\cos x = 1$. Solve for $\cos x = 1$:

$$x = 2n\pi, \quad n \in \mathbb{Z}$$

Thus, $1 - \cos x > 0$ for all $x \in \mathbb{R} \setminus \{2n\pi : n \in \mathbb{Z}\}$.

$$\text{Domain}(f) = \mathbb{R} \setminus \{2n\pi : n \in \mathbb{Z}\}$$

Final Answer

(i) Domain = $\mathbb{R}$

(ii) Domain = $\mathbb{R} \setminus \{2n\pi : n \in \mathbb{Z}\}$

Question 22

Find the range of the following functions: (i) $f(x) = 2 - 3\cos x$ (ii) $f(x) = 2 + 5\sin 3x$

TYPE 21 Finding Domain and Range of Trigonometric Functions

Given: (i) $f(x) = 2 - 3\cos x$, (ii) $f(x) = 2 + 5\sin 3x$

To Find: Range of each function

Concept / Formula Used: $-1 \leq \cos x \leq 1$ $-1 \leq \sin \theta \leq 1$

Solution:

(i) For $f(x) = 2 - 3 \cos x$: For all $x \in \mathbb{R}$:

$$-1 \leq \cos x \leq 1$$

Multiply by -3 (reversing the inequality signs):

$$\begin{aligned} 3 &\geq -3 \cos x \geq -3 \\ -3 &\leq -3 \cos x \leq 3 \end{aligned}$$

Add 2 throughout:

$$\begin{aligned} 2 - 3 &\leq 2 - 3 \cos x \leq 2 + 3 \\ -1 &\leq f(x) \leq 5 \end{aligned}$$

$$\text{Range}(f) = [-1, 5]$$

(ii) For $f(x) = 2 + 5 \sin 3x$: For all $x \in \mathbb{R}$, $\theta = 3x \in \mathbb{R}$, so:

$$-1 \leq \sin 3x \leq 1$$

Multiply by 5:

$$-5 \leq 5 \sin 3x \leq 5$$

Add 2 throughout:

$$\begin{aligned} 2 - 5 &\leq 2 + 5 \sin 3x \leq 2 + 5 \\ -3 &\leq f(x) \leq 7 \end{aligned}$$

$$\text{Range}(f) = [-3, 7]$$

Final Answer

(i) $\text{Range} = [-1, 5]$, (ii) $\text{Range} = [-3, 7]$

1.3 – One-One, Onto and Bijective Functions

Question 1

Show that the function $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(x) = 3x - 2$ is one-one.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(x) = 3x - 2$ is one-one (injective).

Concept / Formula Used: $f : A \rightarrow B$ $f(x_1) = f(x_2) \implies x_1 = x_2$ $x_1, x_2 \in A$

Solution:

Let $x_1, x_2 \in \mathbb{Q}$ such that $f(x_1) = f(x_2)$.

By definition of $f(x)$:

$$3x_1 - 2 = 3x_2 - 2$$

Adding 2 to both sides:

$$3x_1 = 3x_2$$

Dividing both sides by 3:

$$x_1 = x_2$$

Since $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in \mathbb{Q}$, the function f is one-one.

Hence Proved

The function $f : \mathbb{Q} \rightarrow \mathbb{Q}$ is one-one.

Question 2

Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 2x - 1$ is not onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 2x - 1$ is not onto (surjective).

Concept / Formula Used: $f : A \rightarrow B$ $y \in B$ $x \in A$ $f(x) = y$

Solution:

Let $y \in \mathbb{N}$ be an element in the codomain.

We set $f(x) = y$:

$$\begin{aligned} 2x - 1 &= y \\ 2x &= y + 1 \\ x &= \frac{y + 1}{2} \end{aligned}$$

Consider $y = 2 \in \mathbb{N}$ (codomain).

Substituting $y = 2$:

$$\begin{aligned} x &= \frac{2 + 1}{2} \\ x &= \frac{3}{2} \notin \mathbb{N} \end{aligned}$$

Thus, there is no element $x \in \mathbb{N}$ (domain) such that $f(x) = 2$.

Hence, f is not onto.

Hence Proved

The function $f : \mathbb{N} \rightarrow \mathbb{N}$ is not onto.

Common Mistake: Assuming $\frac{y+1}{2}$ is always a natural number. For even values of y , $\frac{y+1}{2}$ is not an integer, so it does not belong to $\mathbb{N}$.

Question 3

Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n^2$ is injective.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n^2$ is injective.

Concept / Formula Used: $f : A \rightarrow B \quad f(n_1) = f(n_2) \implies n_1 = n_2 \quad n_1, n_2 \in A$

Solution:

Let $n_1, n_2 \in \mathbb{N}$ such that $f(n_1) = f(n_2)$.

By definition of $f(n)$:

$$n_1^2 = n_2^2$$

Taking the square root on both sides:

$$n_1 = \pm n_2$$

Since $n_1, n_2 \in \mathbb{N}$, both $n_1 > 0$ and $n_2 > 0$.

Therefore, $n_1 = -n_2$ is impossible for natural numbers.

Hence:

$$n_1 = n_2$$

Thus, $f(n_1) = f(n_2) \implies n_1 = n_2$, so f is injective.

Hence Proved

The function $f : \mathbb{N} \rightarrow \mathbb{N}$ is injective.

Question 4

Is the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(m) = m^2 + m + 2$ one-one? Justify your answer.

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: The function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(m) = m^2 + m + 2$.

Concept / Formula Used: $f : A \rightarrow B \quad f(m_1) = f(m_2) \implies m_1 = m_2 \quad m_1, m_2 \in A$

Solution:

Let $m_1, m_2 \in \mathbb{N}$ such that $f(m_1) = f(m_2)$.

Substitute into the function definition:

$$m_1^2 + m_1 + 2 = m_2^2 + m_2 + 2$$

Subtract 2 from both sides:

$$m_1^2 + m_1 = m_2^2 + m_2$$

Rearranging terms to one side:

$$\begin{aligned}(m_1^2 - m_2^2) + (m_1 - m_2) &= 0 \\(m_1 - m_2)(m_1 + m_2) + (m_1 - m_2) &= 0 \\(m_1 - m_2)(m_1 + m_2 + 1) &= 0\end{aligned}$$

Since $m_1, m_2 \in \mathbb{N}$, we have $m_1 \geq 1$ and $m_2 \geq 1$, which implies:

$$m_1 + m_2 + 1 \geq 3 \neq 0$$

Dividing both sides by $(m_1 + m_2 + 1)$:

$$\begin{aligned}m_1 - m_2 &= 0 \\m_1 &= m_2\end{aligned}$$

Therefore, $f(m_1) = f(m_2) \implies m_1 = m_2$ for all $m_1, m_2 \in \mathbb{N}$.

Hence, the function f is one-one.

Final Answer

Yes, the function f is one-one.

Question 5

What is the range of the function $f(x) = \frac{|x-1|}{x-1}$?

TYPE 15 Finding Domain and Range of Modulus Functions

To Find: The range of the function $f(x) = \frac{|x-1|}{x-1}$.

Concept / Formula Used: $|x-1| = \begin{cases} x-1, & \text{if } x > 1 \\ -(x-1), & \text{if } x < 1 \end{cases}$

Solution:

The function is defined for all $x \in \mathbb{R} \setminus \{1\}$.

Case 1: When $x > 1$:

$$\begin{aligned}|x - 1| &= x - 1 \\ f(x) &= \frac{x - 1}{x - 1} = 1\end{aligned}$$

Case 2: When $x < 1$:

$$\begin{aligned}|x - 1| &= -(x - 1) \\ f(x) &= \frac{-(x - 1)}{x - 1} = -1\end{aligned}$$

Thus, the only values taken by $f(x)$ are -1 and 1 .

Hence, the range of $f(x)$ is $\{-1, 1\}$.

Final Answer

Range of $f = \{-1, 1\}$

Common Mistake: Including $x = 1$ in the domain. At $x = 1$, the expression is $\frac{0}{0}$ which is undefined.

Question 6

Find the domain of the function $f(x) = \frac{1}{5 - 3 \sin x}$.

TYPE 21 Finding Domain and Range of Trigonometric Functions

To Find: The domain of $f(x) = \frac{1}{5 - 3 \sin x}$.

Concept / Formula Used: $\frac{1}{g(x)} \quad g(x) \neq 0 \quad -1 \leq \sin x \leq 1$

Solution:

The given function is:

$$f(x) = \frac{1}{5 - 3 \sin x}$$

For $f(x)$ to be defined, the denominator must not be zero:

$$\begin{aligned}5 - 3 \sin x &\neq 0 \\ 3 \sin x &\neq 5 \\ \sin x &\neq \frac{5}{3}\end{aligned}$$

Using the standard result $-1 \leq \sin x \leq 1$ for all real x :

Multiplying by -3 :

$$-3 \leq -3 \sin x \leq 3$$

Adding 5 to all parts:

$$\begin{aligned} 5 - 3 &\leq 5 - 3 \sin x \leq 5 + 3 \\ 2 &\leq 5 - 3 \sin x \leq 8 \end{aligned}$$

Since $5 - 3 \sin x \geq 2 > 0$ for all real numbers x , the denominator is never zero for any $x \in \mathbb{R}$.

Thus, $f(x)$ is defined for all $x \in \mathbb{R}$.

Final Answer

Domain of $f = \mathbb{R}$

Question 7

Find the range of the function $f(x) = \frac{1}{2 - \cos x}$.

TYPE 21 Finding Domain and Range of Trigonometric Functions

To Find: The range of $f(x) = \frac{1}{2 - \cos x}$.

Concept / Formula Used: $x \quad -1 \leq \cos x \leq 1$

Solution:

Using the standard result $-1 \leq \cos x \leq 1$ for all $x \in \mathbb{R}$:

Multiply the inequality by -1 (which reverses inequality signs):

$$-1 \leq -\cos x \leq 1$$

Add 2 to all parts of the inequality:

$$\begin{aligned} 2 - 1 &\leq 2 - \cos x \leq 2 + 1 \\ 1 &\leq 2 - \cos x \leq 3 \end{aligned}$$

Taking the reciprocal of all terms (since all terms are positive):

$$\begin{aligned} \frac{1}{3} &\leq \frac{1}{2 - \cos x} \leq \frac{1}{1} \\ \frac{1}{3} &\leq f(x) \leq 1 \end{aligned}$$

Therefore, the range of $f(x)$ is $\left[\frac{1}{3}, 1\right]$.

Final Answer

Range of $f = \left[\frac{1}{3}, 1\right]$

Question 8

Find the range of the following functions:

- (i) $f(x) = \sin x + \cos x$
 (ii) $f(x) = 3 \sin x + 4 \cos x$

TYPE 21 Finding Domain and Range of Trigonometric Functions

To Find: The range of (i) $f(x) = \sin x + \cos x$ and (ii) $f(x) = 3 \sin x + 4 \cos x$.

Concept / Formula Used: $a \sin x + b \cos x - \sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$

Solution:

Part (i): $f(x) = \sin x + \cos x$

Here, $a = 1$ and $b = 1$.

Calculate $\sqrt{a^2 + b^2}$:

$$\begin{aligned}\sqrt{a^2 + b^2} &= \sqrt{1^2 + 1^2} \\ &= \sqrt{1 + 1} \\ &= \sqrt{2}\end{aligned}$$

Using the standard result $-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$:

$$-\sqrt{2} \leq \sin x + \cos x \leq \sqrt{2}$$

Thus, the range of $f(x) = \sin x + \cos x$ is $[-\sqrt{2}, \sqrt{2}]$.

Part (ii): $f(x) = 3 \sin x + 4 \cos x$

Here, $a = 3$ and $b = 4$.

Calculate $\sqrt{a^2 + b^2}$:

$$\begin{aligned}\sqrt{a^2 + b^2} &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

Using the standard result $-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$:

$$-5 \leq 3 \sin x + 4 \cos x \leq 5$$

Thus, the range of $f(x) = 3 \sin x + 4 \cos x$ is $[-5, 5]$.

Final Answer

- (i) Range of $f = [-\sqrt{2}, \sqrt{2}]$
 (ii) Range of $f = [-5, 5]$

Question 9

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = |2x - 3|$.

- (i) Is f one-one?
- (ii) Is f onto?

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = |2x - 3|$.

Concept / Formula Used: $f : A \rightarrow B \quad f(x_1) = f(x_2) \implies x_1 = x_2 \quad f \quad B$

Solution:

Part (i): Checking for One-One

Let us find two distinct elements in $\mathbb{R}$ with the same image.

$$\text{Set } 2x - 3 = 1 \implies 2x = 4 \implies x = 2.$$

$$\text{Set } 2x - 3 = -1 \implies 2x = 2 \implies x = 1.$$

Now evaluate $f(x)$ at these points:

$$f(2) = |2(2) - 3| = |4 - 3| = |1| = 1$$

$$f(1) = |2(1) - 3| = |2 - 3| = |-1| = 1$$

Since $f(2) = f(1) = 1$ but $2 \neq 1$, the function f is not one-one.

Part (ii): Checking for Onto

By definition of absolute value, $|2x - 3| \geq 0$ for all $x \in \mathbb{R}$.

Thus:

$$\text{Range}(f) = [0, \infty)$$

However, the codomain is $\mathbb{R}$ (which includes negative real numbers).

For example, for $y = -2 \in \mathbb{R}$ (codomain), there is no $x \in \mathbb{R}$ such that $f(x) = -2$.

Since $\text{Range}(f) = [0, \infty) \neq \mathbb{R}$ (codomain), the function f is not onto.

Final Answer

- (i) f is not one-one.
- (ii) f is not onto.

Question 10

If $A = \{1, 2, 3\}$ and $B = \{3, 5\}$, then write the number of functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{1, 2, 3\}$ and $B = \{3, 5\}$.

Concept / Formula Used: $|A| = m$ $|B| = n$ A B n^m

Solution:

Number of elements in set A :

$$m = |A| = 3$$

Number of elements in set B :

$$n = |B| = 2$$

Using the standard result for total number of functions $N = n^m$:

$$\begin{aligned} N &= 2^3 \\ &= 8 \end{aligned}$$

Final Answer

Number of functions from A to $B = 8$

Question 11

If $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$, then write the number of functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$.

Concept / Formula Used: $|A| = m$ $|B| = n$ A B n^m

Solution:

Number of elements in set A :

$$m = |A| = 3$$

Number of elements in set B :

$$n = |B| = 4$$

Using the formula $N = n^m$:

$$\begin{aligned} N &= 4^3 \\ &= 64 \end{aligned}$$

Final Answer

Number of functions from A to $B = 64$

Question 12

If $A = \{a, b\}$ and $B = \{1, 3, 5\}$, then write the number of one-one functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{a, b\}$ and $B = \{1, 3, 5\}$.

Concept / Formula Used: $|A| = m \quad |B| = n \quad A \rightarrow B \quad {}^n P_m = \frac{n!}{(n-m)!} \quad m \leq n \quad 0 \quad m > n$

Solution:

Number of elements in set A :

$$m = |A| = 2$$

Number of elements in set B :

$$n = |B| = 3$$

Since $m \leq n$, using the formula $N = {}^n P_m$:

$$\begin{aligned} N &= {}^3 P_2 \\ &= \frac{3!}{(3-2)!} \\ &= \frac{6}{1!} \\ &= 6 \end{aligned}$$

Final Answer

Number of one-one functions from A to $B = 6$

Question 13

If $A = \{a, b, c\}$ and $B = \{p, q\}$, then write the number of onto functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{a, b, c\}$ and $B = \{p, q\}$.

Concept / Formula Used: $|A| = m \geq 2 \quad |B| = 2 \quad A \rightarrow B \quad 2^m - 2$

Solution:

Number of elements in A :

$$m = |A| = 3$$

Number of elements in B :

$$n = |B| = 2$$

Using the standard formula for $n = 2$:

$$\begin{aligned}\text{Number of onto functions} &= 2^m - 2 \\ &= 2^3 - 2 \\ &= 8 - 2 \\ &= 6\end{aligned}$$

Final Answer

Number of onto functions from A to $B = 6$

Question 14

If $A = \{1, 2, 3, 4\}$ and $B = \{-1, 3\}$, then write the number of onto functions from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{1, 2, 3, 4\}$ and $B = \{-1, 3\}$.

Concept / Formula Used: $|A| = m \geq 2$ $|B| = 2$ A B $2^m - 2$

Solution:

Number of elements in A :

$$m = |A| = 4$$

Number of elements in B :

$$n = |B| = 2$$

Using the standard formula $N = 2^m - 2$:

$$\begin{aligned}N &= 2^4 - 2 \\ &= 16 - 2 \\ &= 14\end{aligned}$$

Final Answer

Number of onto functions from A to $B = 14$

Question 15

If $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, then write the number of

- (i) one-one onto functions from A to B
- (ii) bijections from A to B .

TYPE 23 Counting Functions Between Finite Sets

Given: $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$.

Concept / Formula Used: $|A| = |B| = n \quad A \quad B \quad n!$

Solution:

Number of elements in set A and set B :

$$m = |A| = 3$$

$$n = |B| = 3$$

Since $m = n = 3$, using the formula $N = n!$:

$$\begin{aligned} N &= 3! \\ &= 3 \times 2 \times 1 \\ &= 6 \end{aligned}$$

Note that "one-one onto functions" and "bijections" refer to the exact same set of functions.

Part (i): Number of one-one onto functions = 6.

Part (ii): Number of bijections = 6.

Final Answer

(i) Number of one-one onto functions = 6

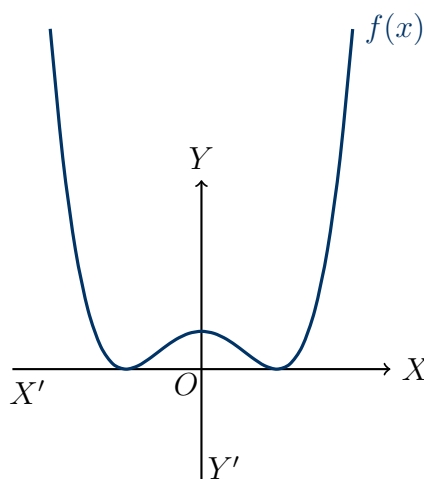
(ii) Number of bijections = 6

Question 16

The given function $f : \mathbb{R} \rightarrow \mathbb{R}$ is not 'onto' function. Give reason. *(ISC Specimen Paper 2025)*

TYPE 22 Testing Injectivity and Surjectivity of Functions

★ Likely Exam Question

Graph of function $f : \mathbb{R} \rightarrow \mathbb{R}$

Given: A graph of a function $f : \mathbb{R} \rightarrow \mathbb{R}$.

Concept / Formula Used: $f : \mathbb{R} \rightarrow \mathbb{R}$ $\mathbb{R}$ $y = c$ $c \in \mathbb{R}$

Solution:

From the given graph, we observe that:

1. As $x \rightarrow \infty$ and $x \rightarrow -\infty$, the curve extends upwards towards $+\infty$.
2. The graph has a global minimum value below which it does not exist (the curve does not extend downwards to $-\infty$).
3. Consequently, for negative values of y below the global minimum, there are no corresponding points (x, y) on the graph.

Therefore:

$$\text{Range}(f) \subset \mathbb{R} \quad \text{and} \quad \text{Range}(f) \neq \mathbb{R}$$

Since the range of the function is not equal to its codomain $\mathbb{R}$, f is not an onto function.

Hence Proved

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is not onto because its range is a proper subset of the codomain $\mathbb{R}$.

Question 17

Let $A = \{1, 2\}$, $B = \{3, 6\}$, function $f : A \rightarrow B$ be defined by $f(x) = 3x$ and function $g : A \rightarrow B$ be defined by $g(x) = x^2 + 2$. Show that $f = g$.

TYPE 24 Testing Equality of Functions

To Prove: $f = g$, where $f(x) = 3x$ and $g(x) = x^2 + 2$ on domain $A = \{1, 2\}$.

Concept / Formula Used: $f : A \rightarrow B$ $g : A \rightarrow B$ $f(x) = g(x)$ $x \in A$

Solution:

Domain of both functions is $A = \{1, 2\}$.

For $x = 1 \in A$:

$$\begin{aligned} f(1) &= 3(1) = 3 \\ g(1) &= (1)^2 + 2 = 1 + 2 = 3 \end{aligned}$$

Thus, $f(1) = g(1) = 3$.

For $x = 2 \in A$:

$$\begin{aligned} f(2) &= 3(2) = 6 \\ g(2) &= (2)^2 + 2 = 4 + 2 = 6 \end{aligned}$$

Thus, $f(2) = g(2) = 6$.

Since $f(x) = g(x)$ for all $x \in A$, the two functions are equal.

Hence Proved

$f = g$, as $f(x) = g(x)$ for all $x \in A$.

Question 18

Check the injectivity and surjectivity of the following functions:

- (i) $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2$
- (ii) $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^2$
- (iii) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$
- (iv) $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^3$
- (v) $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x^3$
- (vi) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Six functions specified on different domains and codomains ($\mathbb{N}, \mathbb{Z}, \mathbb{R}$).

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2$ f

Solution:

Part (i): $f : \mathbb{N} \rightarrow \mathbb{N}, f(x) = x^2$

Injectivity: Let $f(x_1) = f(x_2) \implies x_1^2 = x_2^2 \implies x_1 = x_2$ (since $x_1, x_2 \in \mathbb{N}$). Thus, f is injective.

Surjectivity: $\text{Range}(f) = \{1, 4, 9, 16, \dots\} \neq \mathbb{N}$. For example, $2 \in \mathbb{N}$ has no pre-image in $\mathbb{N}$ since $x = \sqrt{2} \notin \mathbb{N}$. Thus, f is not surjective.

Part (ii): $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^2$

Injectivity: $f(-1) = (-1)^2 = 1$ and $f(1) = 1^2 = 1$. Since $f(-1) = f(1)$ but $-1 \neq 1$, f is not injective.

Surjectivity: $\text{Range}(f) = \{0, 1, 4, 9, \dots\} \neq \mathbb{Z}$. Negative integers like $-2 \in \mathbb{Z}$ have no pre-image. Thus, f is not surjective.

Part (iii): $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$

Injectivity: $f(-2) = 4$ and $f(2) = 4$. Since $f(-2) = f(2)$ but $-2 \neq 2$, f is not injective.

Surjectivity: $\text{Range}(f) = [0, \infty) \neq \mathbb{R}$. Negative real numbers have no pre-image. Thus, f is not surjective.

Part (iv): $f : \mathbb{N} \rightarrow \mathbb{N}, f(x) = x^3$

Injectivity: $f(x_1) = f(x_2) \implies x_1^3 = x_2^3 \implies x_1 = x_2$. Thus, f is injective.

Surjectivity: $\text{Range}(f) = \{1, 8, 27, \dots\} \neq \mathbb{N}$. For example, $2 \in \mathbb{N}$ has no pre-image in $\mathbb{N}$. Thus, f is not surjective.

Part (v): $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^3$

Injectivity: $f(x_1) = f(x_2) \implies x_1^3 = x_2^3 \implies x_1 = x_2$ (over real/integer numbers). Thus, f is injective.

Surjectivity: $\text{Range}(f) = \{\dots, -8, -1, 0, 1, 8, \dots\} \neq \mathbb{Z}$. For example, $2 \in \mathbb{Z}$ has no pre-image in $\mathbb{Z}$ ($x = \sqrt[3]{2} \notin \mathbb{Z}$). Thus, f is not surjective.

Part (vi): $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$

Injectivity: $f(x_1) = f(x_2) \implies x_1^3 = x_2^3 \implies x_1 = x_2$. Thus, f is injective.

Surjectivity: For any $y \in \mathbb{R}$, taking $x = y^{1/3} \in \mathbb{R}$, we get $f(x) = (y^{1/3})^3 = y$. Thus, $\text{Range}(f) = \mathbb{R}$, so f is surjective.

Final Answer

- (i) Injective, not surjective.
- (ii) Neither injective nor surjective.
- (iii) Neither injective nor surjective.
- (iv) Injective, not surjective.
- (v) Injective, not surjective.
- (vi) Both injective and surjective (bijective).

Question 19

Let $\mathbb{I}$ be the set of all integers. Show that the function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = |x|$ is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = |x|$ is neither one-one nor onto.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad f$

Solution:

1. Injectivity (One-One) Test:

Consider $x_1 = 1 \in \mathbb{I}$ and $x_2 = -1 \in \mathbb{I}$.

Evaluating $f(x)$ at these points:

$$\begin{aligned} f(1) &= |1| = 1 \\ f(-1) &= |-1| = 1 \end{aligned}$$

Here $f(1) = f(-1) = 1$, but $1 \neq -1$.

Hence, f is not one-one.

2. Surjectivity (Onto) Test:

By definition, $|x| \geq 0$ for all $x \in \mathbb{I}$.

Thus:

$$\text{Range}(f) = \{0, 1, 2, 3, \dots\}$$

Consider $-3 \in \mathbb{I}$ (codomain).

There is no integer $x \in \mathbb{I}$ such that $f(x) = |x| = -3$.

Hence, $\text{Range}(f) \neq \mathbb{I}$, so f is not onto.

Hence Proved

The function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = |x|$ is neither one-one nor onto.

Question 20

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x^2 - 2$ is many-one into.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x^2 - 2$ is many-one into.

Concept / Formula Used: $\neq$

Solution:

1. Proving Many-One:

Consider $x = 1 \in \mathbb{R}$ and $x = -1 \in \mathbb{R}$:

$$\begin{aligned} f(1) &= 3(1)^2 - 2 = 3(1) - 2 = 1 \\ f(-1) &= 3(-1)^2 - 2 = 3(1) - 2 = 1 \end{aligned}$$

Since $f(1) = f(-1) = 1$ but $1 \neq -1$, f is not one-one, which means f is many-one.

2. Proving Into:

For any $x \in \mathbb{R}$, we know $x^2 \geq 0$.

Multiplying by 3:

$$3x^2 \geq 0$$

Subtracting 2:

$$\begin{aligned} 3x^2 - 2 &\geq -2 \\ f(x) &\geq -2 \end{aligned}$$

Thus, $\text{Range}(f) = [-2, \infty)$.

Since $\text{Range}(f) = [-2, \infty) \neq \mathbb{R}$ (codomain), negative numbers less than -2 (such as $-5 \in \mathbb{R}$) have no pre-image in $\mathbb{R}$.

Hence, f is an into function.

Combining both properties, f is many-one into.

Hence Proved

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x^2 - 2$ is many-one into.

Question 21

Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and $f = \{(1, 4), (2, 5), (3, 6)\}$ be a function from A to B . Show that f is one-one but not onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: The function $f = \{(1, 4), (2, 5), (3, 6)\}$ from A to B is one-one but not onto.

Concept / Formula Used: f A B f f B

Solution:

Given function mapping as ordered pairs:

$$f(1) = 4$$

$$f(2) = 5$$

$$f(3) = 6$$

1. Injectivity (One-One):

Distinct elements in domain $A = \{1, 2, 3\}$ have distinct images in codomain B :

$$f(1) \neq f(2) \neq f(3)$$

Therefore, f is one-one.

2. Surjectivity (Onto):

The range of f is:

$$\text{Range}(f) = \{4, 5, 6\}$$

The codomain is $B = \{4, 5, 6, 7\}$.

Element $7 \in B$ has no pre-image in A .

Since $\text{Range}(f) = \{4, 5, 6\} \neq B$, f is not onto.

Hence Proved

The function f is one-one but not onto.

Question 22

In each of the following cases, state whether the function is bijective or not. Justify your answer.

- (i) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 1$
- (ii) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3 - 4x$
- (iii) $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1 + x^2$

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Three functions defined on $\mathbb{R} \rightarrow \mathbb{R}$.

Solution:

Part (i): $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 1$

Injectivity: Let $f(x_1) = f(x_2) \implies 2x_1 + 1 = 2x_2 + 1 \implies 2x_1 = 2x_2 \implies x_1 = x_2$. So f is one-one.

Surjectivity: Let $y \in \mathbb{R}$. Set $2x + 1 = y \implies x = \frac{y-1}{2} \in \mathbb{R}$. Then $f(x) = 2\left(\frac{y-1}{2}\right) + 1 = y$. So f is onto.

Since f is both one-one and onto, f is bijective.

Part (ii): $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3 - 4x$

Injectivity: Let $f(x_1) = f(x_2) \implies 3 - 4x_1 = 3 - 4x_2 \implies -4x_1 = -4x_2 \implies x_1 = x_2$. So f is one-one.

Surjectivity: Let $y \in \mathbb{R}$. Set $3 - 4x = y \implies 4x = 3 - y \implies x = \frac{3-y}{4} \in \mathbb{R}$. Then $f(x) = 3 - 4\left(\frac{3-y}{4}\right) = y$. So f is onto.

Since f is both one-one and onto, f is bijective.

Part (iii): $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 1 + x^2$

Injectivity: $f(1) = 1 + 1^2 = 2$ and $f(-1) = 1 + (-1)^2 = 2$. Since $f(1) = f(-1) = 2$ but $1 \neq -1$, f is not one-one.

Since it is not one-one, f is not bijective.

Final Answer

- (i) Bijective.
- (ii) Bijective.
- (iii) Not bijective.

Question 23

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^4$. Show that f is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^4$ is neither one-one nor onto.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad f \quad \mathbb{R}$

Solution:**1. Injectivity (One-One) Test:**

Consider $x_1 = 1 \in \mathbb{R}$ and $x_2 = -1 \in \mathbb{R}$.

$$\begin{aligned} f(1) &= (1)^4 = 1 \\ f(-1) &= (-1)^4 = 1 \end{aligned}$$

Since $f(1) = f(-1) = 1$ but $1 \neq -1$, f is not one-one.

2. Surjectivity (Onto) Test:

For any real number x , $x^4 \geq 0$.

$$\text{Range}(f) = [0, \infty)$$

Consider $-2 \in \mathbb{R}$ (codomain).

There exists no real number $x \in \mathbb{R}$ such that $x^4 = -2$.

Since $\text{Range}(f) = [0, \infty) \neq \mathbb{R}$, f is not onto.

Hence, f is neither one-one nor onto.

Hence Proved

The function $f(x) = x^4$ is neither one-one nor onto.

Question 24

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$ is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$ is neither one-one nor onto.

Concept / Formula Used: $\mathbb{R} \quad -1 \leq \cos x \leq 1$

Solution:**1. Injectivity (One-One) Test:**

Consider $x_1 = 0 \in \mathbb{R}$ and $x_2 = 2\pi \in \mathbb{R}$:

$$\begin{aligned} f(0) &= \cos(0) = 1 \\ f(2\pi) &= \cos(2\pi) = 1 \end{aligned}$$

Since $f(0) = f(2\pi) = 1$ but $0 \neq 2\pi$, f is not one-one.

2. Surjectivity (Onto) Test:

Using the standard result $-1 \leq \cos x \leq 1$ for all $x \in \mathbb{R}$:

$$\text{Range}(f) = [-1, 1]$$

Consider $y = 2 \in \mathbb{R}$ (codomain).

There exists no $x \in \mathbb{R}$ such that $\cos x = 2$.

Since $\text{Range}(f) = [-1, 1] \neq \mathbb{R}$, f is not onto.

Hence, f is neither one-one nor onto.

Hence Proved

The function $f(x) = \cos x$ is neither one-one nor onto.

Question 25

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin^2 x + \cos^2 x$ is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin^2 x + \cos^2 x$ is neither one-one nor onto.

Concept / Formula Used: $\sin^2 x + \cos^2 x = 1 \quad x \in \mathbb{R}$

Solution:

Using the standard identity $\sin^2 x + \cos^2 x = 1$:

$$f(x) = 1 \quad \text{for all } x \in \mathbb{R}$$

Thus, $f(x)$ is a constant function.

1. Injectivity (One-One) Test:

Take $x_1 = 0$ and $x_2 = \frac{\pi}{2}$:

$$\begin{aligned} f(0) &= 1 \\ f\left(\frac{\pi}{2}\right) &= 1 \end{aligned}$$

Since $f(0) = f\left(\frac{\pi}{2}\right) = 1$ but $0 \neq \frac{\pi}{2}$, f is not one-one.

2. Surjectivity (Onto) Test:

The range of f contains only a single element:

$$\text{Range}(f) = \{1\}$$

Since $\text{Range}(f) = \{1\} \neq \mathbb{R}$ (codomain), any value $y \neq 1$ in $\mathbb{R}$ has no pre-image.

Hence, f is not onto.

Thus, f is neither one-one nor onto.

Hence Proved

The constant function $f(x) = \sin^2 x + \cos^2 x = 1$ is neither one-one nor onto.

Question 26

Show that the function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = x^2 + x$ is neither one-one nor onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: Function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = x^2 + x$ is neither one-one nor onto.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad f \quad \mathbb{I}$

Solution:

1. Injectivity (One-One) Test:

Consider $x_1 = 0 \in \mathbb{I}$ and $x_2 = -1 \in \mathbb{I}$:

$$\begin{aligned} f(0) &= (0)^2 + 0 = 0 \\ f(-1) &= (-1)^2 + (-1) = 1 - 1 = 0 \end{aligned}$$

Since $f(0) = f(-1) = 0$ but $0 \neq -1$, f is not one-one.

2. Surjectivity (Onto) Test:

Rewrite $f(x)$:

$$f(x) = x(x + 1)$$

For any integer x , one of x or $x + 1$ must be even.

Thus, $x(x + 1)$ is always an even integer.

Therefore, $f(x)$ takes only even integer values, so $\text{Range}(f) \subseteq \{\text{even integers}\}$.

Odd integers such as $1 \in \mathbb{I}$ have no pre-image in $\mathbb{I}$.

For example, setting $x^2 + x = 1$:

$$\begin{aligned} x^2 + x - 1 &= 0 \\ x &= \frac{-1 \pm \sqrt{1 - 4(1)(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2} \notin \mathbb{I} \end{aligned}$$

Since $\text{Range}(f) \neq \mathbb{I}$, f is not onto.

Hence, f is neither one-one nor onto.

Hence Proved

The function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined by $f(x) = x^2 + x$ is neither one-one nor onto.

Question 27

If the function $f : \mathbb{I} \rightarrow \mathbb{I}$ is defined by

$$f(x) = \begin{cases} 2x + 1, & \text{if } x \text{ is odd} \\ 2x - 1, & \text{if } x \text{ is even} \end{cases}$$

(i) Find $f(3)$.

- (ii) Find $f(4)$.
(iii) Find x , if $f(x) = -9$.
Is this function one-one?

TYPE 25 Testing Injectivity and Surjectivity of Piecewise Functions

Given: Function $f : \mathbb{I} \rightarrow \mathbb{I}$ defined piecewise based on parity of x .

Concept / Formula Used: $f(x) \quad x$

Solution:

Part (i): Finding $f(3)$

Since $x = 3$ is odd, we use the branch $f(x) = 2x + 1$:

$$\begin{aligned} f(3) &= 2(3) + 1 \\ &= 6 + 1 \\ &= 7 \end{aligned}$$

Part (ii): Finding $f(4)$

Since $x = 4$ is even, we use the branch $f(x) = 2x - 1$:

$$\begin{aligned} f(4) &= 2(4) - 1 \\ &= 8 - 1 \\ &= 7 \end{aligned}$$

Part (iii): Finding x when $f(x) = -9$

Case 1: If x is odd:

$$\begin{aligned} 2x + 1 &= -9 \\ 2x &= -10 \\ x &= -5 \quad (\text{which is odd, valid solution}) \end{aligned}$$

Case 2: If x is even:

$$\begin{aligned} 2x - 1 &= -9 \\ 2x &= -8 \\ x &= -4 \quad (\text{which is even, valid solution}) \end{aligned}$$

Thus, $x = -5$ or $x = -4$.

Checking if f is one-one:

From parts (i) and (ii), we have:

$$\begin{aligned} f(3) &= 7 \\ f(4) &= 7 \end{aligned}$$

Since $f(3) = f(4) = 7$ but $3 \neq 4$, the function f is not one-one.

Final Answer

(i) $f(3) = 7$

(ii) $f(4) = 7$

(iii) $x = -5$ or $x = -4$

Is f one-one? No, f is not one-one.**Question 28**(i) Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by

$$f(n) = \begin{cases} n+1, & \text{if } n \text{ is odd} \\ n-1, & \text{if } n \text{ is even} \end{cases}$$

is both one-one and onto.

(ii) Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n - (-1)^n$ for all $n \in \mathbb{N}$ is bijective.**TYPE 25 Testing Injectivity and Surjectivity of Piecewise Functions****★ Likely Exam Question****To Prove:** Both given representations define a bijective function $f : \mathbb{N} \rightarrow \mathbb{N}$.**Solution:****Part (i):****1. Injectivity (One-One):**Let $n_1, n_2 \in \mathbb{N}$ such that $f(n_1) = f(n_2)$.

- Case A: Both n_1, n_2 are odd. Then $n_1 + 1 = n_2 + 1 \implies n_1 = n_2$.
- Case B: Both n_1, n_2 are even. Then $n_1 - 1 = n_2 - 1 \implies n_1 = n_2$.
- Case C: n_1 is odd and n_2 is even. Then $f(n_1) = n_1 + 1$ (which is even) and $f(n_2) = n_2 - 1$ (which is odd). An even number cannot equal an odd number, so $f(n_1) = f(n_2)$ is impossible in this case.

Thus, $f(n_1) = f(n_2) \implies n_1 = n_2$ in all valid cases. So f is one-one.**2. Surjectivity (Onto):**Let $m \in \mathbb{N}$ (codomain).

- If m is odd, then $m + 1$ is even. Then $f(m + 1) = (m + 1) - 1 = m$.
- If m is even, then $m - 1$ is odd (and since $m \geq 2$, $m - 1 \in \mathbb{N}$). Then $f(m - 1) = (m - 1) + 1 = m$.

Thus, every $m \in \mathbb{N}$ has a pre-image in $\mathbb{N}$. So f is onto.Hence, f is both one-one and onto.**Part (ii):**

For $n \in \mathbb{N}$:

$$\text{If } n \text{ is odd, } (-1)^n = -1 \implies f(n) = n - (-1) = n + 1$$

$$\text{If } n \text{ is even, } (-1)^n = 1 \implies f(n) = n - 1$$

This is identical to the function defined in Part (i).

Since the function in Part (i) was shown to be both one-one and onto, $f(n) = n - (-1)^n$ is also bijective.

Hence Proved

Both functions are proved to be bijective on $\mathbb{N}$.

Question 29

Find the number of all one-one functions from the set $\{1, 2, 3, 4\}$ to itself.

TYPE 23 Counting Functions Between Finite Sets

To Find: The number of one-one functions from $A = \{1, 2, 3, 4\}$ to A .

Concept / Formula Used: $A \quad n \quad n!$

Solution:

Let $A = \{1, 2, 3, 4\}$.

The number of elements in set A is:

$$n = |A| = 4$$

A one-one function from a finite set to itself is a permutation of the set elements.

Using the formula $N = n!$:

$$\begin{aligned} N &= 4! \\ &= 4 \times 3 \times 2 \times 1 \\ &= 24 \end{aligned}$$

Final Answer

Number of one-one functions = 24

Question 30

Show that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{5x-3}{4}$ is both one-one and onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

To Prove: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{5x-3}{4}$ is both one-one and onto.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad y \in \mathbb{R} \quad \exists x \in \mathbb{R} \quad f(x) = y$

Solution:

1. Injectivity (One-One) Proof:

Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$.

Substitute into the function definition:

$$\frac{5x_1 - 3}{4} = \frac{5x_2 - 3}{4}$$

Multiply both sides by 4:

$$5x_1 - 3 = 5x_2 - 3$$

Add 3 to both sides:

$$5x_1 = 5x_2$$

Divide both sides by 5:

$$x_1 = x_2$$

Therefore, f is one-one.

2. Surjectivity (Onto) Proof:

Let $y \in \mathbb{R}$ be an arbitrary element in the codomain.

Set $f(x) = y$:

$$\begin{aligned} \frac{5x - 3}{4} &= y \\ 5x - 3 &= 4y \\ 5x &= 4y + 3 \\ x &= \frac{4y + 3}{5} \end{aligned}$$

Since $y \in \mathbb{R}$, we have $x = \frac{4y+3}{5} \in \mathbb{R}$ (domain).

Now evaluate $f(x)$:

$$\begin{aligned} f(x) &= f\left(\frac{4y + 3}{5}\right) \\ &= \frac{5\left(\frac{4y+3}{5}\right) - 3}{4} \\ &= \frac{(4y + 3) - 3}{4} \\ &= \frac{4y}{4} \\ &= y \end{aligned}$$

Thus, for every $y \in \mathbb{R}$, there exists an $x \in \mathbb{R}$ such that $f(x) = y$.

Therefore, f is onto.

Hence, f is both one-one and onto.

Hence Proved

The function $f(x) = \frac{5x-3}{4}$ is both one-one and onto.

Question 31

Let $f: \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \frac{4x}{3x+4}$. Show that f is a one-one function. Also check whether f is an onto function or not.

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: A function $f: \mathbb{R} - \{-\frac{4}{3}\} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{4x}{3x+4}$.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad x_1, x_2 \in \text{Domain}$
 $y \in \text{Codomain} \quad x \in \text{Domain} \quad f(x) = y$

Solution:

Step 1: Proof for One-One (Injectivity)

Let $x_1, x_2 \in \mathbb{R} - \{-\frac{4}{3}\}$ such that:

$$f(x_1) = f(x_2)$$

Substituting the function definition:

$$\frac{4x_1}{3x_1+4} = \frac{4x_2}{3x_2+4}$$

Dividing both sides by 4:

$$\frac{x_1}{3x_1+4} = \frac{x_2}{3x_2+4}$$

Cross-multiplying:

$$\begin{aligned} x_1(3x_2+4) &= x_2(3x_1+4) \\ 3x_1x_2+4x_1 &= 3x_1x_2+4x_2 \end{aligned}$$

Subtracting $3x_1x_2$ from both sides:

$$\begin{aligned} 4x_1 &= 4x_2 \\ x_1 &= x_2 \end{aligned}$$

Since $f(x_1) = f(x_2) \implies x_1 = x_2$, the function f is a **one-one (injective)** function.

Step 2: Check for Onto (Surjectivity)

Let $y \in \mathbb{R}$ (codomain). Set $f(x) = y$:

$$\begin{aligned}y &= \frac{4x}{3x+4} \\y(3x+4) &= 4x \\3xy+4y &= 4x \\4x-3xy &= 4y \\x(4-3y) &= 4y \\x &= \frac{4y}{4-3y}\end{aligned}$$

For x to be a well-defined real number in the domain $\mathbb{R} - \{-\frac{4}{3}\}$, the denominator cannot be zero:

$$\begin{aligned}4-3y &\neq 0 \\3y &\neq 4 \\y &\neq \frac{4}{3}\end{aligned}$$

If $y = \frac{4}{3} \in \mathbb{R}$ (codomain), let us check if there exists any pre-image x :

$$\begin{aligned}\frac{4x}{3x+4} &= \frac{4}{3} \\12x &= 4(3x+4) \\12x &= 12x+16 \\0 &= 16 \quad (\text{which is impossible})\end{aligned}$$

Thus, the element $y = \frac{4}{3}$ in the codomain has no pre-image in the domain $\mathbb{R} - \{-\frac{4}{3}\}$.

Therefore, f is **not** an onto function.

Final Answer

$f(x)$ is a **one-one** function, but it is **not** an onto function because $y = \frac{4}{3} \in \mathbb{R}$ has no pre-image in the domain.

Question 32

If $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$, then show that the function $f: A \rightarrow B$ defined by $f(x) = 2x$ is both one-one and onto.

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Sets $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$, and function $f: A \rightarrow B$ defined by $f(x) = 2x$.

To Prove: The function f is both one-one and onto.

Concept / Formula Used: $f: A \rightarrow B$ $A \subseteq B$ $f: A \rightarrow B$
Range of f = Codomain B

Solution:

Evaluating the image $f(x) = 2x$ for every element $x \in A$:

$$\text{For } x = 1, \quad f(1) = 2(1) = 2$$

$$\text{For } x = 2, \quad f(2) = 2(2) = 4$$

$$\text{For } x = 3, \quad f(3) = 2(3) = 6$$

$$\text{For } x = 4, \quad f(4) = 2(4) = 8$$

1. Proof of One-One (Injectivity): The images of the elements 1, 2, 3, 4 under f are 2, 4, 6, 8 respectively. Different elements in the domain A have different images in the codomain B :

$$1 \neq 2 \neq 3 \neq 4 \implies f(1) \neq f(2) \neq f(3) \neq f(4)$$

Alternatively, for any $x_1, x_2 \in A$:

$$f(x_1) = f(x_2)$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

Hence, f is a one-one function.

2. Proof of Onto (Surjectivity): The range of f is the set of all image values:

$$\text{Range}(f) = \{f(1), f(2), f(3), f(4)\} = \{2, 4, 6, 8\}$$

The given codomain is:

$$\text{Codomain } B = \{2, 4, 6, 8\}$$

Since $\text{Range}(f) = B$, every element in B has a pre-image in A . Hence, f is an onto function.

Hence Proved

Since f is both one-one and onto, $f: A \rightarrow B$ is a bijective function. Hence proved.

Question 33

(a) If a real-valued function is given by $f(x) = \sqrt{25 - x^2}$ is an onto function, then find the codomain for $f(x)$.

(b) If the domain is given to be $[-5, 5]$, is $f(x)$ a one-one function?

(c) Find all possible values of 'a' for which $f(a) = 4$.

TYPE 22 Testing Injectivity and Surjectivity of Functions

Given: Real-valued function $f(x) = \sqrt{25 - x^2}$ defined on domain $[-5, 5]$.

Concept / Formula Used: Codomain = Range $f(x)$ $x_1 \neq x_2$ $f(x_1) = f(x_2)$

Solution:

Part (a): Finding the codomain for $f(x)$ to be onto

Given domain is $[-5, 5]$, which means:

$$-5 \leq x \leq 5$$

Squaring all terms in the domain range:

$$\begin{aligned} 0 &\leq x^2 \leq 25 \\ -25 &\leq -x^2 \leq 0 \\ 25 - 25 &\leq 25 - x^2 \leq 25 - 0 \\ 0 &\leq 25 - x^2 \leq 25 \end{aligned}$$

Taking the non-negative principal square root:

$$\begin{aligned} \sqrt{0} &\leq \sqrt{25 - x^2} \leq \sqrt{25} \\ 0 &\leq f(x) \leq 5 \end{aligned}$$

Thus, the range of $f(x)$ is $[0, 5]$. For $f(x)$ to be an onto function, the codomain must equal its range:

$$\text{Codomain} = \text{Range} = [0, 5]$$

Part (b): Checking if $f(x)$ is a one-one function on $[-5, 5]$

Consider two distinct points $x_1 = 3$ and $x_2 = -3$ in the domain $[-5, 5]$. Evaluating $f(x)$ at these points:

$$\begin{aligned} f(3) &= \sqrt{25 - (3)^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \\ f(-3) &= \sqrt{25 - (-3)^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \end{aligned}$$

Here, $3 \neq -3$, but $f(3) = f(-3) = 4$. Since distinct elements in the domain have the same image, $f(x)$ is **not** a one-one function.

Part (c): Finding all possible values of 'a' for which $f(a) = 4$

Given equation:

$$f(a) = 4$$

Substituting the expression for $f(a)$:

$$\sqrt{25 - a^2} = 4$$

Squaring both sides:

$$\begin{aligned} 25 - a^2 &= 4^2 \\ 25 - a^2 &= 16 \\ a^2 &= 25 - 16 \\ a^2 &= 9 \\ a &= \pm\sqrt{9} \\ a &= \pm 3 \end{aligned}$$

Both values $a = 3$ and $a = -3$ belong to the domain $[-5, 5]$. Thus, the possible values of a are $\{-3, 3\}$.

Final Answer

- (a) Codomain $= [0, 5]$
- (b) No, $f(x)$ is not a one-one function.
- (c) $a \in \{-3, 3\}$

Common Mistake: In part (c), taking only the positive square root $a = 3$ and forgetting $a = -3$ is a common error; both values lie within the domain $[-5, 5]$.

1.4 – Composition of Functions

Question 1

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = (3 - x^3)^{1/3}$, then find $(f \circ f)(x)$.

TYPE 26 Finding Composition of Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = (3 - x^3)^{1/3}$.

Concept / Formula Used: $f \quad (f \circ f)(x) = f(f(x))$

Solution:

By the definition of composite functions:

$$(f \circ f)(x) = f(f(x))$$

Substitute the expression for $f(x)$:

$$(f \circ f)(x) = f((3 - x^3)^{1/3})$$

Apply the rule $f(y) = (3 - y^3)^{1/3}$ with $y = (3 - x^3)^{1/3}$:

$$\begin{aligned}(f \circ f)(x) &= \left[3 - ((3 - x^3)^{1/3})^3 \right]^{1/3} \\ &= [3 - (3 - x^3)]^{1/3} \\ &= [3 - 3 + x^3]^{1/3} \\ &= [x^3]^{1/3} \\ &= x\end{aligned}$$

Final Answer

$(f \circ f)(x) = x$ for all $x \in \mathbb{R}$.

Question 2

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = (5 - x^5)^{1/5}$, then find $f \circ f$.

TYPE 26 Finding Composition of Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = (5 - x^5)^{1/5}$.

Concept / Formula Used: $f \quad (f \circ f)(x) = f(f(x))$

Solution:

By definition of composite functions:

$$(f \circ f)(x) = f(f(x))$$

Substitute $f(x) = (5 - x^5)^{1/5}$ into the function:

$$(f \circ f)(x) = f((5 - x^5)^{1/5})$$

Applying the definition $f(y) = (5 - y^5)^{1/5}$ with $y = (5 - x^5)^{1/5}$:

$$\begin{aligned}(f \circ f)(x) &= \left[5 - ((5 - x^5)^{1/5})^5 \right]^{1/5} \\ &= [5 - (5 - x^5)]^{1/5} \\ &= [5 - 5 + x^5]^{1/5} \\ &= [x^5]^{1/5} \\ &= x\end{aligned}$$

Final Answer

$(f \circ f)(x) = x$ for all $x \in \mathbb{R}$.

Question 3

If $f(x) = |x|$ and $g(x) = [x]$ for all $x \in \mathbb{R}$, find

(i) $(g \circ f)\left(-\frac{5}{3}\right)$

(ii) $(f \circ g)\left(-\frac{5}{3}\right)$

TYPE 27 Evaluating Composite Functions at a Point

Given: $f(x) = |x|$ and $g(x) = [x]$ (Greatest Integer Function) for all $x \in \mathbb{R}$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x))$ $(f \circ g)(x) = f(g(x))$

Solution:

(i) Evaluation of $(g \circ f)\left(-\frac{5}{3}\right)$: By definition of composite functions:

$$(g \circ f)\left(-\frac{5}{3}\right) = g\left(f\left(-\frac{5}{3}\right)\right)$$

First, evaluate $f\left(-\frac{5}{3}\right)$:

$$f\left(-\frac{5}{3}\right) = \left|-\frac{5}{3}\right| = \frac{5}{3}$$

Now evaluate $g\left(\frac{5}{3}\right)$:

$$g\left(\frac{5}{3}\right) = \left[\frac{5}{3}\right] = [1.666\dots] = 1$$

Therefore, $(g \circ f)\left(-\frac{5}{3}\right) = 1$.

(ii) Evaluation of $(f \circ g)\left(-\frac{5}{3}\right)$: By definition of composite functions:

$$(f \circ g)\left(-\frac{5}{3}\right) = f\left(g\left(-\frac{5}{3}\right)\right)$$

First, evaluate $g\left(-\frac{5}{3}\right)$:

$$g\left(-\frac{5}{3}\right) = \left[-\frac{5}{3}\right] = [-1.666\dots] = -2$$

Now evaluate $f(-2)$:

$$f(-2) = |-2| = 2$$

Therefore, $(f \circ g)\left(-\frac{5}{3}\right) = 2$.

Final Answer

$$(i) (g \circ f) \left(-\frac{5}{3} \right) = 1$$

$$(ii) (f \circ g) \left(-\frac{5}{3} \right) = 2$$

Question 4

If $f(x) = x + 7$ and $g(x) = x - 7$, find $(f \circ g)(7)$.

TYPE 27 Evaluating Composite Functions at a Point

Given: $f(x) = x + 7$ and $g(x) = x - 7$.

Concept / Formula Used: $(f \circ g)(x) = f(g(x))$

Solution:

By definition of function composition:

$$(f \circ g)(7) = f(g(7))$$

First, calculate $g(7)$:

$$\begin{aligned} g(7) &= 7 - 7 \\ &= 0 \end{aligned}$$

Now substitute $g(7) = 0$ into f :

$$\begin{aligned} f(g(7)) &= f(0) \\ &= 0 + 7 \\ &= 7 \end{aligned}$$

Final Answer

$$(f \circ g)(7) = 7$$

Question 5

If $f : \{2, 3, 4, 5\} \rightarrow \{3, 4, 5, 9\}$ and $g : \{3, 4, 5, 9\} \rightarrow \{7, 11, 15\}$ are functions defined as $f(2) = 3$, $f(3) = 4$, $f(4) = f(5) = 5$ and $g(3) = g(4) = 7$, $g(5) = g(9) = 11$, find $g \circ f$.

TYPE 28 Composition of Functions on Finite Sets

Given: $f : \{2, 3, 4, 5\} \rightarrow \{3, 4, 5, 9\}$ with $f(2) = 3$, $f(3) = 4$, $f(4) = 5$, $f(5) = 5$.
 $g : \{3, 4, 5, 9\} \rightarrow \{7, 11, 15\}$ with $g(3) = 7$, $g(4) = 7$, $g(5) = 11$, $g(9) = 11$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x)) \quad x \in \text{Domain}(f) = \{2, 3, 4, 5\}$

Solution:

We evaluate $(g \circ f)(x)$ for each element in the domain $\{2, 3, 4, 5\}$:

$$(g \circ f)(2) = g(f(2)) = g(3) = 7$$

$$(g \circ f)(3) = g(f(3)) = g(4) = 7$$

$$(g \circ f)(4) = g(f(4)) = g(5) = 11$$

$$(g \circ f)(5) = g(f(5)) = g(5) = 11$$

Expressing $(g \circ f)$ as a set of ordered pairs:

$$(g \circ f) = \{(2, 7), (3, 7), (4, 11), (5, 11)\}$$

Final Answer

$$g \circ f = \{(2, 7), (3, 7), (4, 11), (5, 11)\}$$

Question 6

Let $A = \{3, 4, 5, 6\}$, $B = \{13, 14, 15, 16\}$ and $C = \{23, 24, 25\}$. If $f : A \rightarrow B$ and $g : B \rightarrow C$ are defined by

$$f = \{(3, 13), (4, 14), (5, 15), (6, 16)\}$$

and

$$g = \{(13, 23), (14, 23), (15, 24), (16, 25)\}$$

then find $g \circ f : A \rightarrow C$.

TYPE 28 Composition of Functions on Finite Sets

Given: $A = \{3, 4, 5, 6\}$, $B = \{13, 14, 15, 16\}$, $C = \{23, 24, 25\}$, $f : A \rightarrow B$, and $g : B \rightarrow C$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x)) \quad x \in A$

Solution:

We evaluate $(g \circ f)(x)$ for each element in the domain set $A = \{3, 4, 5, 6\}$:

$$(g \circ f)(3) = g(f(3)) = g(13) = 23$$

$$(g \circ f)(4) = g(f(4)) = g(14) = 23$$

$$(g \circ f)(5) = g(f(5)) = g(15) = 24$$

$$(g \circ f)(6) = g(f(6)) = g(16) = 25$$

Writing $g \circ f$ as a set of ordered pairs:

$$g \circ f = \{(3, 23), (4, 23), (5, 24), (6, 25)\}$$

Final Answer

$$g \circ f = \{(3, 23), (4, 23), (5, 24), (6, 25)\}$$

Question 7

Let $A = \{1, 2, 3, 4\}$. If the functions $f : A \rightarrow A$ and $g : A \rightarrow A$ are defined by

$$f = \{(1, 4), (2, 1), (3, 3), (4, 2)\}$$

and

$$g = \{(1, 3), (2, 1), (3, 2), (4, 4)\}$$

then find

- (i) $g \circ f$
- (ii) $f \circ g$
- (iii) $f \circ f$

TYPE 28 Composition of Functions on Finite Sets

Given: $A = \{1, 2, 3, 4\}$, $f = \{(1, 4), (2, 1), (3, 3), (4, 2)\}$, $g = \{(1, 3), (2, 1), (3, 2), (4, 4)\}$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x))$ $(f \circ g)(x) = f(g(x))$ $(f \circ f)(x) = f(f(x))$

Solution:

(i) Finding $g \circ f$: Evaluating $(g \circ f)(x) = g(f(x))$ for all $x \in A$:

$$\begin{aligned}(g \circ f)(1) &= g(f(1)) = g(4) = 4 \\(g \circ f)(2) &= g(f(2)) = g(1) = 3 \\(g \circ f)(3) &= g(f(3)) = g(3) = 2 \\(g \circ f)(4) &= g(f(4)) = g(2) = 1\end{aligned}$$

Thus, $g \circ f = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$.

(ii) Finding $f \circ g$: Evaluating $(f \circ g)(x) = f(g(x))$ for all $x \in A$:

$$\begin{aligned}(f \circ g)(1) &= f(g(1)) = f(3) = 3 \\(f \circ g)(2) &= f(g(2)) = f(1) = 4 \\(f \circ g)(3) &= f(g(3)) = f(2) = 1 \\(f \circ g)(4) &= f(g(4)) = f(4) = 2\end{aligned}$$

Thus, $f \circ g = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$.

(iii) Finding $f \circ f$: Evaluating $(f \circ f)(x) = f(f(x))$ for all $x \in A$:

$$\begin{aligned}(f \circ f)(1) &= f(f(1)) = f(4) = 2 \\(f \circ f)(2) &= f(f(2)) = f(1) = 4 \\(f \circ f)(3) &= f(f(3)) = f(3) = 3 \\(f \circ f)(4) &= f(f(4)) = f(2) = 1\end{aligned}$$

Thus, $f \circ f = \{(1, 2), (2, 4), (3, 3), (4, 1)\}$.

Final Answer

- (i) $g \circ f = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$
 (ii) $f \circ g = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$
 (iii) $f \circ f = \{(1, 2), (2, 4), (3, 3), (4, 1)\}$

Question 8

If the functions f and g are given by $f = \{(1, 2), (3, 5), (4, 1)\}$ and $g = \{(2, 3), (5, 1), (1, 3)\}$,

- (i) find the domain of f and g
 (ii) find the range of f and g
 (iii) write $f \circ g$ and $g \circ f$ as sets of ordered pairs

TYPE 28 Composition of Functions on Finite Sets

Given: $f = \{(1, 2), (3, 5), (4, 1)\}$ and $g = \{(2, 3), (5, 1), (1, 3)\}$.

Concept / Formula Used: $R = \{(x, y)\}$ Domain = $\{x : (x, y) \in R\}$
 Range = $\{y : (x, y) \in R\}$

Solution:

(i) **Domain of f and g :** The domain consists of the first elements of each ordered pair:

$$\text{Domain of } f = \{1, 3, 4\}$$

$$\text{Domain of } g = \{2, 5, 1\} = \{1, 2, 5\}$$

(ii) **Range of f and g :** The range consists of the second elements of each ordered pair:

$$\text{Range of } f = \{2, 5, 1\} = \{1, 2, 5\}$$

$$\text{Range of } g = \{3, 1, 3\} = \{1, 3\}$$

(iii) **Writing $f \circ g$ and $g \circ f$ as sets of ordered pairs:** For $f \circ g$, the domain is $\text{Domain}(g) = \{1, 2, 5\}$:

$$(f \circ g)(1) = f(g(1)) = f(3) = 5$$

$$(f \circ g)(2) = f(g(2)) = f(5) = 1$$

$$(f \circ g)(5) = f(g(5)) = f(1) = 2$$

Hence, $f \circ g = \{(1, 5), (2, 1), (5, 2)\}$.

For $g \circ f$, the domain is $\text{Domain}(f) = \{1, 3, 4\}$:

$$(g \circ f)(1) = g(f(1)) = g(2) = 3$$

$$(g \circ f)(3) = g(f(3)) = g(5) = 1$$

$$(g \circ f)(4) = g(f(4)) = g(1) = 3$$

Hence, $g \circ f = \{(1, 3), (3, 1), (4, 3)\}$.

Final Answer

- (i) $\text{Domain}(f) = \{1, 3, 4\}$, $\text{Domain}(g) = \{1, 2, 5\}$
(ii) $\text{Range}(f) = \{1, 2, 5\}$, $\text{Range}(g) = \{1, 3\}$
(iii) $f \circ g = \{(1, 5), (2, 5), (5, 2)\}$, $g \circ f = \{(1, 3), (3, 1), (4, 3)\}$

Question 9

Let $\mathbb{R}$ be the set of all real numbers. Find $g \circ f$ and $f \circ g$ when $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by

- (i) $f(x) = 2x - 1$ and $g(x) = x^2 + 3$ for all $x \in \mathbb{R}$
(ii) $f(x) = x + 1$ and $g(x) = |x|$ for all $x \in \mathbb{R}$
(iii) $f(x) = x^3$ and $g(x) = 2x^2 + 1$ for all $x \in \mathbb{R}$
(iv) $f(x) = \sin x$ and $g(x) = 5x^2$ for all $x \in \mathbb{R}$

TYPE 26 Finding Composition of Functions**★ Likely Exam Question**

Given: Functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x))$ $(f \circ g)(x) = f(g(x))$

Solution:

(i) $f(x) = 2x - 1$ and $g(x) = x^2 + 3$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(2x - 1) \\ &= (2x - 1)^2 + 3 \\ &= 4x^2 - 4x + 1 + 3 \\ &= 4x^2 - 4x + 4\end{aligned}$$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(x^2 + 3) \\ &= 2(x^2 + 3) - 1 \\ &= 2x^2 + 6 - 1 \\ &= 2x^2 + 5\end{aligned}$$

(ii) $f(x) = x + 1$ and $g(x) = |x|$:

$$(g \circ f)(x) = g(f(x)) = g(x + 1) = |x + 1|$$

$$(f \circ g)(x) = f(g(x)) = f(|x|) = |x| + 1$$

(iii) $f(x) = x^3$ and $g(x) = 2x^2 + 1$ (ISC 2019):

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(x^3) \\ &= 2(x^3)^2 + 1 \\ &= 2x^6 + 1\end{aligned}$$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(2x^2 + 1) \\ &= (2x^2 + 1)^3\end{aligned}$$

(iv) $f(x) = \sin x$ and $g(x) = 5x^2$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(\sin x) \\ &= 5(\sin x)^2 \\ &= 5 \sin^2 x\end{aligned}$$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(5x^2) \\ &= \sin(5x^2)\end{aligned}$$

Final Answer

(i) $(g \circ f)(x) = 4x^2 - 4x + 4$, $(f \circ g)(x) = 2x^2 + 5$

(ii) $(g \circ f)(x) = |x + 1|$, $(f \circ g)(x) = |x| + 1$

(iii) $(g \circ f)(x) = 2x^6 + 1$, $(f \circ g)(x) = (2x^2 + 1)^3$

(iv) $(g \circ f)(x) = 5 \sin^2 x$, $(f \circ g)(x) = \sin(5x^2)$

Question 10

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = x^2 - 3x + 2$ for all $x \in \mathbb{R}$, then find $f \circ f$.

TYPE 26 Finding Composition of Functions

★ Likely Exam Question

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2 - 3x + 2$.

Concept / Formula Used: $(f \circ f)(x) = f(f(x))$

Solution:

By definition of composite functions:

$$(f \circ f)(x) = f(f(x))$$

Substitute $f(x) = x^2 - 3x + 2$:

$$(f \circ f)(x) = (x^2 - 3x + 2)^2 - 3(x^2 - 3x + 2) + 2$$

Expanding $(x^2 - 3x + 2)^2$ using the expansion formula $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$:

$$\begin{aligned}(x^2 - 3x + 2)^2 &= (x^2)^2 + (-3x)^2 + (2)^2 + 2(x^2)(-3x) + 2(-3x)(2) + 2(2)(x^2) \\ &= x^4 + 9x^2 + 4 - 6x^3 - 12x + 4x^2 \\ &= x^4 - 6x^3 + 13x^2 - 12x + 4\end{aligned}$$

Expanding the second part:

$$-3(x^2 - 3x + 2) = -3x^2 + 9x - 6$$

Substitute back into the expression:

$$\begin{aligned}
 (f \circ f)(x) &= (x^4 - 6x^3 + 13x^2 - 12x + 4) + (-3x^2 + 9x - 6) + 2 \\
 &= x^4 - 6x^3 + (13x^2 - 3x^2) + (-12x + 9x) + (4 - 6 + 2) \\
 &= x^4 - 6x^3 + 10x^2 - 3x + 0 \\
 &= x^4 - 6x^3 + 10x^2 - 3x
 \end{aligned}$$

Final Answer

$$(f \circ f)(x) = x^4 - 6x^3 + 10x^2 - 3x$$

Common Mistake: A common error occurs in expanding $(x^2 - 3x + 2)^2$; remember to keep track of signs when evaluating cross-terms like $2(x^2)(-3x) = -6x^3$.

Question 11

Let $\mathbb{R}$ be the set of all real numbers. If the functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are given by $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$ for all $x \in \mathbb{R}$, then find

- (i) $f \circ g$
- (ii) $g \circ f$
- (iii) $f \circ f$
- (iv) $g \circ g$

TYPE 26 Finding Composition of Functions

Given: $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $(f \circ g)(x) = f(g(x))$ $(g \circ f)(x) = g(f(x))$ $(f \circ f)(x) = f(f(x))$ $(g \circ g)(x) = g(g(x))$

Solution:

(i) **Finding $(f \circ g)(x)$:**

$$\begin{aligned}
 (f \circ g)(x) &= f(g(x)) = f(2x - 3) \\
 &= (2x - 3)^2 + 3(2x - 3) + 1 \\
 &= (4x^2 - 12x + 9) + (6x - 9) + 1 \\
 &= 4x^2 - 12x + 6x + 9 - 9 + 1 \\
 &= 4x^2 - 6x + 1
 \end{aligned}$$

(ii) **Finding $(g \circ f)(x)$:**

$$\begin{aligned}
 (g \circ f)(x) &= g(f(x)) = g(x^2 + 3x + 1) \\
 &= 2(x^2 + 3x + 1) - 3 \\
 &= 2x^2 + 6x + 2 - 3 \\
 &= 2x^2 + 6x - 1
 \end{aligned}$$

(iii) Finding $(f \circ f)(x)$:

$$\begin{aligned}
 (f \circ f)(x) &= f(f(x)) = f(x^2 + 3x + 1) \\
 &= (x^2 + 3x + 1)^2 + 3(x^2 + 3x + 1) + 1 \\
 &= [(x^2)^2 + (3x)^2 + (1)^2 + 2(x^2)(3x) + 2(3x)(1) + 2(1)(x^2)] + 3x^2 + 9x + 3 + 1 \\
 &= [x^4 + 9x^2 + 1 + 6x^3 + 6x + 2x^2] + 3x^2 + 9x + 4 \\
 &= x^4 + 6x^3 + (9 + 2 + 3)x^2 + (6 + 9)x + (1 + 4) \\
 &= x^4 + 6x^3 + 14x^2 + 15x + 5
 \end{aligned}$$

(iv) Finding $(g \circ g)(x)$:

$$\begin{aligned}
 (g \circ g)(x) &= g(g(x)) = g(2x - 3) \\
 &= 2(2x - 3) - 3 \\
 &= 4x - 6 - 3 \\
 &= 4x - 9
 \end{aligned}$$

Final Answer

- (i) $(f \circ g)(x) = 4x^2 - 6x + 1$
(ii) $(g \circ f)(x) = 2x^2 + 6x - 1$
(iii) $(f \circ f)(x) = x^4 + 6x^3 + 14x^2 + 15x + 5$
(iv) $(g \circ g)(x) = 4x - 9$

Question 12

If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^2$ and $g(x) = x + 1$ for all $x \in \mathbb{R}$, show that $g \circ f \neq f \circ g$.

TYPE 26 Finding Composition of Functions**To Prove:** $g \circ f \neq f \circ g$

Concept	/	Formula	Used:	h_1	h_2	$h_1(x)$	$=$	$h_2(x)$	x	$\in$
$\mathbb{R}$		x	$h_1(x) \neq h_2(x)$							

Solution:First, calculate $(g \circ f)(x)$:

$$\begin{aligned}
 (g \circ f)(x) &= g(f(x)) \\
 &= g(x^2) \\
 &= x^2 + 1
 \end{aligned}$$

Next, calculate $(f \circ g)(x)$:

$$\begin{aligned}
 (f \circ g)(x) &= f(g(x)) \\
 &= f(x + 1) \\
 &= (x + 1)^2 \\
 &= x^2 + 2x + 1
 \end{aligned}$$

Comparing $(g \circ f)(x)$ and $(f \circ g)(x)$:

$$(g \circ f)(x) = x^2 + 1$$

$$(f \circ g)(x) = x^2 + 2x + 1$$

For example, evaluate both at $x = 1$:

$$(g \circ f)(1) = 1^2 + 1 = 2$$

$$(f \circ g)(1) = (1 + 1)^2 = 4$$

Since $(g \circ f)(1) \neq (f \circ g)(1)$, the two composition functions are not equal.

Hence Proved

$(g \circ f)(x) \neq (f \circ g)(x)$, hence $g \circ f \neq f \circ g$ is proved.

Question 13

If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x + 1$ and $g(x) = x - 1$ for all $x \in \mathbb{R}$, show that $g \circ f = f \circ g = I_{\mathbb{R}}$.

TYPE 26 Finding Composition of Functions

To Prove: $g \circ f = f \circ g = I_{\mathbb{R}}$, where $I_{\mathbb{R}}(x) = x$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $\mathbb{R} \quad I_{\mathbb{R}}(x) = x \quad x \in \mathbb{R}$

Solution:

Step 1: Evaluate $(g \circ f)(x)$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(x + 1) \\ &= (x + 1) - 1 \\ &= x \\ &= I_{\mathbb{R}}(x)\end{aligned}$$

Thus, $g \circ f = I_{\mathbb{R}}$.

Step 2: Evaluate $(f \circ g)(x)$:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(x - 1) \\ &= (x - 1) + 1 \\ &= x \\ &= I_{\mathbb{R}}(x)\end{aligned}$$

Thus, $f \circ g = I_{\mathbb{R}}$.

From Step 1 and Step 2, we conclude $g \circ f = f \circ g = I_{\mathbb{R}}$.

Hence Proved

$g \circ f = f \circ g = I_{\mathbb{R}}$, hence proved.

Question 14

If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = 3x^2 - 2$ and $g(x) = \sin 2x$ respectively, find

- (i) $g \circ f$
- (ii) $f \circ g$
- (iii) $f \circ f$

TYPE 26 Finding Composition of Functions

Given: $f(x) = 3x^2 - 2$ and $g(x) = \sin 2x$ for all $x \in \mathbb{R}$.

Concept / Formula Used: $(g \circ f)(x) = g(f(x))$ $(f \circ g)(x) = f(g(x))$ $(f \circ f)(x) = f(f(x))$

Solution:

(i) Finding $(g \circ f)(x)$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(3x^2 - 2) \\ &= \sin(2(3x^2 - 2)) \\ &= \sin(6x^2 - 4)\end{aligned}$$

(ii) Finding $(f \circ g)(x)$:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(\sin 2x) \\ &= 3(\sin 2x)^2 - 2 \\ &= 3 \sin^2 2x - 2\end{aligned}$$

(iii) Finding $(f \circ f)(x)$:

$$\begin{aligned}(f \circ f)(x) &= f(f(x)) \\ &= f(3x^2 - 2) \\ &= 3(3x^2 - 2)^2 - 2 \\ &= 3(9x^4 - 12x^2 + 4) - 2 \\ &= 27x^4 - 36x^2 + 12 - 2 \\ &= 27x^4 - 36x^2 + 10\end{aligned}$$

Final Answer

- (i) $(g \circ f)(x) = \sin(6x^2 - 4)$
- (ii) $(f \circ g)(x) = 3 \sin^2 2x - 2$
- (iii) $(f \circ f)(x) = 27x^4 - 36x^2 + 10$

Question 15

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{x^2 + 1}$, for all $x \in \mathbb{R}$ is neither one-one nor onto. Also if $g : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $g(x) = 2x - 1$, find $(f \circ g)(x)$.

TYPE 22 Testing Injectivity and Surjectivity of Functions

★ **Likely Exam Question**

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = \frac{x}{x^2 + 1}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ where $g(x) = 2x - 1$.

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2$ $\text{Range}(f) = \text{Codomain of } f = \mathbb{R}$

Solution:

Part 1: Show f is not one-one (injective): Consider two distinct domain values $x = 2$ and $x = \frac{1}{2}$ in $\mathbb{R}$:

$$f(2) = \frac{2}{2^2 + 1} = \frac{2}{4 + 1} = \frac{2}{5}$$

$$f\left(\frac{1}{2}\right) = \frac{\frac{1}{2}}{\left(\frac{1}{2}\right)^2 + 1} = \frac{\frac{1}{2}}{\frac{1}{4} + 1} = \frac{\frac{1}{2}}{\frac{5}{4}} = \frac{2}{5}$$

Since $f(2) = f\left(\frac{1}{2}\right) = \frac{2}{5}$ for $2 \neq \frac{1}{2}$, f is **not one-one**.

Part 2: Show f is not onto (surjective): Let $y = f(x) = \frac{x}{x^2 + 1}$. Cross-multiplying:

$$y(x^2 + 1) = x \implies yx^2 - x + y = 0$$

For x to be a real number, the discriminant D of this quadratic equation in x must be non-negative ($D \geq 0$):

$$D = (-1)^2 - 4(y)(y)$$

$$= 1 - 4y^2 \geq 0$$

Solving $1 - 4y^2 \geq 0$:

$$(1 - 2y)(1 + 2y) \geq 0 \implies -\frac{1}{2} \leq y \leq \frac{1}{2}$$

Thus, $\text{Range}(f) = \left[-\frac{1}{2}, \frac{1}{2}\right] \neq \mathbb{R}$ (Codomain). For instance, $y = 2 \in \mathbb{R}$ has no real pre-image because $1 - 4(2)^2 = -15 < 0$. Hence, f is **not onto**.

Part 3: Evaluate $(f \circ g)(x)$:

$$\begin{aligned}
 (f \circ g)(x) &= f(g(x)) \\
 &= f(2x - 1) \\
 &= \frac{2x - 1}{(2x - 1)^2 + 1} \\
 &= \frac{2x - 1}{(4x^2 - 4x + 1) + 1} \\
 &= \frac{2x - 1}{4x^2 - 4x + 2}
 \end{aligned}$$

Final Answer

Function f is neither one-one nor onto.

$$(f \circ g)(x) = \frac{2x - 1}{4x^2 - 4x + 2}$$

Common Mistake: When checking surjectivity for rational functions, always set up $y = f(x)$, solve for real x via the discriminant, and verify if the Range equals the Codomain.

1.5 – Invertible Functions and Inverse of a Function

Question 1

If $f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$, find f^{-1} .

TYPE 29 Finding Inverse Function on Finite Sets

Given: $f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$

To Find: f^{-1}

Concept / Formula Used: $f = \{(x, y)\}$ $f^{-1} = \{(y, x) \mid (x, y) \in f\}$

Solution:

Given the function defined as a set of ordered pairs:

$$f = \{(1, 5), (2, 3), (3, 0), (4, -2)\}$$

To find the inverse function f^{-1} , reverse the components of each ordered pair $(x, y) \mapsto$

(y, x) :

$$(1, 5) \implies (5, 1)$$

$$(2, 3) \implies (3, 2)$$

$$(3, 0) \implies (0, 3)$$

$$(4, -2) \implies (-2, 4)$$

Combining these inverted pairs gives:

$$f^{-1} = \{(5, 1), (3, 2), (0, 3), (-2, 4)\}$$

Final Answer

$$f^{-1} = \{(5, 1), (3, 2), (0, 3), (-2, 4)\}$$

Question 2

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = 3x + 5$, then find f^{-1} .

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x + 5$

To Find: f^{-1}

Concept / Formula Used: $f^{-1}(x)$ $y = f(x)$ x y

Solution:

Let $y = f(x)$.

$$y = 3x + 5$$

Subtract 5 from both sides:

$$y - 5 = 3x$$

Divide both sides by 3:

$$x = \frac{y - 5}{3}$$

Since $y = f(x) \iff x = f^{-1}(y)$, we have:

$$f^{-1}(y) = \frac{y - 5}{3}$$

Replacing y with x gives the inverse function:

$$f^{-1}(x) = \frac{x - 5}{3}$$

Final Answer

$$f^{-1}(x) = \frac{x-5}{3} \text{ for all } x \in \mathbb{R}$$

Question 3

If $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{3x-2}{5}$ is invertible, find the inverse of f .

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{3x-2}{5}$

To Find: f^{-1}

Concept / Formula Used: $y = f(x) \quad f^{-1}(y) = x$

Solution:

Let $y = f(x)$.

$$y = \frac{3x-2}{5}$$

Multiply both sides by 5:

$$5y = 3x - 2$$

Add 2 to both sides:

$$5y + 2 = 3x$$

Divide both sides by 3:

$$x = \frac{5y+2}{3}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{5y+2}{3}$$

Replacing the variable y with x :

$$f^{-1}(x) = \frac{5x+2}{3}$$

Final Answer

$$f^{-1}(x) = \frac{5x+2}{3}$$

Question 4

If $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x-7}{4}$ is invertible, find f^{-1} .

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{2x-7}{4}$

To Find: f^{-1}

Concept / Formula Used: $y = f(x) \quad x \quad y$

Solution:

Let $y = f(x)$.

$$y = \frac{2x - 7}{4}$$

Multiply both sides by 4:

$$4y = 2x - 7$$

Add 7 to both sides:

$$4y + 7 = 2x$$

Divide both sides by 2:

$$x = \frac{4y + 7}{2}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{4y + 7}{2}$$

Replacing y with x :

$$f^{-1}(x) = \frac{4x + 7}{2}$$

Final Answer

$$f^{-1}(x) = \frac{4x+7}{2}$$

Question 5

If f is an invertible function defined as $f(x) = \frac{3x-4}{5}$, write $f^{-1}(x)$.

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = \frac{3x - 4}{5}$, f invertible

To Find: $f^{-1}(x)$

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = \frac{3x - 4}{5}$$

Multiply both sides by 5:

$$5y = 3x - 4$$

Add 4 to both sides:

$$5y + 4 = 3x$$

Divide both sides by 3:

$$x = \frac{5y + 4}{3}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{5y + 4}{3}$$

Replacing y with x :

$$f^{-1}(x) = \frac{5x + 4}{3}$$

Final Answer

$$f^{-1}(x) = \frac{5x + 4}{3}$$

Question 6

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = ax - b$, $a, b \in \mathbb{R}$, $a \neq 0$, then find f^{-1} .

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = ax - b$, $a \neq 0$

To Find: f^{-1}

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = ax - b$$

Add b to both sides:

$$y + b = ax$$

Divide both sides by a (valid since $a \neq 0$):

$$x = \frac{y + b}{a}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{y + b}{a}$$

Final Answer

$$f^{-1} : \mathbb{R} \rightarrow \mathbb{R} \text{ is given by } f^{-1}(y) = \frac{y + b}{a}$$

Question 7

If $f : \mathbb{R} - \left\{-\frac{3}{5}\right\} \rightarrow \mathbb{R}$ is a function defined by $f(x) = \frac{2x}{5x + 3}$, then find $f^{-1} : \text{range of } f \rightarrow \mathbb{R} - \left\{-\frac{3}{5}\right\}$.

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = \frac{2x}{5x+3}, x \neq -\frac{3}{5}$

To Find: $f^{-1} : \text{range of } f \rightarrow \mathbb{R} - \left\{-\frac{3}{5}\right\}$

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = \frac{2x}{5x+3}$$

Cross-multiply:

$$y(5x+3) = 2x$$

Expand the left side:

$$5xy + 3y = 2x$$

Collect all x -terms on one side:

$$5xy - 2x = -3y$$

Factor out x :

$$x(5y - 2) = -3y$$

Divide both sides by $(5y - 2)$:

$$x = \frac{-3y}{5y-2} = \frac{3y}{2-5y}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{3y}{2-5y}$$

Final Answer

$f^{-1} : \text{range of } f \rightarrow \mathbb{R} - \left\{-\frac{3}{5}\right\}$ is given by $f^{-1}(y) = \frac{3y}{2-5y}$

Question 8

If $f : \mathbb{R} - \left\{-\frac{4}{3}\right\} \rightarrow \mathbb{R}$ is a function defined by $f(x) = \frac{4x}{3x+4}$, then find f^{-1} .
(NCERT)

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = \frac{4x}{3x+4}, x \neq -\frac{4}{3}$

To Find: f^{-1}

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = \frac{4x}{3x + 4}$$

Cross-multiply:

$$y(3x + 4) = 4x$$

Expand the left side:

$$3xy + 4y = 4x$$

Collect all x -terms on one side:

$$3xy - 4x = -4y$$

Factor out x :

$$x(3y - 4) = -4y$$

Divide both sides by $(3y - 4)$:

$$x = \frac{-4y}{3y - 4} = \frac{4y}{4 - 3y}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{4y}{4 - 3y}$$

Final Answer

$f^{-1} : \text{range of } f \rightarrow \mathbb{R} - \left\{-\frac{4}{3}\right\}$ is given by $f^{-1}(y) = \frac{4y}{4 - 3y}$

Question 9

If $f(x) = [4 - (x - 7)^3]^{1/5}$ is a real invertible function, then find $f^{-1}(x)$. (ISC 2023)

TYPE 30 Finding Inverse of Algebraic Functions

Given: $f(x) = [4 - (x - 7)^3]^{1/5}$

To Find: $f^{-1}(x)$

Concept / Formula Used: $y = f(x) \iff x = f^{-1}(y)$

Solution:

Let $y = f(x)$.

$$y = [4 - (x - 7)^3]^{1/5}$$

Raise both sides to the power 5:

$$y^5 = 4 - (x - 7)^3$$

Rearrange to isolate the cube term:

$$(x - 7)^3 = 4 - y^5$$

Take the cube root of both sides:

$$x - 7 = (4 - y^5)^{1/3}$$

Add 7 to both sides:

$$x = 7 + (4 - y^5)^{1/3}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \sqrt[3]{4 - y^5} + 7$$

Replacing y with x :

$$f^{-1}(x) = \sqrt[3]{4 - x^5} + 7$$

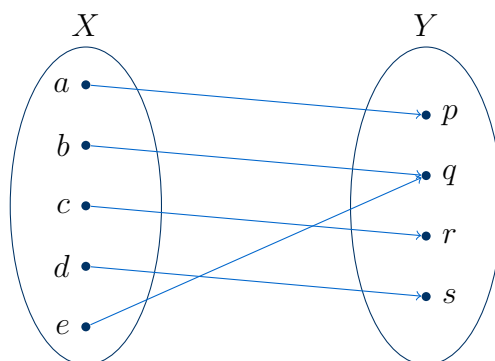
Final Answer

$$f^{-1}(x) = \sqrt[3]{4 - x^5} + 7$$

Question 10

The function $f : X \rightarrow Y$ is not invertible. State the reason. (ISC Specimen Paper 2024)

TYPE 31 Testing Invertibility of Functions



$f : X \rightarrow Y$ mapping five elements of X onto only four elements of Y , with two arrows landing on the same image

Given: $f : X \rightarrow Y$, $n(X) = 5$, $n(Y) = 4$ (as shown in the given arrow diagram)

Solution:

From the arrow diagram, X has more elements than Y . By the pigeonhole principle, at least two distinct elements of X must map to the same element of Y .

In the diagram, two arrows point to the same element of Y , i.e. there exist $x_1, x_2 \in X$ with $x_1 \neq x_2$ such that

$$f(x_1) = f(x_2)$$

This means distinct elements of X have the same image, so f is not one-one.

Since a function must be both one-one and onto (bijective) to be invertible, and f fails to be one-one, f cannot be invertible.

Final Answer

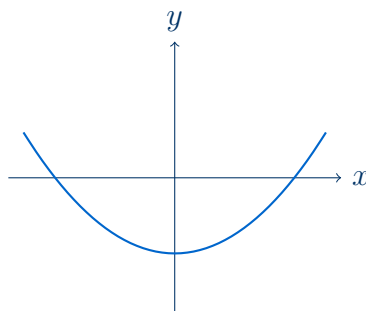
f is not invertible because f is not one-one.

Question 11

Which one of the following graphs is a function of x ?

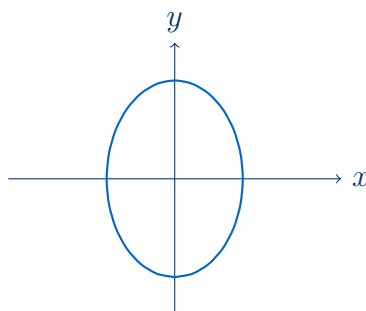
(ISC 2024)

TYPE 32 Vertical Line Test for a Function



Graph A

Graph A: a single continuous curve, dipping down and rising again – every vertical line meets it exactly once



Graph B

Graph B: a sideways loop – a vertical line through the loop meets the curve more than once

Solution:

By the definition of a function, $f : X \rightarrow Y$ is a function iff every element $x \in X$ has *exactly one* image $y \in Y$. Graphically, this is tested using the **vertical line test**: a curve

represents a function of x iff every vertical line $x = k$ intersects the curve in *at most one* point.

Graph A: Every vertical line drawn on Graph A crosses the curve exactly once. So for each x , there is a unique y .

$\implies$ Graph A represents a function of x

Graph B: Graph B is a closed loop. A vertical line drawn through the middle of the loop crosses the curve at *two* points, giving two different y -values for the same x .

$\implies$ Graph B does *not* represent a function of x

Final Answer

Graph A is the function of x .

Question 12

Let $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$ be a function defined by $f(1) = a$, $f(2) = b$ and $f(3) = c$. Show that f is invertible and find f^{-1} . Also show that $(f^{-1})^{-1} = f$. (NCERT)

TYPE 33a Proving Invertibility & Finding Inverse – Roster/Finite Functions

To Prove: f is invertible, and $(f^{-1})^{-1} = f$

Given: $f = \{(1, a), (2, b), (3, c)\}$

Solution:

One-one: The images a, b, c are all distinct (given), so distinct elements of the domain have distinct images.

$\implies f$ is one-one

Onto: The range of f is $\{a, b, c\}$, which equals the codomain.

$\implies f$ is onto

Since f is both one-one and onto, f is bijective, and hence invertible.

To find f^{-1} , reverse each ordered pair of f :

$$(1, a) \implies (a, 1)$$

$$(2, b) \implies (b, 2)$$

$$(3, c) \implies (c, 3)$$

$$f^{-1} = \{(a, 1), (b, 2), (c, 3)\}$$

Now, to show $(f^{-1})^{-1} = f$, reverse each ordered pair of f^{-1} :

$$(a, 1) \implies (1, a)$$

$$(b, 2) \implies (2, b)$$

$$(c, 3) \implies (3, c)$$

$$(f^{-1})^{-1} = \{(1, a), (2, b), (3, c)\} = f$$

Hence Proved

f is invertible, $f^{-1} = \{(a, 1), (b, 2), (c, 3)\}$, and $(f^{-1})^{-1} = f$.

Question 13

Let $S = \{1, 2, 3\}$. Determine whether the functions $f : S \rightarrow S$ defined as below have inverse. Find f^{-1} , if it exists.

- (i) $f = \{(1, 1), (2, 2), (3, 3)\}$
- (ii) $f = \{(1, 2), (2, 1), (3, 1)\}$
- (iii) $f = \{(1, 3), (3, 2), (2, 1)\}$
- (iv) $f = \{(1, 3), (2, 2), (3, 1)\}$

TYPE 33a Proving Invertibility & Finding Inverse – Roster/Finite Functions

Given: $S = \{1, 2, 3\}$

Solution:

(i) $f = \{(1, 1), (2, 2), (3, 3)\}$

Images 1, 2, 3 are all distinct $\implies f$ is one-one. Range = $\{1, 2, 3\} = S \implies f$ is onto. So f is bijective and invertible.

Reversing the pairs: $(1, 1) \implies (1, 1), (2, 2) \implies (2, 2), (3, 3) \implies (3, 3)$.

$$f^{-1} = \{(1, 1), (2, 2), (3, 3)\}$$

(ii) $f = \{(1, 2), (2, 1), (3, 1)\}$

Check one-one: $f(2) = 1$ and $f(3) = 1$, i.e. two distinct elements 2, 3 have the same image 1.

$\implies f$ is not one-one $\implies f$ is not invertible

(iii) $f = \{(1, 3), (3, 2), (2, 1)\}$

Images 3, 2, 1 (for $x = 1, 3, 2$ respectively) are all distinct $\implies f$ is one-one. Range = $\{3, 2, 1\} = S \implies f$ is onto. So f is bijective and invertible.

Reversing the pairs: $(1, 3) \implies (3, 1), (3, 2) \implies (2, 3), (2, 1) \implies (1, 2)$.

$$f^{-1} = \{(3, 1), (2, 3), (1, 2)\}$$

(iv) $f = \{(1, 3), (2, 2), (3, 1)\}$

Images 3, 2, 1 are all distinct $\implies f$ is one-one. Range = $\{3, 2, 1\} = S \implies f$ is onto. So f is bijective and invertible.

Reversing the pairs: $(1, 3) \implies (3, 1), (2, 2) \implies (2, 2), (3, 1) \implies (1, 3)$.

$$f^{-1} = \{(3, 1), (2, 2), (1, 3)\}$$

Final Answer

- (i) Invertible, $f^{-1} = \{(1, 1), (2, 2), (3, 3)\}$
- (ii) Not invertible, since f is not one-one
- (iii) Invertible, $f^{-1} = \{(3, 1), (2, 3), (1, 2)\}$
- (iv) Invertible, $f^{-1} = \{(3, 1), (2, 2), (1, 3)\}$

Question 14

Explain why for the following functions $f : X \rightarrow Y$, we cannot define f^{-1} :

- (i) $X = \mathbb{R} - \{0\}, Y = \mathbb{R}$ and $f(x) = \frac{1}{x}$ for all $x \in X$
- (ii) $X = \{-1, 0, 1, 2\}, Y = \{2, 3\}$ and $f(x) = 2$ for $x = -1, 0, 1$ and $f(2) = 3$
- (iii) $X = \mathbb{R} = Y$ and $f(x) = x^2 + 1$ for all $x \in \mathbb{R}$

TYPE 31 Testing Invertibility of Functions

To Prove: Explain why f^{-1} cannot be defined for each of the given functions.

Concept / Formula Used: $f : X \rightarrow Y$

Solution:

- (i) For $X = \mathbb{R} - \{0\}, Y = \mathbb{R}, f(x) = \frac{1}{x}$:

$$\text{Range of } f = \left\{ \frac{1}{x} \mid x \in \mathbb{R} - \{0\} \right\} = \mathbb{R} - \{0\}$$

Since $0 \in Y$ has no pre-image in X (as $\frac{1}{x} = 0$ has no solution), the range of f does not equal the codomain $Y = \mathbb{R}$. Thus, f is not onto. Therefore, f^{-1} cannot be defined.

- (ii) For $X = \{-1, 0, 1, 2\}, Y = \{2, 3\}$ with $f(-1) = 2, f(0) = 2, f(1) = 2, f(2) = 3$:
Observe that:

$$f(-1) = f(0) = f(1) = 2$$

Since distinct domain elements $-1, 0, 1$ share the same image 2, f is not one-one. Therefore, f^{-1} cannot be defined.

- (iii) For $X = \mathbb{R}, Y = \mathbb{R}, f(x) = x^2 + 1$: Take $x_1 = 1$ and $x_2 = -1$:

$$\begin{aligned} f(1) &= 1^2 + 1 = 2 \\ f(-1) &= (-1)^2 + 1 = 2 \end{aligned}$$

Since $f(1) = f(-1) = 2$ while $1 \neq -1$, f is not one-one. Furthermore, $x^2 + 1 \geq 1$ for all $x \in \mathbb{R}$, so negative numbers in Y have no pre-images, making f not onto as well. Therefore, f^{-1} cannot be defined.

Hence Proved

In all three cases, f fails to be bijective (either not one-one or not onto), so f^{-1} cannot be defined.

Question 15

Let $S = \{a, b, c\}$ and $T = \{1, 2, 3\}$. Find f^{-1} of the following functions f from S to T , if it exists.

- (i) $f = \{(a, 3), (b, 2), (c, 1)\}$
 (ii) $f = \{(a, 2), (b, 1), (c, 1)\}$

TYPE 29 Finding Inverse Function on Finite Sets

Given: $S = \{a, b, c\}$, $T = \{1, 2, 3\}$, and functions $f : S \rightarrow T$

To Find: f^{-1} if it exists

Concept / Formula Used: f^{-1} f

Solution:

(i) For $f = \{(a, 3), (b, 2), (c, 1)\}$: - Injectivity: $f(a) = 3, f(b) = 2, f(c) = 1$. All images are distinct, so f is one-one. - Surjectivity: Range of $f = \{3, 2, 1\} = T$, so f is onto. Since f is bijective, f^{-1} exists. Reversing the ordered pairs $(x, y) \mapsto (y, x)$:

$$f^{-1} = \{(3, a), (2, b), (1, c)\} \quad \text{or} \quad \{(1, c), (2, b), (3, a)\}$$

(ii) For $f = \{(a, 2), (b, 1), (c, 1)\}$: Observe that $f(b) = 1$ and $f(c) = 1$. Since $f(b) = f(c)$ with $b \neq c$, f is not one-one. Since f is not one-one, f is not bijective. Therefore, f^{-1} does not exist.

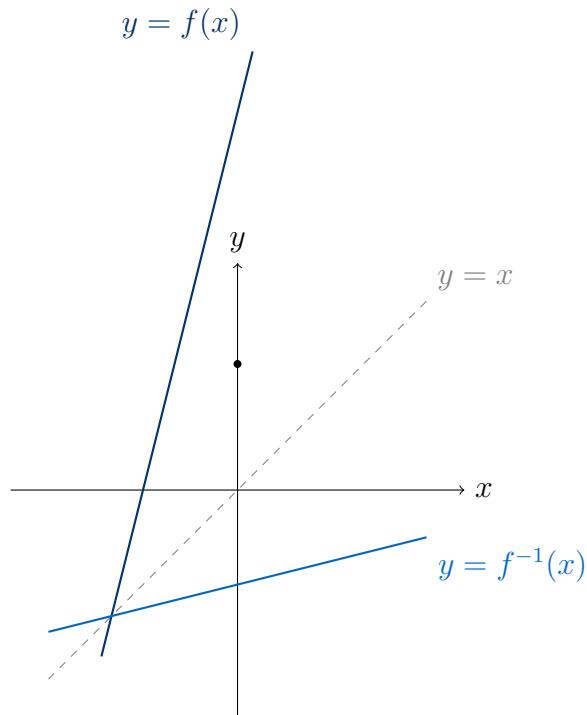
Final Answer

- (i) $f^{-1} = \{(1, c), (2, b), (3, a)\}$
 (ii) f^{-1} does not exist

Question 16

Let $\mathbb{R}$ be the set of all real numbers. Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 4x + 5$ is invertible. Also find inverse of f . Draw their sketch on one graph.

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions



Graphs of $f(x) = 4x + 5$, $f^{-1}(x) = \frac{x-5}{4}$, and symmetry line $y = x$

To Prove: f is invertible, find $f^{-1}(x)$, and sketch both graphs

Concept / Formula Used: f is invertible $\iff f(x_1) = f(x_2) \implies x_1 = x_2 \quad \forall y \in \text{Codomain}, \exists x \in \text{Domain} \text{ such that } f(x) = y$

Solution:

1. Proof of One-One (Injective): Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$.

$$4x_1 + 5 = 4x_2 + 5$$

$$4x_1 = 4x_2$$

$$x_1 = x_2$$

Hence, f is one-one.

2. Proof of Onto (Surjective): Let $y \in \mathbb{R}$ (codomain). Set $y = f(x)$:

$$y = 4x + 5$$

$$y - 5 = 4x$$

$$x = \frac{y - 5}{4}$$

For any $y \in \mathbb{R}$, $x = \frac{y-5}{4} \in \mathbb{R}$ (domain) and:

$$f(x) = f\left(\frac{y-5}{4}\right) = 4\left(\frac{y-5}{4}\right) + 5 = y - 5 + 5 = y$$

Hence, f is onto.

3. Finding f^{-1} : Since f is bijective, it is invertible. The inverse function $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ is:

$$f^{-1}(y) = \frac{y-5}{4} \implies f^{-1}(x) = \frac{x-5}{4}$$

Hence Proved

f is invertible with inverse $f^{-1}(x) = \frac{x-5}{4}$.

Question 17

Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x-1}{3}$ is one-one and onto. Also find the inverse of the function f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

To Prove: f is one-one and onto, and find f^{-1}

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad \forall y \in \mathbb{R}, \exists x \in \mathbb{R}$
 $f(x) = y$

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{R}$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned} \frac{2x_1-1}{3} &= \frac{2x_2-1}{3} \\ 2x_1-1 &= 2x_2-1 \\ 2x_1 &= 2x_2 \\ x_1 &= x_2 \end{aligned}$$

Therefore, f is one-one.

2. Proof of Onto: Let $y \in \mathbb{R}$ (codomain). Set $y = f(x)$:

$$\begin{aligned} y &= \frac{2x-1}{3} \\ 3y &= 2x-1 \\ 2x &= 3y+1 \\ x &= \frac{3y+1}{2} \end{aligned}$$

For every $y \in \mathbb{R}$, $x = \frac{3y+1}{2} \in \mathbb{R}$ (domain) and:

$$f(x) = f\left(\frac{3y+1}{2}\right) = \frac{2\left(\frac{3y+1}{2}\right)-1}{3} = \frac{3y+1-1}{3} = \frac{3y}{3} = y$$

Therefore, f is onto.

3. Finding f^{-1} : Since f is bijective, $f^{-1}(y) = x = \frac{3y+1}{2}$. Replacing y with x :

$$f^{-1}(x) = \frac{3x+1}{2}$$

Hence Proved

f is one-one and onto, with $f^{-1}(x) = \frac{3x+1}{2}$.

Question 18

Consider a bijective function $f : \mathbb{R}_0^+ \rightarrow (7, \infty)$ given by $f(x) = 16x^2 + 24x + 7$, where $\mathbb{R}_0^+$ is the set of positive real numbers. Find the inverse function of f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

Given: $f : \mathbb{R}_0^+ \rightarrow (7, \infty)$, $f(x) = 16x^2 + 24x + 7$, where $\mathbb{R}_0^+ = (0, \infty)$

To Find: $f^{-1}(x)$

Concept / Formula Used: $ax^2 + bx + c$ x $y = f(x)$

Solution:

Let $y = f(x)$.

$$y = 16x^2 + 24x + 7$$

Complete the square on the right-hand side:

$$\begin{aligned} y &= (4x)^2 + 2(4x)(3) + 3^2 - 9 + 7 \\ y &= (4x + 3)^2 - 2 \end{aligned}$$

Add 2 to both sides:

$$y + 2 = (4x + 3)^2$$

Since $x \in \mathbb{R}_0^+$ (i.e. $x > 0$), we have $4x + 3 > 3 > 0$. Taking the positive square root on both sides:

$$\sqrt{y + 2} = 4x + 3$$

Subtract 3 from both sides:

$$4x = \sqrt{y + 2} - 3$$

Divide by 4:

$$x = \frac{\sqrt{y + 2} - 3}{4}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{\sqrt{y + 2} - 3}{4}$$

Replacing dummy variable y with x :

$$f^{-1}(x) = \frac{\sqrt{x + 2} - 3}{4}$$

Final Answer

$$f^{-1}(x) = \frac{\sqrt{x+2}-3}{4} \text{ for } x \in (7, \infty)$$

Question 19

If $A = \mathbb{R} - \{-3\}$ and $B = \mathbb{R} - \{2\}$ and a function $f : A \rightarrow B$ be given by $f(x) = \frac{2x+1}{x+3}$, then show that the function f is one-one and onto. Hence, find f^{-1} .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

To Prove: f is one-one and onto, and find f^{-1}

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2 \quad \forall y \in B, \exists x \in A$
 $f(x) = y$

Solution:

1. Proof of One-One: Let $x_1, x_2 \in A = \mathbb{R} - \{-3\}$ such that $f(x_1) = f(x_2)$.

$$\frac{2x_1 + 1}{x_1 + 3} = \frac{2x_2 + 1}{x_2 + 3}$$

Cross-multiplying:

$$(2x_1 + 1)(x_2 + 3) = (2x_2 + 1)(x_1 + 3)$$

Expanding both sides:

$$2x_1x_2 + 6x_1 + x_2 + 3 = 2x_1x_2 + 6x_2 + x_1 + 3$$

Subtract $2x_1x_2 + 3$ from both sides:

$$6x_1 + x_2 = 6x_2 + x_1$$

Rearranging terms:

$$6x_1 - x_1 = 6x_2 - x_2$$

$$5x_1 = 5x_2$$

$$x_1 = x_2$$

Thus, f is one-one.

2. Proof of Onto: Let $y \in B = \mathbb{R} - \{2\}$. Set $y = f(x)$:

$$y = \frac{2x + 1}{x + 3}$$

$$y(x + 3) = 2x + 1$$

$$xy + 3y = 2x + 1$$

$$xy - 2x = 1 - 3y$$

$$x(y - 2) = 1 - 3y$$

$$x = \frac{1 - 3y}{y - 2}$$

Since $y \neq 2$, $x = \frac{1-3y}{y-2}$ is a well-defined real number. Check if $x \in A$ (i.e., $x \neq -3$): If $x = -3$:

$$\frac{1 - 3y}{y - 2} = -3 \implies 1 - 3y = -3y + 6 \implies 1 = 6 \quad (\text{Impossible})$$

Thus, $x \neq -3$, so $x \in A$. Evaluating $f(x)$:

$$f(x) = f\left(\frac{1-3y}{y-2}\right) = \frac{2\left(\frac{1-3y}{y-2}\right) + 1}{\left(\frac{1-3y}{y-2}\right) + 3} = \frac{2-6y+y-2}{1-3y+3y-6} = \frac{-5y}{-5} = y$$

Thus, f is onto.

3. Finding f^{-1} : Since f is bijective, $f^{-1} : B \rightarrow A$ is:

$$f^{-1}(y) = \frac{1-3y}{y-2} \implies f^{-1}(x) = \frac{1-3x}{x-2}$$

Hence Proved

f is one-one and onto, with $f^{-1}(x) = \frac{1-3x}{x-2}$.

Question 20

If $f : A \rightarrow A$ and $A = \mathbb{R} - \left\{\frac{8}{5}\right\}$, show that the function $f(x) = \frac{8x+3}{5x-8}$ is one-one onto. Hence, find f^{-1} .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

★ Likely Exam Question

To Prove: f is one-one and onto, and find f^{-1}

Concept / Formula Used: $f(x_1) = f(x_2) \implies x_1 = x_2$ $y = f(x)$ x

Solution:

1. Proof of One-One: Let $x_1, x_2 \in A$ such that $f(x_1) = f(x_2)$.

$$\frac{8x_1+3}{5x_1-8} = \frac{8x_2+3}{5x_2-8}$$

Cross-multiplying:

$$(8x_1+3)(5x_2-8) = (8x_2+3)(5x_1-8)$$

Expanding both sides:

$$40x_1x_2 - 64x_1 + 15x_2 - 24 = 40x_1x_2 - 64x_2 + 15x_1 - 24$$

Cancel $40x_1x_2 - 24$ from both sides:

$$-64x_1 + 15x_2 = -64x_2 + 15x_1$$

Rearranging terms:

$$15x_2 + 64x_2 = 15x_1 + 64x_1$$

$$79x_2 = 79x_1$$

$$x_1 = x_2$$

Thus, f is one-one.

2. Proof of Onto: Let $y \in A = \mathbb{R} - \{\frac{8}{5}\}$. Set $y = f(x)$:

$$\begin{aligned} y &= \frac{8x+3}{5x-8} \\ y(5x-8) &= 8x+3 \\ 5xy-8y &= 8x+3 \\ 5xy-8x &= 8y+3 \\ x(5y-8) &= 8y+3 \\ x &= \frac{8y+3}{5y-8} \end{aligned}$$

Since $y \neq \frac{8}{5}$, $5y-8 \neq 0$, so x is well-defined. If $x = \frac{8}{5}$, then $\frac{8y+3}{5y-8} = \frac{8}{5} \implies 40y+15 = 40y-64 \implies 15 = -64$, which is impossible. Hence $x \in A$. Evaluating $f(x)$:

$$f(x) = f\left(\frac{8y+3}{5y-8}\right) = \frac{8\left(\frac{8y+3}{5y-8}\right) + 3}{5\left(\frac{8y+3}{5y-8}\right) - 8} = \frac{64y+24+15y-24}{40y+15-40y+64} = \frac{79y}{79} = y$$

Thus, f is onto.

3. Finding f^{-1} :

$$f^{-1}(y) = \frac{8y+3}{5y-8} \implies f^{-1}(x) = \frac{8x+3}{5x-8}$$

Notice that $f^{-1}(x) = f(x)$ for all $x \in A$.

Hence Proved

f is one-one and onto, and $f^{-1}(x) = \frac{8x+3}{5x-8}$.

Question 21

Consider the function $f : \mathbb{R}_+ \rightarrow [9, \infty)$ defined by $f(x) = x^2 + 9$, where $\mathbb{R}_+$ is the set of all non-negative real numbers. Show that f is invertible. Also find the inverse of f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

To Prove: f is invertible and find f^{-1}

Concept / Formula Used: f

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{R}_+ = [0, \infty)$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned} x_1^2 + 9 &= x_2^2 + 9 \\ x_1^2 &= x_2^2 \end{aligned}$$

Taking square roots on both sides, since $x_1, x_2 \geq 0$:

$$x_1 = x_2$$

Thus, f is one-one.

2. Proof of Onto: Let $y \in [9, \infty)$ (codomain). Set $y = f(x)$:

$$\begin{aligned} y &= x^2 + 9 \\ y - 9 &= x^2 \\ x &= \sqrt{y - 9} \quad (\text{since } x \geq 0 \text{ and } y \geq 9) \end{aligned}$$

Since $y \geq 9$, $y - 9 \geq 0$, so $x = \sqrt{y - 9} \in \mathbb{R}_+$. Evaluating $f(x)$:

$$f(\sqrt{y - 9}) = (\sqrt{y - 9})^2 + 9 = (y - 9) + 9 = y$$

Thus, f is onto.

3. Finding f^{-1} : Since f is one-one and onto, f is invertible:

$$f^{-1}(y) = \sqrt{y - 9} \implies f^{-1}(x) = \sqrt{x - 9}$$

Hence Proved

f is invertible with inverse $f^{-1}(x) = \sqrt{x - 9}$.

Question 22

If $\mathbb{R}_0$ is the set of all non-zero real numbers, show that the function $f : \mathbb{R}_0 \rightarrow \mathbb{R}_0$ defined by $f(x) = \frac{3}{x}$ is invertible and it is the inverse of itself.

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

To Prove: f is invertible and $f^{-1}(x) = f(x)$

Concept / Formula Used: $f \quad f^{-1}(x) = f(x)$

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{R}_0$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned} \frac{3}{x_1} &= \frac{3}{x_2} \\ 3x_2 &= 3x_1 \\ x_1 &= x_2 \end{aligned}$$

Hence, f is one-one.

2. Proof of Onto: Let $y \in \mathbb{R}_0$ (codomain). Set $y = f(x)$:

$$\begin{aligned} y &= \frac{3}{x} \\ x &= \frac{3}{y} \end{aligned}$$

Since $y \neq 0$, $x = \frac{3}{y} \in \mathbb{R}_0$ (domain). Evaluating $f(x)$:

$$f\left(\frac{3}{y}\right) = \frac{3}{\frac{3}{y}} = y$$

Thus, f is onto.

3. Finding f^{-1} : Since f is bijective, f^{-1} exists. From $x = \frac{3}{y}$:

$$f^{-1}(y) = \frac{3}{y} \implies f^{-1}(x) = \frac{3}{x}$$

Therefore, $f^{-1}(x) = f(x) = \frac{3}{x}$ for all $x \in \mathbb{R}_0$. Thus, f is the inverse of itself.

Hence Proved

f is invertible and $f^{-1}(x) = f(x) = \frac{3}{x}$.

Question 23

Prove that the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x^2 + x + 1$ is one-one but not onto. Find the inverse of $f : \mathbb{N} \rightarrow S$ where S is the range of f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

★ Likely Exam Question

To Prove: $f : \mathbb{N} \rightarrow \mathbb{N}$ is one-one, not onto, and $f^{-1} : S \rightarrow \mathbb{N}$ is $f^{-1}(x) = \frac{\sqrt{4x-3}-1}{2}$

Concept / Formula Used: $f(x_1) - f(x_2) = 0$

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{N}$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned} x_1^2 + x_1 + 1 &= x_2^2 + x_2 + 1 \\ (x_1^2 - x_2^2) + (x_1 - x_2) &= 0 \\ (x_1 - x_2)(x_1 + x_2) + (x_1 - x_2) &= 0 \\ (x_1 - x_2)(x_1 + x_2 + 1) &= 0 \end{aligned}$$

Since $x_1, x_2 \in \mathbb{N}$, $x_1 \geq 1$ and $x_2 \geq 1$, which implies $(x_1 + x_2 + 1) \geq 3 > 0$. Dividing by $(x_1 + x_2 + 1)$:

$$x_1 - x_2 = 0 \implies x_1 = x_2$$

Thus, f is one-one.

2. Proof that f is not onto on $\mathbb{N}$: For $x = 1 \in \mathbb{N}$, $f(1) = 1^2 + 1 + 1 = 3$. For $x = 2 \in \mathbb{N}$, $f(2) = 2^2 + 2 + 1 = 7$. Consider $2 \in \mathbb{N}$ (codomain). There is no $x \in \mathbb{N}$ such that $f(x) = 2$, since $f(x) \geq 3$ for all $x \in \mathbb{N}$. Thus, $f : \mathbb{N} \rightarrow \mathbb{N}$ is not onto.

3. Inverse of $f : \mathbb{N} \rightarrow S$ (where $S = \text{Range of } f$): By restricting the codomain to range S , $f : \mathbb{N} \rightarrow S$ becomes bijective and invertible. Let $y = f(x) = x^2 + x + 1$. Multiply both sides by 4:

$$\begin{aligned} 4y &= 4x^2 + 4x + 4 \\ 4y &= (2x + 1)^2 + 3 \\ (2x + 1)^2 &= 4y - 3 \end{aligned}$$

Since $x \in \mathbb{N}$, $2x + 1 > 0$, take the positive square root:

$$\begin{aligned} 2x + 1 &= \sqrt{4y - 3} \\ 2x &= \sqrt{4y - 3} - 1 \\ x &= \frac{\sqrt{4y - 3} - 1}{2} \end{aligned}$$

Therefore, $f^{-1} : S \rightarrow \mathbb{N}$ is:

$$f^{-1}(x) = \frac{\sqrt{4x - 3} - 1}{2}$$

Hence Proved

f is one-one but not onto on $\mathbb{N}$, and $f^{-1}(x) = \frac{\sqrt{4x-3}-1}{2}$ for domain S .

Question 24

Let $f : \mathbb{N} \rightarrow S$ be a function defined by $f(x) = 4x^2 + 12x + 15$, where S is the range of f . Show that f is invertible and find the inverse of f .

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

★ Likely Exam Question

To Prove: f is invertible and find f^{-1}

Concept / Formula Used: S

Solution:

1. Proof of One-One: Let $x_1, x_2 \in \mathbb{N}$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned} 4x_1^2 + 12x_1 + 15 &= 4x_2^2 + 12x_2 + 15 \\ 4(x_1^2 - x_2^2) + 12(x_1 - x_2) &= 0 \\ 4(x_1 - x_2)(x_1 + x_2) + 12(x_1 - x_2) &= 0 \\ 4(x_1 - x_2)(x_1 + x_2 + 3) &= 0 \end{aligned}$$

Since $x_1, x_2 \in \mathbb{N}$, $x_1 \geq 1$ and $x_2 \geq 1$, so $(x_1 + x_2 + 3) \geq 5 > 0$. Dividing by $4(x_1 + x_2 + 3)$:

$$x_1 - x_2 = 0 \implies x_1 = x_2$$

Thus, f is one-one.

2. Proof of Onto: Since S is defined explicitly as the range of f , every element $y \in S$ has at least one pre-image $x \in \mathbb{N}$ such that $f(x) = y$. Hence, $f : \mathbb{N} \rightarrow S$ is onto. Since f is both one-one and onto, f is invertible.

3. Finding f^{-1} : Let $y = f(x) = 4x^2 + 12x + 15$. Complete the square on the right side:

$$\begin{aligned}y &= (2x)^2 + 2(2x)(3) + 3^2 - 9 + 15 \\y &= (2x + 3)^2 + 6\end{aligned}$$

Subtract 6 from both sides:

$$(2x + 3)^2 = y - 6$$

Since $x \in \mathbb{N}$, $2x + 3 > 0$. Taking the positive square root:

$$\begin{aligned}2x + 3 &= \sqrt{y - 6} \\2x &= \sqrt{y - 6} - 3 \\x &= \frac{\sqrt{y - 6} - 3}{2}\end{aligned}$$

Since $x = f^{-1}(y)$:

$$f^{-1}(y) = \frac{\sqrt{y - 6} - 3}{2}$$

Replacing y with x :

$$f^{-1}(x) = \frac{\sqrt{x - 6} - 3}{2}$$

Hence Proved

f is invertible and $f^{-1}(x) = \frac{\sqrt{x-6}-3}{2}$.

Question 25

Let $f : \mathbb{N} \rightarrow S$ be a function defined by $f(x) = 9x^2 + 6x - 5$, where S is the range of f . Show that f is invertible. Find the inverse of f and hence find $f^{-1}(43)$ and $f^{-1}(163)$.

TYPE 33b Proving Invertibility & Finding Inverse – Algebraic Functions

Given: $f : \mathbb{N} \rightarrow S$ defined by $f(x) = 9x^2 + 6x - 5$, where $S = \text{Range}(f)$.

To Find: Show that f is invertible, find $f^{-1}(y)$, and calculate $f^{-1}(43)$ and $f^{-1}(163)$.

Concept / Formula Used: $f : X \rightarrow Y$ $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$

Solution:

Step 1: Proof of Injectivity (One-One)

Let $x_1, x_2 \in \mathbf{N}$ such that $f(x_1) = f(x_2)$.

$$\begin{aligned} 9x_1^2 + 6x_1 - 5 &= 9x_2^2 + 6x_2 - 5 \\ 9x_1^2 - 9x_2^2 + 6x_1 - 6x_2 &= 0 \\ 9(x_1^2 - x_2^2) + 6(x_1 - x_2) &= 0 \\ 9(x_1 - x_2)(x_1 + x_2) + 6(x_1 - x_2) &= 0 \\ (x_1 - x_2)[9(x_1 + x_2) + 6] &= 0 \end{aligned}$$

Since $x_1, x_2 \in \mathbf{N}$, we have $x_1 \geq 1$ and $x_2 \geq 1$, which implies:

$$9(x_1 + x_2) + 6 \geq 9(1 + 1) + 6 = 24 > 0$$

Since $9(x_1 + x_2) + 6 \neq 0$, we must have:

$$x_1 - x_2 = 0 \implies x_1 = x_2$$

Therefore, f is a one-one (injective) function.

Step 2: Proof of Surjectivity (Onto)

The codomain of f is given to be S , where $S = \text{Range}(f)$. By definition of range, for every element $y \in S$, there exists an element $x \in \mathbf{N}$ such that $f(x) = y$. Therefore, f is an onto (surjective) function.

Since f is both one-one and onto, f is invertible.

Step 3: Finding the Inverse Function f^{-1}

Let $y \in S$ such that $y = f(x)$ for some $x \in \mathbf{N}$.

$$\begin{aligned} y &= 9x^2 + 6x - 5 \\ y &= (3x)^2 + 2(3x)(1) + 1^2 - 1 - 5 \\ y &= (3x + 1)^2 - 6 \\ y + 6 &= (3x + 1)^2 \\ \sqrt{y + 6} &= 3x + 1 \quad (\text{since } x \in \mathbf{N} \implies 3x + 1 > 0) \\ 3x &= \sqrt{y + 6} - 1 \\ x &= \frac{\sqrt{y + 6} - 1}{3} \end{aligned}$$

Since $f(x) = y \iff x = f^{-1}(y)$, we have:

$$f^{-1}(y) = \frac{\sqrt{y + 6} - 1}{3}$$

or expressed in terms of variable x :

$$f^{-1}(x) = \frac{\sqrt{x + 6} - 1}{3}$$

Step 4: Evaluating $f^{-1}(43)$ and $f^{-1}(163)$

For $y = 43$:

$$\begin{aligned} f^{-1}(43) &= \frac{\sqrt{43+6}-1}{3} \\ &= \frac{\sqrt{49}-1}{3} \\ &= \frac{7-1}{3} \\ &= \frac{6}{3} = 2 \end{aligned}$$

For $y = 163$:

$$\begin{aligned} f^{-1}(163) &= \frac{\sqrt{163+6}-1}{3} \\ &= \frac{\sqrt{169}-1}{3} \\ &= \frac{13-1}{3} \\ &= \frac{12}{3} = 4 \end{aligned}$$

Final Answer

The inverse function is $f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}$.

The values are $f^{-1}(43) = 2$ and $f^{-1}(163) = 4$.

Common Mistake: Taking both positive and negative roots when solving $(3x+1)^2 = y+6$ is a common error. Since the domain is $\mathbf{N}$, $x > 0 \implies 3x+1 > 0$, so only the positive root is valid.

Question 26

If function $f : \mathbf{R} \rightarrow \mathbf{R}$ is defined by $f(x) = 5x - 3$, then find function $g : \mathbf{R} \rightarrow \mathbf{R}$ such that $g \circ f = I_{\mathbf{R}} = f \circ g$.

TYPE 34 Finding Function using Composite Identity

Given: $f : \mathbf{R} \rightarrow \mathbf{R}$ defined by $f(x) = 5x - 3$, and $g \circ f = I_{\mathbf{R}} = f \circ g$.

To Find: The function $g : \mathbf{R} \rightarrow \mathbf{R}$.

Concept / Formula Used: $g \circ f = I_{\mathbf{R}} \quad f \circ g = I_{\mathbf{R}} \quad g \quad f \quad g(x) = f^{-1}(x)$

Solution:

By definition of inverse function, since $g \circ f = I_{\mathbf{R}}$ and $f \circ g = I_{\mathbf{R}}$, g must be the inverse of f .

Let $y = f(x)$.

$$\begin{aligned}y &= 5x - 3 \\y + 3 &= 5x \\x &= \frac{y + 3}{5}\end{aligned}$$

Since $f(x) = y \iff x = f^{-1}(y)$, we get:

$$f^{-1}(y) = \frac{y + 3}{5}$$

Replacing the dummy variable y with x , the inverse function is:

$$g(x) = \frac{x + 3}{5}$$

Verification of Composite Properties:

$$(g \circ f)(x) = g(f(x)) = g(5x - 3) = \frac{(5x - 3) + 3}{5} = \frac{5x}{5} = x = I_{\mathbf{R}}(x)$$

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{x + 3}{5}\right) = 5\left(\frac{x + 3}{5}\right) - 3 = x + 3 - 3 = x = I_{\mathbf{R}}(x)$$

Final Answer

The required function $g : \mathbf{R} \rightarrow \mathbf{R}$ is $g(x) = \frac{x + 3}{5}$.

Question 27

A teacher wrote $f(x) = x^9$ on the board and asked her student to examine whether the following equation is true or false:

$$[f(x)]^{1/3} = [f^{-1}(x)]^3$$

Raghu said, "It is true". Is he correct? Justify your answer.

TYPE 35 Verifying Properties of Inverse Functions

Given: Function $f(x) = x^9$ on $\mathbf{R}$, and the equation $[f(x)]^{1/3} = [f^{-1}(x)]^3$.

To Find: Determine whether Raghu's claim that the equation is true is correct, with justification.

Concept / Formula Used: $f(x) = x^n$ $f^{-1}(x) = x^{1/n}$ $(x^a)^b = x^{a \cdot b}$

Solution:

First, let us find the inverse function $f^{-1}(x)$. Let $y = f(x) = x^9$. Taking the 9th root on both sides:

$$\begin{aligned}y &= x^9 \\x &= y^{1/9}\end{aligned}$$

Thus, the inverse function is:

$$f^{-1}(x) = x^{1/9}$$

Now, let us evaluate the Left Hand Side (LHS) of the given equation:

$$\begin{aligned}\text{LHS} &= [f(x)]^{1/3} \\ &= (x^9)^{1/3} \\ &= x^{9 \cdot \frac{1}{3}} \\ &= x^3\end{aligned}$$

Next, let us evaluate the Right Hand Side (RHS) of the given equation:

$$\begin{aligned}\text{RHS} &= [f^{-1}(x)]^3 \\ &= (x^{1/9})^3 \\ &= x^{\frac{1}{9} \cdot 3} \\ &= x^{1/3}\end{aligned}$$

Comparing LHS and RHS:

$$\text{LHS} = x^3 \quad \text{and} \quad \text{RHS} = x^{1/3}$$

For general values of $x \in \mathbf{R}$ (e.g., for $x = 8$, $\text{LHS} = 8^3 = 512$ while $\text{RHS} = 8^{1/3} = 2$), we have:

$$\text{LHS} \neq \text{RHS}$$

Therefore, the statement $[f(x)]^{1/3} = [f^{-1}(x)]^3$ is **FALSE**. Raghu is incorrect.

Final Answer

Raghu is **incorrect**. The given equation is **false** because $[f(x)]^{1/3} = x^3$, whereas $[f^{-1}(x)]^3 = x^{1/3}$, which are not equal for all $x \in \mathbf{R}$.

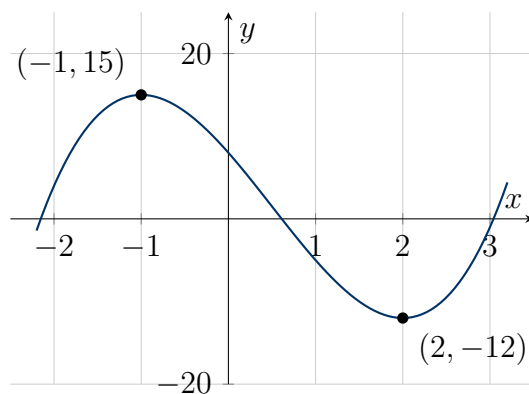
Common Mistake: Confusing exponent rules by assuming $(x^9)^{1/3}$ equals $(x^{1/9})^3$ is a common error; $(x^9)^{1/3} = x^3$ while $(x^{1/9})^3 = x^{1/3}$.

Question 28

A part of the graph of the function $f(x) = 2x^3 - 3x^2 - 12x + 8, x \in \mathbf{R}$ is shown alongside: Answer the following questions:

- Explain why ' f ' does not have an inverse.
- The domain of ' f ' is now restricted to $a \leq x \leq b$ where $a < 0$ and $b > 0$. a and b are chosen so that f has an inverse and the interval $[a, b]$ is as large as possible. Find the domain and range of f^{-1} .

TYPE 36 Invertibility and Inverse under Restricted Domain



Graph of $f(x) = 2x^3 - 3x^2 - 12x + 8$ showing local extrema at $x = -1$ and $x = 2$.

Given: Function $f(x) = 2x^3 - 3x^2 - 12x + 8$, $x \in \mathbf{R}$, with restricted domain $a \leq x \leq b$ where $a < 0 < b$.

To Find: (a) Explanation for why f lacks an inverse on $\mathbf{R}$.
(b) The domain and range of f^{-1} for the maximal interval $[a, b]$.

Concept / Formula Used: $f \quad f'(x) \geq 0 \quad f'(x) \leq 0$

Solution:

Differentiating $f(x)$ with respect to x :

$$\begin{aligned} f'(x) &= \frac{d}{dx} (2x^3 - 3x^2 - 12x + 8) \\ &= 6x^2 - 6x - 12 \\ &= 6(x^2 - x - 2) \\ &= 6(x - 2)(x + 1) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points at $x = -1$ and $x = 2$.

Part (a): Why f does not have an inverse on $\mathbf{R}$

For $x < -1$, $f'(x) > 0$ (function is strictly increasing).

For $-1 < x < 2$, $f'(x) < 0$ (function is strictly decreasing).

For $x > 2$, $f'(x) > 0$ (function is strictly increasing).

Since $f'(x)$ changes sign, $f(x)$ has a local maximum at $x = -1$ with value $f(-1) = 2(-1)^3 - 3(-1)^2 - 12(-1) + 8 = 15$, and a local minimum at $x = 2$ with value $f(2) = 2(2)^3 - 3(2)^2 - 12(2) + 8 = -12$.

Because the function changes direction, horizontal lines $y = c$ (for $-12 < c < 15$) intersect the graph at three distinct points. Thus, multiple distinct x -values map to the same y -value (f is many-one). Since f is not one-one on $\mathbf{R}$, it is not bijective and does not have an inverse on $\mathbf{R}$.

Part (b): Finding domain and range of f^{-1}

We require $a < 0 < b$ such that f is strictly monotonic on $[a, b]$ and $[a, b]$ is maximal.

Since $a < 0 < b$, the interval $[a, b]$ contains $x = 0$.

The continuous interval containing $x = 0$ on which $f'(x) \leq 0$ (strictly decreasing) is $[-1, 2]$.

Thus, $a = -1 < 0$ and $b = 2 > 0$, giving the largest possible interval $[a, b] = [-1, 2]$.

On the restricted domain $[-1, 2]$:

$$f(-1) = 15 \quad (\text{maximum value})$$

$$f(2) = -12 \quad (\text{minimum value})$$

Since f is strictly decreasing on $[-1, 2]$, the range of f on this interval is $[-12, 15]$.

By the properties of inverse functions:

$$\text{Domain of } f^{-1} = \text{Range of } f = [-12, 15]$$

$$\text{Range of } f^{-1} = \text{Domain of } f = [-1, 2]$$

Final Answer

(a) f lacks an inverse on $\mathbf{R}$ because it has local extrema at $x = -1$ and $x = 2$, making it many-one.

(b) For the maximal interval $[-1, 2]$:

$$\text{Domain of } f^{-1} = [-12, 15]$$

$$\text{Range of } f^{-1} = [-1, 2].$$

Question 29

"A function f is called a self-inverse function if $f^{-1}(x) = f(x)$ for all values of x in the domain." Let $f(x) = \frac{\pi^2}{x}$, where $x \neq 0, x \in \mathbf{R}$. Show that $f(x)$ is a self-inverse function.

TYPE 37 Proving Self-Inverse Functions

To Prove: Show that $f(x) = \frac{\pi^2}{x}$ satisfies $f^{-1}(x) = f(x)$ for all $x \in \mathbf{R} \setminus \{0\}$.

Concept / Formula Used: $f \quad (f \circ f)(x) = x \quad f^{-1}(x) = f(x) \quad x \in \text{Domain}(f)$

Solution:

Method 1: By Composite Function $(f \circ f)(x)$

Let $x \in \mathbf{R} \setminus \{0\}$.

$$\begin{aligned}
 (f \circ f)(x) &= f(f(x)) \\
 &= f\left(\frac{\pi^2}{x}\right) \\
 &= \frac{\pi^2}{\left(\frac{\pi^2}{x}\right)} \\
 &= \pi^2 \cdot \frac{x}{\pi^2} \\
 &= x
 \end{aligned}$$

Since $(f \circ f)(x) = x$ for all $x \neq 0$, f is equal to its own inverse f^{-1} .

Method 2: By Finding $f^{-1}(x)$ Explicitly

Let $y = f(x)$.

$$\begin{aligned}
 y &= \frac{\pi^2}{x} \\
 y \cdot x &= \pi^2 \\
 x &= \frac{\pi^2}{y}
 \end{aligned}$$

Since $f(x) = y \iff x = f^{-1}(y)$, we have:

$$f^{-1}(y) = \frac{\pi^2}{y}$$

Replacing y with x :

$$f^{-1}(x) = \frac{\pi^2}{x}$$

Comparing $f^{-1}(x)$ with $f(x)$:

$$f^{-1}(x) = f(x)$$

Hence, $f(x)$ is a self-inverse function.

Hence Proved

Since $f^{-1}(x) = f(x) = \frac{\pi^2}{x}$ for all $x \in \mathbf{R} \setminus \{0\}$, $f(x)$ is proved to be a self-inverse function.